



Motion analysis lab

Investigating inertial motion capture for monitoring gait differences: A case study on simulating ankle-foot orthoses with a novel active device in stroke survivors

Master Thesis

Author:

Jade THERRAS

Supervisors:

Prof. Silvestro MICERA

Prof. Paolo BONATO

Federico BAUDUCCO

Motion Analysis Lab
Spaulding Rehabilitation Hospital
Harvard Medical School

Boston, MA
11th September 2025

Contents

1	Introduction	1
1.1	Study Background	1
1.2	State of the art	4
1.3	Research objectives	7
2	Methods	9
2.1	Study design	9
2.2	Data Collection	10
2.3	Data extraction	11
2.4	Motion capture systems and biomechanical models	12
2.5	Reference frame alignment	14
2.6	Validation of the inertial motion capture system inside the laboratory	16
2.7	Outdoor gait analysis in stroke patients using a motorized ankle-foot Orthosis	20
3	Results	26
3.1	Dataset available after processing	26
3.2	Validation of the inertial motion capture system inside the laboratory	27
3.3	Comparing the habitual and simulated AFO inside of the laboratory	31
3.4	Assessment of the DEPHY performance outside of the laboratory	41
4	Discussion	49
4.1	Motion capture system estimation differences	49
4.2	Reliability of Xsens joint angle trajectories compared to the golden standard indoor	50
4.3	Reliability of Xsens for detecting gait differences across sessions indoor	53
4.4	Comparing the habitual and simulated AFO, inside and outside of the laboratory	54
5	Conclusion	58
5.1	General findings	58
5.2	Next steps	60
A	Comparison of the habitual and simulated AFOs across tasks	i
	Additional project : Characterization of gait progression in Adrenomyeloneuropathy	vi

Abstract

Post-stroke individuals frequently experience asymmetric lower limb impairments, significantly affecting their gait. Ankle-foot orthoses (AFOs) are commonly prescribed to support mobility during rehabilitation and in daily life, aiming to provide mechanical assistance and promote functional movement.

Most often, mechanically passive AFOs are used. These rely on elements such as springs or dampers to store and release energy, supporting balance and forward propulsion. A wide variety of passive AFO technologies exist, differing in characteristics such as stiffness, equilibrium position and dynamic. Choosing the right AFO for a patient is difficult, and being able to test, or simulate, different orthosis could be useful.

The MDxx prototype from Dephy Inc. is a novel motorized ankle-foot orthosis capable of simulating the stiffness behavior of passive devices. This study is part of a broader investigation into the potential of the Dephy system for rehabilitation and focuses on its ability to replicate a patient's habitual orthosis across multiple ambulation tasks, both in and outside of the laboratory.

To analyze gait differences, optical motion capture is considered the gold standard. However, its use is limited to controlled laboratory environments. To assess gait patterns outside the lab, we employed inertial motion capture (Xsens).

The primary objective of this study was to evaluate the reliability of Xsens for estimating joint angles during gait, especially in detecting gait pattern differences. The secondary objective was to assess the capacity of the Dephy to simulate various AFOs across multiple ambulation tasks : straight walk in a hallway, stairs, and ramp.

Our findings show that Xsens provides highly reliable joint angle estimates and successfully captures complex gait patterns. However, it is subject to session-specific offsets. they may falsify the result when comparing across sessions.

The Dephy performed well in simulating flexible to moderately stiff AFOs, across all tasks. However, it showed limitations when attempting to replicate the behavior of very stiff, dynamic, or mechanically complex orthoses such as hinged AFOs.

Acknowledgments

My deepest gratitude to Professor Paolo Bonato and Professor Silvestro Micera.

A big thank you to all the Motion Analysis Lab crew! Thank you for all the sharing, all the knowledge, and all the laughter. A particular acknowledgement to Federico Bauducco for his advice and mentoring along the way. I learned so much from your side. Thanks to the EPFL and to the Neuro-X section, which made this journey possible. Thanks to Spaulding Rehabilitation Hospital for this opportunity to work as a researcher and for all their hard work.

Merci à tout mes amis, vos presences m'honore. Merci à ceux qui m'ont soutenu, de près comme virtuellement, tout au long de ce voyage.

Merci, merci, merci à ma famille et à mes parents. Que de chance de vous avoir à mes côtés ! je vous admire et vous aime de tout mon cœur. Cette aventure (pour ne pas dire cette Aventure) n'aurait pas possible sans vous.

Preface

Dear reader,

I had the incredible opportunity to work at the Motion Analysis Lab, within the Spaulding Rehabilitation Hospital, in Boston. I am extremely thankful for everything I discovered, learned, and contributed to during my time here. Without a doubt, it was the perfect conclusion to six years of studies in engineering at EPFL, the end of one chapter and the beginning of a new one.

Working directly in a hospital truly changed the vision I had of our role as engineers. It made me realize, in a much deeper way, why our work matters and reminded me who should always remain at the center of our efforts: the patients.

During my time here, I contributed to multiple projects, including data analysis, gait analysis, optical and inertial motion capture, biomechanical modeling, and even a few data collections. I was first introduced to optical gait analysis, along with the associated processing and biomechanics, while working on a project aimed at characterizing gait progression in Adrenomyeloneuropathy, a neurodegenerative disease. This project was particularly valuable for developing the work presented in this report, as it provided my first experience with optical motion capture. A detailed report of this project can be found in the appendix.

After working with optical motion capture, my thesis focused on inertial motion capture, which made me realize the importance of accuracy in motion capture systems and their implications. I learned a lot along the way, and I hope you will find it as interesting as I did.

I wish you a pleasant reading,

Best,

Jade Terras

Chapter 1

Introduction

1.1 Study Background

Stroke (cerebrovascular accident) is a major cause of complex disability worldwide and remains a leading cause of death and disability combined. According to the World Stroke Organization and other recent reports, stroke is consistently ranked as the second-leading cause of death and the third-leading cause of death and disability combined globally. [1, 2]

The cost of stroke is a significant global burden, estimated to be over 721 billion US\$ worldwide [1]. While age-standardized stroke incidence rates have declined in many high-income countries due to better prevention and treatment, the absolute number of people experiencing a stroke has increased substantially worldwide over the last three decades, particularly in low- and middle-income countries. This rise is driven by factors such as an aging population and an increase in risk factors like high blood pressure, obesity, and diabetes. [1].

Post-stroke individuals frequently experience hemiparesis, or weakness on one side of the body. Most patients require long-term rehabilitation, and many continue to struggle with an asymmetric lower-limb impairment, a condition known as hemiparetic gait [3], even six months after the stroke. According to Jorgensen et al. [4] half of the patients are unable to walk, even with personal assistance, in the early post-stroke period. Following rehabilitation, 14% remain dependent on assistance, and 22% do not regain walking function.

This gait is characterized by a reduced walking speed, energy-inefficient gait mechanics [5, 6], significant asymmetry between the two sides of the body [7], and foot drop during both the stance and swing phases. There is a notable reduction in the range of motion at the ankle and knee joints, and insufficient muscle activity during the stance phase, especially at the level of plantarflexors [8]. These impairments further complicate stability and movement. [9, 3, 10].

To address the motor deficits associated with hemiparetic gait, Ankle-Foot Orthoses (AFOs) are commonly prescribed for both rehabilitation and daily life [11]. Studies have shown that AFO use can significantly improve a patient's ambulatory function in the short term, particularly at the ankle joint,

regardless of the AFO category [12, 13, 14] .

Most AFOs are mechanically passive, meaning they rely on mechanical components to store and release energy, without using any motor and electronics. Passive AFOs can be divided into 4 different categories : [13, 15, 14].

- (a) **solid or rigid AFOs (non articulated)** : Provide maximum support and control.
- (b) **Articulated AFOs (Hinged)**: Allow for some ankle movement, often in a controlled range.
- (c) **Flexible AFOS (Posterior Leaf Spring)**: More flexible and are designed to assist with dorsiflexion.
- (d) **Dynamic AFOs (Carbon fiber/fiberglass)**: Designed to enhance or resist ankle motion while simultaneously allowing for controlled movement and energy storage.



Figure 1.1: Representation of the 4 most common type of passive AFOs

Within these categories, specific characteristics such as stiffness, neutral angle, or dynamic release (especially during push-off) can be customized to the patient's needs.

Choosing, and afterward tuning AFO to match a patient's individual requirements is a critical concern in the rehabilitation process. Even if all type of AFOs have been shown to impact positively patient's gait, the choose of the right AFO can profoundly impact a patient's comfort and mobility. [14, 20, 21]. This selection must balance the need for support with the goal of promoting rehabilitation.

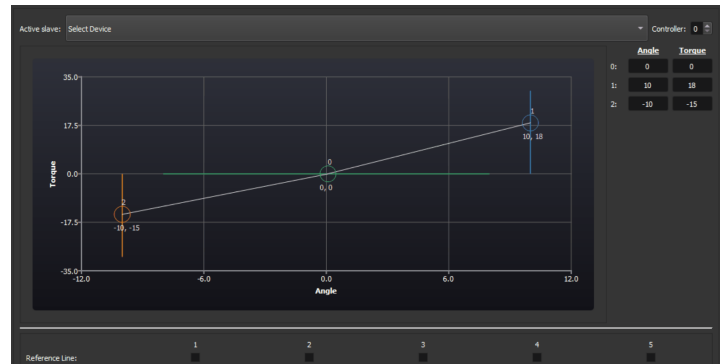
On the other hand, semi-active and active AFOs rely on an external power source, such as a motor, to provide assistance, especially during the push-off phase.

In this context, Dephy Inc. developed the MDxx prototype, a novel powered ankle-foot orthosis based

on research from the MIT Media Lab. The stiffness of the ExoBoot can be tuned both in plantarflexion and in dorsiflexion.



(a) The Dephy device [22]



(b) Screen controller used to tune the device stiffness and choose the neutral angle

This thesis contributes to a broader investigation into the utility of the Dephy device (referred to as the Dephy or DAFO) for rehabilitation 1.2. Specifically, it addresses the device's capacity to simulate multiple AFO types.

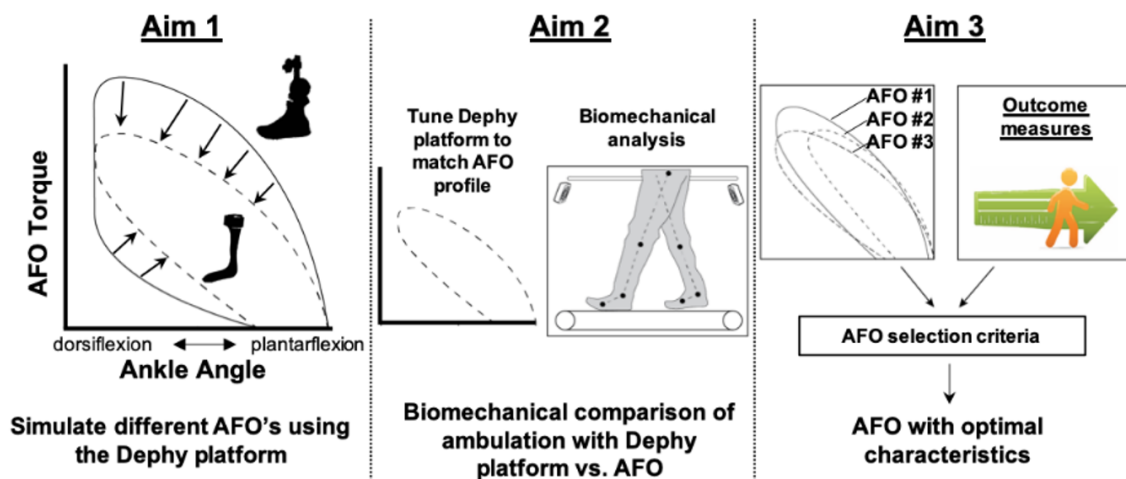


Figure 1. Proposed Research activities

Figure 1.2: Diagram of the research aims of the original study

The ability to simulate various AFO types is particularly valuable. It allows clinicians to test different orthotic characteristics directly with a patient, which can guide the selection of a more appropriate

AFO. Furthermore, this capability enables an experienced clinician to optimize the device's parameters, ensuring a better fit and function for the individual's needs.

1.2 State of the art

1.2.1 Current motion capture technologies

Motion capture technology offers an objective and precise approach to movement analysis. It encompasses technologies designed to precisely track the movement of a body in three-dimensional space.

Nowadays, most motion capture methods used in clinical research follow a similar workflow. This process typically begins with tracking landmarks from the target body, often through the precise placement of markers on anatomical points. Subsequently, a biomechanical model of the patient is created using these landmarks and additional information to fit the patient's body, providing a digital representation of the movement. Finally, motion characteristics such as spatiotemporal parameters and joint kinematics are calculated from this model. Additional technologies, such as force plates, can be integrated to measure ground reaction forces and infer kinetics using methods such as inverse kinetics .

Multiple motion capture methods are employed in clinical research to extract precise kinematic and kinetic gait information about patients in various settings. The main types of motion capture systems commonly used in clinical settings are optical, inertial, and markerless methods. [23] Each type has distinct advantages and limitations.

Marker-based optical motion capture

Marker-based optical systems, such as Vicon [24] and Optitrack [25], rely on the detection of small physical markers placed on the body, either passively reflecting light or actively transmitting light, by high-performance cameras.

They are considered state-of-the-art in terms of precision, providing high accuracy and detailed kinematic data. However, they require controlled environments, especially in the case of passive markers, which need specific lighting conditions. Marker placement can be a time-consuming process and requires trained personnel to ensure consistency and accuracy. These systems also face limitations regarding restricted capture volume, complex setup procedures, and potential occlusion issues, where markers may be obscured from a camera's view. Furthermore, these systems can be expensive and require advanced hardware, often limiting their use to specialized laboratories in clinical research.

Inertial motion capture

Inertial Measurement Unit (IMU) systems, such as Xsens [26] and Shimmer [27] employ wearable sensors to estimate the acceleration, orientation, and angular velocity of body segments.

IMU data, combined with a biomechanical model, allows for the reconstruction of 3D skeletons of people in motion without requiring external visual sensing, offering a distinct alternative to camera-based approaches. [28]

Each IMU embeds a triaxial accelerometer which measures the specific force (i.e., the vector difference between the coordinate and the gravity accelerations), a triaxial gyroscope which measures the angular rate, and a triaxial magnetometer which senses the local magnetic field [29].

The integration of the angular velocity from the gyroscope give a first approximate of the sensor orientation in movement. However, gyroscope usually show bias (ie. if stationnary, they still measure a non-zero angular velocity). In consequence, integrating the angular velocity will lead to a building error in the orientation estimate with time.

The accelerometer work as an inclinometer if stationnary, meaning it can be used to estimate the angle of the sensor with the vertical using the gravity. However, when the sensor is in movement the gravity cannot be separated from the sensor linear acceleration and the inclinaison cannot be estimated. Additionnaly, the accelerometer is also subject to bias and nose.

Finally, the magnetometer act like a compass, giving the orientation in the horizontal plane that the accelerometer is not, but is subject to magnetic disturbance. The magnetometer and the accelerometer combined to give a second estimation of the sensor orientation when the sensor is static.

A sensor fusion algorithm is used to combine the three sensor into a single orientation estimate while limiting the error. Most fusion algorithms can be divided into two categorise, complementary and kalman filter. The gyroscope estimation is use as the primary estimation, and the second estimation is use to update and correct the estimation. The parameter of the fusion algorithm will influence widely the result accuracy [29].

These systems are portable, can be used in multiple environments, and are less susceptible occlusion issues compared to optical systems. They are also generally more affordable than traditional optical motion capture setups. However, IMU systems typically exhibit slightly lower absolute position accuracy when compared to optical systems. The magnetometer is sensible to interference from surrounding metallic objects or electronic devices, which can compromise data accuracy. They are susceptible to drift errors over extended periods, necessitating periodic recalibration to maintain accuracy. The fusion algorithms used to infer IMU trajectories from raw data, and the processing used to account for drift and error, are essential to ensure data reliability. [29]

Markerless optical motion capture

Markerless optical systems typically imply standard video cameras with softwares and machine learning models to track human movement without the need for physical markers on the subject. This

technique offers quick setup times and allows for more natural and unrestricted movement, as subjects are not encumbered by physical markers. It is often considered cost-effective and is suitable for motion capture analysis in outdoor and less contained environments. However, markerless systems generally offer lower precision compared to marker-based optical systems and IMU for biomechanical research. Continuous improvements in tracking accuracy are still required for broader clinical adoption, especially for complex or rapid movements. [30, 31, 32]

1.2.2 Choosing the right motion capture system : optical vs inertial

In the field of motion analysis, a trade-off exists between precision of the measurements, control of the experimental environment and price.

Within the laboratory, highly precise motion capture systems, such as marker-based optical systems combined with force plates, can be utilized, leading to high-quality data.

However, physical constraints, such as the size of the laboratory, limited camera coverage, or the placement of force plates, restrict the number and variety of tasks that can be assessed. In addition, such setup is costly. Finally, a limitation of this approach is that individuals often exhibit different walking patterns inside a lab compared to their natural environments. This is due to several factors, including the artificiality of the setting, the psychological effect of being observed, and the physical discomfort associated with the measurement setup. . These limitations collectively impact the validity of gait pattern assessments and their generalizability to real-world environments.

Inertial motion capture systems offer an interesting compromise between the limitations of existing approaches. Because IMUs are portable, measurements can be performed outside the laboratory, enabling the assessment of a wider range of tasks, and approaching more closely real life conditions. They are also now considered as reliable as markerless motion capture systems [33], do not require cameras, and are less expensive than traditional marker-based optical systems.

Several studies have demonstrated the efficacy of IMU-based motion capture compared to the optical gold standard[33, 34, 35, 36, 37, 38, 39, 40]. While still less precise than marker-based optical systems, IMUs consistently show good reliability in the sagittal plane, regardless of the task assessed.

However, their reliability is influenced by environmental disturbances, the data processing pipeline, and the biomechanical model used. In particular, several studies have reported systematic offsets between optical motion capture and Xsens [38, 41]. One key source of offset is the sensor-to-segment alignment required by IMUs. This alignment is typically established through a static calibration pose (N-pose or T-pose). If the pose is not performed correctly, misalignment is introduced into the model. As Schepers et al. explain: "*In Xsens MVN we need to assume a known pose during calibration, in which the person is asked to stand in the N-pose or T-pose. During this static pose, body segments are assumed to be aligned with the reference pose, whereas the static pose from the optical system is directly*

measured using the positions of the markers attached to the bony landmarks." [41].

As a result, while the Xsens system is consistent within a single session, artificial offsets may occur between sessions if the calibration pose is performed incorrectly. This issue is especially relevant when working with populations with gait or posture impairments, who may not be able to reproduce the calibration pose consistently.

Therefore, its reliability across sessions must be carefully assessed. In particular, when comparing two different conditions recorded at separate time points, it is crucial to verify the system's capacity to detect real differences rather than calibration-related offsets. For this reason, before applying IMU-based motion capture to gait analysis in rehabilitation, its reliability should be validated in the specific context of the study.

1.3 Research objectives

1.3.1 Problem Statement

So far, the DEPHY's capacity to simulate classic AFOs has only been assessed within a laboratory setting, specifically during level-ground walking. To achieve a more general assessment of its performance, a wider variety of ambulation tasks encountered in daily life have been recorded outside of the laboratory, with inertial motion capture only.

Given the trade-off between measurement precision and environmental control in gait analysis, this thesis aims to validate the use of an inertial motion capture system as a reliable tool for accurately quantifying joint angles in the context of this study. An optical motion capture system will serve as the gold standard for validation. Based on these findings, the inertial motion system will then be used to assess DEPHY's performance across a broader range of real-world tasks.

Since stroke survivors often show reduced muscle control and adaptability compared to healthy individuals ??, we expect that any functional discrepancies between the simulated DEPHY and a real orthosis will not be compensated for and will therefore be reflected in patient movement and joint kinematics. However, to obtain a more complete understanding, it would also be beneficial to examine joint kinetics. A significant limitation of the current outdoor experimental setup is the absence of force plates. To address this, an exploratory component of the thesis will focus on estimating kinetics from IMU-based pose data.

1.3.2 Objectives

The objectives of this thesis are as follows:

1. **Validation IMU-based joint angle estimation:** Assess the accuracy of the Xsens inertial motion

capture system by comparing joint kinematics against an optical motion capture gold standard (Vicon) during standardized laboratory gait assessment.

2. **Evaluation of Xsens performance in detecting orthosis-related gait changes:** Determine whether the Xsens system can reliably capture gait differences between walking with the habitual and the simulated ankle-foot orthosis , in comparison to the optical reference system.
3. **Extension of the DEPHY performance assessment outside of the laboratory:** If the inertial motion capture system is found to be sufficiently reliable compared to the optical gold changes between the habitual and the simulated AFO across a broader range of daily-life tasks (e.g., ramps, stairs, hallways) out of the laboratory setting.

1.3.3 Research questions

- How reliable is the Xsens inertial motion capture system for measuring gait kinematics in comparison to an optical motion capture gold standard?
- What are the differences in joint angle estimations between inertial and optical motion capture systems, and which factors contribute most to these discrepancies?
- Can the Xsens system reliably detect gait differences between habitual walking and walking with a simulated AFO, comparable to the optical system?
- To what extent does the DEPHY device replicate the biomechanical effects of conventional passive AFOs across both laboratory and real-world ambulation tasks?

Chapter 2

Methods

2.1 Study design

2.1.1 Participant

The study enrolled patients who had experienced a stroke at least three months before inclusion and who presented unilateral lower-extremity weakness. All participants were able to walk unassisted and used a commercially available AFO in daily life.

Fifteen patients were recruited between 2022 and 2023. The protocol included four in-lab sessions (see Study procedure, Section 2.1.2), but not all participants completed every session. For three participants, a motion-capture system other than Xsens was used for the outdoor recordings, rendering their outdoor data unusable for this analysis; consequently, the analyzed sample was reduced to 12. Of these 12 participants, eight had left-side impairments and four had right-side impairments, resulting in an unbalanced dataset.

All AFO types introduced earlier are represented in the cohort: solid AFO, hinged AFO, posterior leaf spring, and dynamic AFO. Carbon-fiber posterior lateral struts (a type of dynamic AFO) are overrepresented (6 of 12 participants). Only one participant used a solid AFO. Table 2.1 summarizes data availability by AFO type.

2.1.2 Study procedure

The study was composed of four sessions following a similar protocol. In each visit, a different condition was assessed. Sessions are referred to as Habitual, Simulated, Optimal, and Active.

Each visit was composed of an indoor and an outdoor part. The indoor part consisted in a classic instrumented gait analysis, while the second part consisted in multiple more complex ambulation tasks. I did not take part in the data collection and study design.

1. **Habitual** : In the first visit, patients were using their habitual AFO.
2. **Simulated** : For the second visit, the patient was equipped with the DAFO. The stiffness of the

Type of AFO	Characteristics	Number of Subjects	Subject IDs
Carbon Fiber Posterior Lateral Strut	Dynamic AFO. Better energy storage, weight, and durability. Increase push-off force. Supports ankle sagittal plane motion.	6	SpryStep: 11, 12, 14, 15 BlueRocker: 13 EZ Stride: 5
Hinged AFO with posterior stop	Allows dorsiflexion while restricting plantar flexion, aiding push-off.	2	7, 10
Posterior Leaf Spring	Semi-rigid, maintains some ankle motion while providing stability.	3	4, 8, 9
Solid AFO	Good support and stability. Reduces ankle range of motion, high stiffness. Favors knee motion by restricting ankle motion.	1	6

Table 2.1: Number of subject per type of AFOs and Their Characteristics

AFO was adapted to mimic their habitual AFO. The goal of this session was to assess the hability of the DAFO to simulate the properties of multiple AFOs.

3. **Optimal** : In the third session, a therapist assessed the gait of the patient to determine optimal parameters for the DAFO starting with the simulation parameters. The goal of this session was to try to improve the gait of the patient using the DEPHY devices.
4. **Active** : Finally, in the last session, the DAFO was tuned and actived to assist the propulsion in the push off phase. The goal of this session was to assess the impact of active assistance on the patient's gait.

2.2 Data Collection

2.2.1 Indoor intrumented gait evaluation

In the first part of the session, a gait evaluation was conducted indoors under controlled laboratory conditions.

Participants were equipped with both Xsens IMUs and a vicon passive markers (following the CAST markerset) on the lower body. Depending of the visit, they were wearing either their habitual AFO or the DAFO. For balance purpose, the patient were offered to wear a second DAFO in transparent mode on the valid limb .

The data collection protocol included synchronization trials, static trials, and walking trials involving two force plates.

Each recording session consisted of the following phases:

1. **Synchronization Trial:** At the beginning and end of each session, a single Vicon marker and one IMU sensor were manually shaken together to allow temporal alignment of both systems in post processing.
2. **Static Trial:** The participant stand statically on A-pose. This trial serve for the calibration of the biomechanical model of the optoelectric system, and as a baseline for angle offset comparison.
3. **Gait Trials:** Participants walked in a straight line, stepping on two force plates. At least three valid gait trials, defined as those with successful force plate contact, were collected per session. The number of gait trials available per session and patient is highly variable.

2.2.2 Outdoor ambulation tasks

In the second part of the session, participant were asked to do several ambulation tasks, inside and outside of the laboratory.

Similarly to the first part, participants were equipped with Xsens IMUs in the lower body, along with either their habitual AFO, or the DAFO on the affected limb, and may have the DAFO in transparent mode on the valid limb. Ideally, the IMUs were not moved between the first and the second part.

Additionally, a button was also used for task segmentation.

Along with synchronisation, thee tasks were recorded :

1. **Synchronization:** At the beginning and end of the session, the button (A shimmer IMU) and one IMU sensor were manually shaken together to allow temporal alignment of both systems in post processing.
2. **Ramp task:** Participant where asked to walk up and down a ramp 5 times.
3. **Hallway task:** Participant where asked to walk on a straight hallway for at least 10 meters.
4. **Stair task:** Participant had to go up and down stairs. The exercice was adapted to the capacity of the patient, some did several floors up and down were other only did one.

2.3 Data extraction

For the Vicon system, separate files were recorded for each trial using Vicon Nexus software. The data were processed in Visual3D, and kinematic and kinetic outputs for each trial were exported as MATLAB structures.

For the Xsens system, a single file per part (indoor, outdoor) was recorded using Xsens MVN Studio Pro (various versions) and processed with the same software (version 2024). The resulting data were exported to Excel and then converted to CSV format using a custom Python script.

All processing and analyses were conducted in MATLAB (MathWorks, Inc.).

2.4 Motion capture systems and biomechanical models

As this project aims to compare the two different motion capture system, it is essential to understand their particularities. In the following part, we will detail each system : marker placement, biomechanical model, signal generation and processing.

2.4.1 Optical motion capture system : Vicon Nexus

System overview

A Vicon Nexus optical motion capture system, with 10 cameras have been use in combinaison with passive markers, along with force plates. Marker labelling,gap filling, and gait event identification have been done using Nexus. Then Visual 3D was use to define the biomechanical model, filtering and signal generation.

The static calibration is performed in A pose.

Marker trajectories were processed as following : interpolation with polynomial order 3, low pass Butterworth Bidirectional Filter with cutting frequency 6Hz. Joint angles are calculated following an intrinsic cardan sequence Sagittal/coronal/transverse.

Hip, knee, and ankle kinematics were generated, along with ankle and knee kinetics.

Biomechanical model

The CAST (Calibrated Anatomical Systems Technique) (Cappozzo et al., 1995 & 1997) [42, 43] have been used and implemented in Visual3D.

The CAST markerset was introduced to improve biomechanical analyses by reducing errors caused by skin movement artifacts. It combines anatomical markers, placed during a static calibration, with technical marker clusters attached rigidly to each segment. The anatomical markers establish anatomically meaningful reference frames, while the clusters allow consistent tracking of segment motion during dynamic trials. This separation makes the approach more robust and reliable compared to using anatomical markers alone.

In this study, a variation of the method was applied: the lateral and medial markers at the knee and ankle were digitized only during the static calibration trial using a pointer wand.

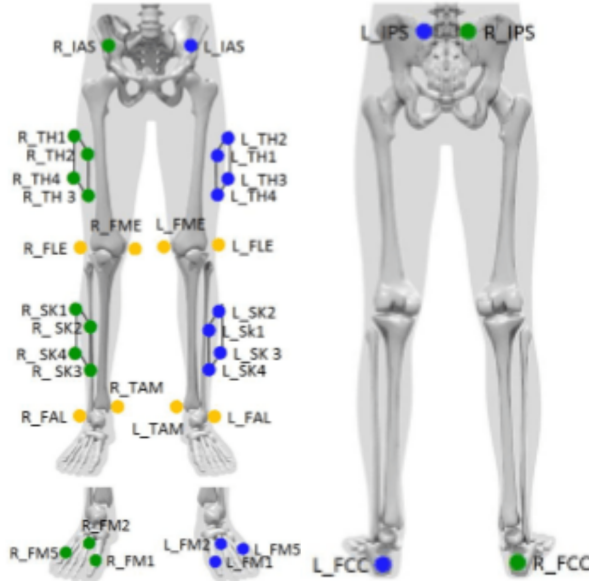


Figure 2.1: CAST markerset [42, 44]

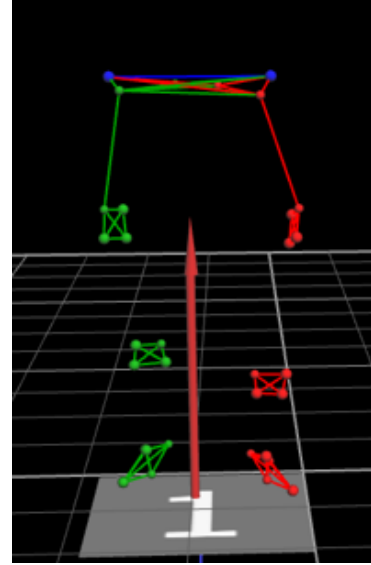


Figure 2.2: Markerset reconstruction in Nexus

2.4.2 Inertial motion capture system : Xsens

System overview

We used MVN Awinda starter from Xsens, with 7 MTw2 IMU sensors placed in the lower body with straps. The recording have been done in MVN Studio Pro (mutiple version) in Multi-Level scenario, with a calibration in N-pose. The data have been HD processed in MVN Studio Pro V2024 before exporting joint angles and output data.

Joint angles are calculated following an intrinsic cardan sequence Sagittal/coronal/transverse similarly as in vicon. We exported the hip, knee, ankle kinematics.

Fusion algorithm

In MVN analyse Pro, the orientation of each sensor is estimated using the proprietary MVN Fusion Engine, a kalman filter.

biomechanical model

We applied the default biomechanical model of Xsens, Biomech 1.0 [41, 45, 46], using only lower-body sensors. The model employs seven sensors placed on the feet, pelvis, upper legs, and lower legs, and is composed of nine segments: pelvis, upper legs, lower legs, feet, and toes.

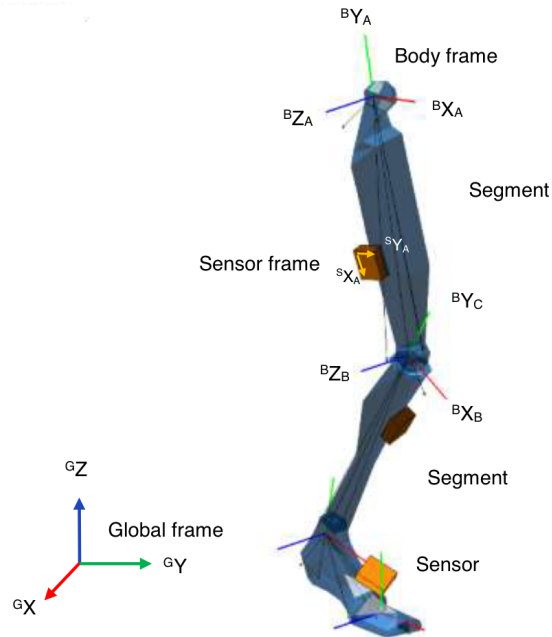
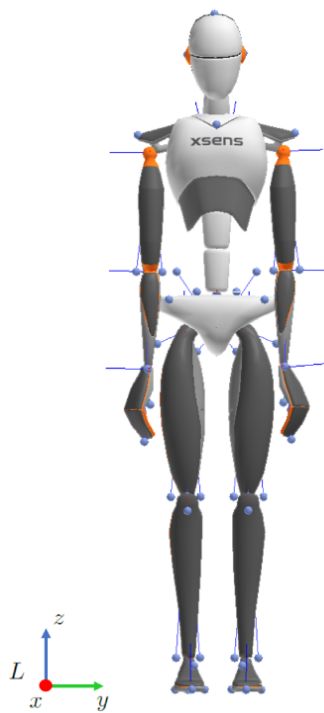


Figure 2.3: Xsens MVN avatar in N-pose (left), all segment coordinate frames are aligned with the common coordinate frame L shown at the bottom left.. [41]

Figure 2.4: Global and local coordinate frames in MVN Studio. Figure from the Xsens User manual. [45]

First, the generic model is scaled to the subject's dimensions based on the provided anthropometric measurements. Next, a sensor-to-segment calibration is performed to establish the alignment between the sensors and the corresponding body segments when the subject is performing a known N-pose. A first estimation of the joint center position is made, and this estimation is adapted using functional calibration when the subject is performing a short walk.

The model operates within two coordinate systems: a global reference frame, in which segment positions, velocities, and related kinematic quantities are expressed, and local coordinate frames, which are used for joint angle estimation.

The model define the hip, knee, ankle and ballfoot joint based on the ISB convention [47]. for joint angles definition [41].

2.5 Reference frame alignment

Previously in the literature, offset between Xsens and optical motion capture systems have been reported, either due to biomechanical model differences [38] or because of poor Xsens calibration and marker placement for the optical system [41]. As we observed similar result, we processed to a

reference frame alignment by using the static trial (A pose) recorded for the vicon calibration as a common baseline for joint angles from both systems. These joint angles (referred to as static angles) were compared visually and quantitatively to assess systematic bias between the two systems. Then, as it was hypothesised that the optical system was the golden standard, the Xsens system have been aligned with the vicon system by adding the difference between the static angles from the two system.

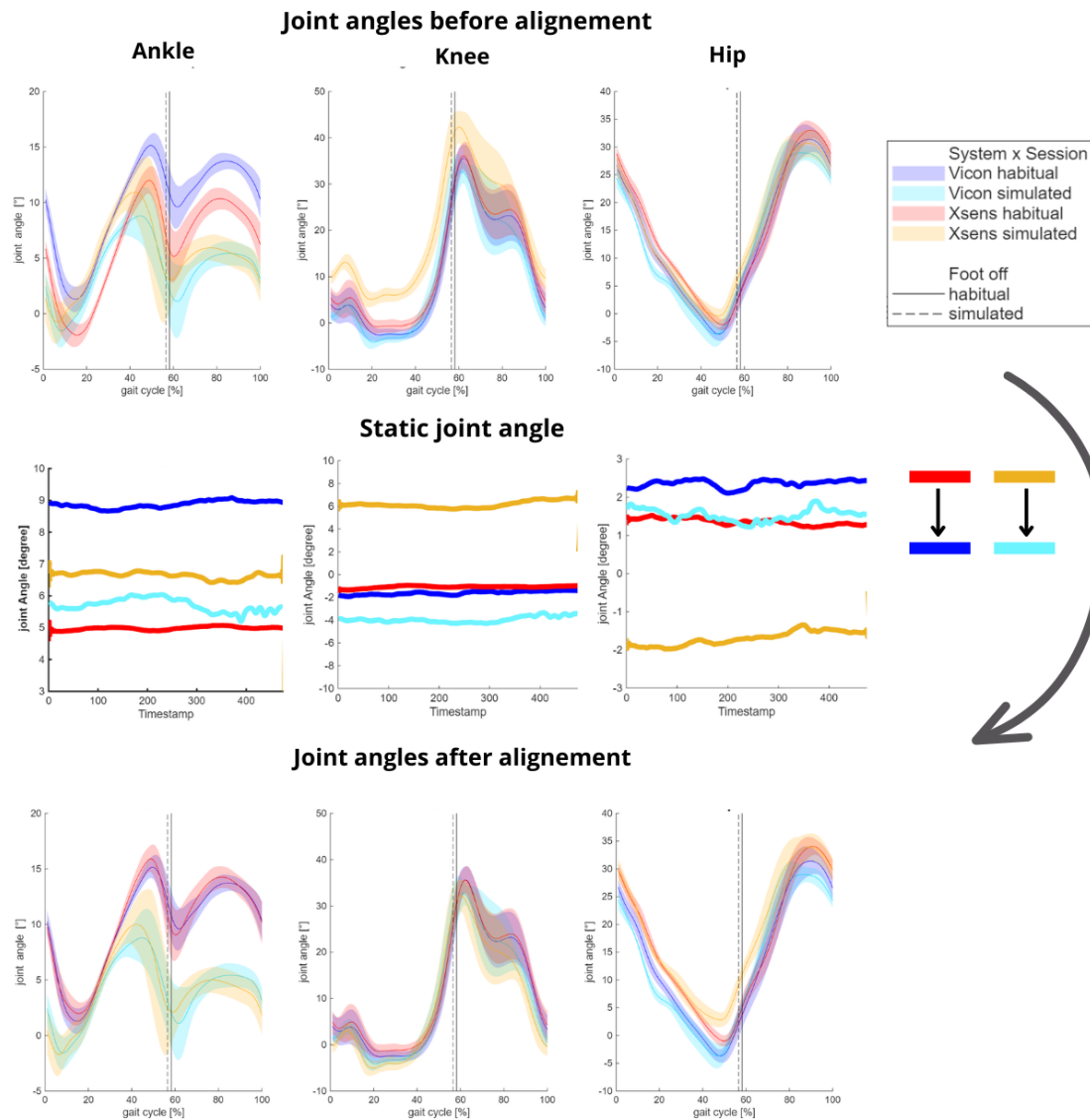


Figure 2.5: Example of the effect of the reference frame alignment. Here, we aligned both session (habitual/simulated) of the Xsens system with the vicon system

Comparing the raw joint angles from both systems assesses the performance of Xsens in measuring the real joint angles, treating Vicon as the gold standard, with both systems referencing the zero of each angles as defined by the ISB convention for joint angles definition [47]. In contrast, comparing joint angles after aligning the reference frames using a common pose evaluates the difference

between the two system without taking into account baseline bias between the two systems. However, even when the zero position corresponds to the same physical pose, differences in reference frame orientation and joint center locations will persist due to biomechanical model variations.

The second approach (reference frame alignment) is less influenced by biases introduced by biomechanical model differences. This is useful when comparing measurements between systems recorded at the same time, but will introduces bias if comparing data recorded from different sessions, because patient may not have exactly the same static pose everytime, especially patient with disabilities.

In the laboratory, the comparison was conducted using both methods: with and without static angle removal. The variability of static offsets across sessions was analyzed to determine whether discrepancies are systematic (assumed constant) or vary between sessions.

2.6 Validation of the inertial motion capture system inside the laboratory

2.6.1 Objective

This part aims to validate the joint angle estimates provided by the Xsens inertial motion capture system, using the Vicon optoelectronic motion capture system as the gold standard. This validation will be conducted within the laboratory. This validation is necessary to ensure that the kinematic data collected only with Xsens can be considered reliable for further analysis.

2.6.2 Data synchronization and segmentation

To align the Vicon and Xsens data streams, accelerometer data from the synchronisation Vicon marker and the corresponding Xsens IMU were extracted. A cross-correlation analysis was performed to compute the temporal delay between the two systems. This time offset was applied to synchronize the Xsens timeframe base on the Vicon reference. Redundant synchronization trials were used for verification and to account for potential missing or noisy data.

Following synchronization, the Xsens recording was segmented using the vicon timestamps associated with each processed trials. The Xsens dataset was reformatted to mirror the Vicon data structure to facilitate direct comparison. Visual inspection was used to validate segmentation accuracy.

To maximize the amount of data available for analysis, full trials were initially included, not just corresponding gait cycle (defined as the interval between two consecutive foot strikes of the same side). However, some outliers appeared at the extremities of the trials on the vicon data, likely due to patient turning. To address this, the metrics were calculated specifically within the gait cycles, reducing the number of data points but ensuring the analysis focused on more relevant and consistent data.

2.6.3 static angles

The joint angles during the static A-pose were extracted for both Vicon and Xsens. These joint angles (referred to as static angles) were compared visually and quantitatively to assess systematic bias between the two systems. Additionally, these angles have been used to calculate the offset between the two systems and align them for both the indoor and the outdoor ambulation tasks. As the Xsens data is not change between

2.6.4 Comparison metrics

To assess the similarity between Vicon and Xsens joint angle estimates, several metrics were computed. A summary is provided in Table 2.2, with detailed explanations given below.

Table 2.2: Summary of comparison metrics between motion capture systems

Metric	Description / Purpose
Static angles	Characterize baseline differences between systems, reflecting true static pose and system-specific errors (e.g., biomechanical model, marker placement). Retained to examine inherent biases.
Difference in static angles	Quantifies system-specific deviations in static poses.
Mean Absolute Error (MAE)	Global measure of agreement; preferred over RMSE due to interpretability of units.
Bias (mean error)	Captures constant errors (systematic offsets).
Standard deviation of the error	Quantifies shape distortion after bias removal.
Pearson correlation coefficient	Assesses directional consistency between systems, independent of offsets or scaling.

Static offsets were analyzed to characterize baseline differences between systems, reflecting both true static pose angles and system-specific errors (e.g., biomechanical model, marker placement). Most comparison metrics were computed after offset removal, but static angles were retained to examine inherent biases.

Mean Absolute Error (MAE) was chosen as the global measure of agreement over Root Mean Square Error (RMSE), as its unit is more easily interpretable [37]. Pearson correlation assessed directional consistency independent of offsets or scaling. Bias captured constant errors, and the standard deviation of the error (also referred to as WDI in previous comparison papers) quantified shape distortion

after bias removal. [37, 38]

The Coefficient of Multiple Correlation (CMC), a correlation coefficient widely used in motion analysis as it accounts for shape, magnitude, and offsets [48, 38], was also calculated. However, it was later omitted from the Results as it provided no additional insight, being largely a combination of Pearson correlation, WDI, and bias. A linear regression slope between systems was also computed to detect amplitude scaling errors but was excluded from the Results due to redundancy with RMSE and bias.

It was hypothesized that within a given session, metrics would remain relatively stable, while variability would be more pronounced across sessions.

Except for static offsets, all metrics were computed per gait trial. Mean and standard deviation of each metric within a session were used to assess intra-session consistency and identify outliers. These results were visualized per aggregated joint using scatter plots and boxplots.

The mean and standard deviation of each metric across all gaits and all sessions were reported in a summary table.

2.6.5 Metric aggregation

Visual inspection of the average joint angles during a gait cycle for both systems was used as an initial assessment of discrepancies between the systems.

For intra-subject analysis, each joint angle can be examined separately. As the objective of this study was also to provide a general characterisation of Xsens reliability, joint data have to be aggregated together. Different aggregation strategies have been examined :

- **By Joint Type:** Left and right joints were combined to calculate global metrics for hip, knee and ankle joint.
- **By Affected Side:** Joint were grouped based on the clinically affected side to test whether range of motion or pathological movement influenced measurement accuracy, ending up with two value per joint, affected and valid limb.

Past study have reported that movement pattern and velocity did not impact significantly XSENS IMU's reliability [36, 49]. However, the range of motion reported may be impacted by the biomechanical model.

To investigate, ttest were performed between affected and affected valid limb Xsens assessment. We looked at significant statistical difference in the MAE and Pearson.

2.6.6 Consistency of Simulation Performance Across Systems

Having assessed the general accuracy of IMU-based joint angle estimation, the next step is to determine if the findings regarding the DAFO's simulation performance are consistent across both the

Vicon and Xsens motion capture systems. If the results are comparable, a similar analysis can then be performed for the outdoor tasks.

Along with visual analysis of joint kinematics, a set of parameters was selected based on previous studies of stroke survivors walking with AFOs [3, 13, 14, 50]. Kinematic parameters can be categorized as (i) joint angles at specific gait events (e.g., ankle angle at foot strike) and (ii) peak values within a gait phase (e.g., maximum knee flexion during swing). Because event detection from kinematics only in the outdoor conditions may be less reliable, priority was given to peak values.

The analysis was restricted to the sagittal plane, where the majority of gait motion occurs. Particular attention was given to the ankle, since AFOs primarily act on ankle kinematics [14], but knee and hip motion were also considered, given evidence that AFOs can indirectly alter their movement patterns.

Previous studies have shown that stroke survivors typically demonstrate: (i) excessive plantarflexion at initial contact, reduced dorsiflexion during stance, and diminished push-off plantarflexion at the ankle; (ii) flexion at initial contact, reduced peak knee flexion in swing and abnormal patterns in stance, resulting in persistent flexion or hyperextension depending on the patient; (iii) reduced hip extension during stance, contributing to shortened stride length [3, 50]. AFO stiffness modulates these patterns, generally reducing ankle range of motion and increasing dorsiflexion at contact [14]. Based on these observations, the following joint angle parameters were selected for analysis.

1. Ankle angle at foot strike
2. Peak ankle dorsiflexion in stance
3. Peak ankle plantarflexion at push-off (late stance)
4. Knee angle at foot strike
5. Peak knee extension during stance
6. Peak knee flexion during swing
7. Peak hip extension in late stance

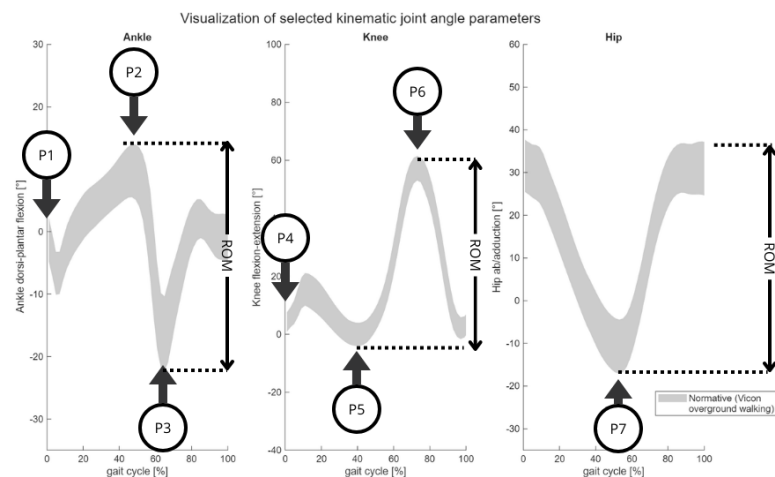


Figure 2.6: All selected joint kinematic parameters

Additionally, the range of motion for the ankle, knee, and hip will be analyzed, as highlighted in Fig. 2.6.

In addition to comparing specific joint parameters, the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Pearson correlation between the joint angle of the habitual and simulated session have been calculated. Since the primary interest is in the influence of the AFO on the gait pattern when the foot is on the ground, the metrics have been calculated in the stance phase.

2.7 Outdoor gait analysis in stroke patients using a motorized ankle-foot Orthosis

2.7.1 Objective

In the following part, the objective is to process and analyze the Xsens data from the outdoor ambulation tasks. Then, the capacity of the dephy to simulated habitual AFOs will be assessed for each tasks.

2.7.2 Task segmentation

To isolate each ambulation task within the continuous Xsens recording, task segmentation was first attempted using a button pressed at the beginning and end of each task. However, the button experienced technical issues and was functional in only 14 of 24 sessions (considering only habitual and simulated sessions).

To address this, an alternative segmentation method was developed based on the pelvis trajectory in a 3D Earth-based reference frame, alongside with validation using the Xsens animation. This method was validated against the sessions where the Shimmer button was operational and subsequently applied to the rest of the dataset, ensuring complete task segmentation for all sessions.

Shimmer Button Synchronization

For the sessions where the Shimmer button was functional, it was shaken alongside an Xsens IMU, mirroring the procedure used in the laboratory to synchronise Xsens and Vicon. The data from both devices were then synchronized by cross-correlating the magnitude of their acceleration signals, aligning the shake events. The order of performed tasks (Ramp,Hallway,Stair) was determined using the clinical reports.

Pelvis Trajectory-Based Segmentation

The pelvis was chosen as a proxy for the patient's position in space due to its relatively stable movement compared to the patient's overall motion in a 3D environment. Using the IMU-based output of the pelvis's 3D motion, its trajectory was preprocessed to extract pattern for each tasks. After Ensuring that all trajectories was positive, a proxy of the trajectory in the horizontal plane ("Horizontal trajectory") was calculated as the sum of vector in the XY plane, while movement in the vertical plane ("Vertical trajectory") was simply the Z-axis component. Both signals were then min-max normalized

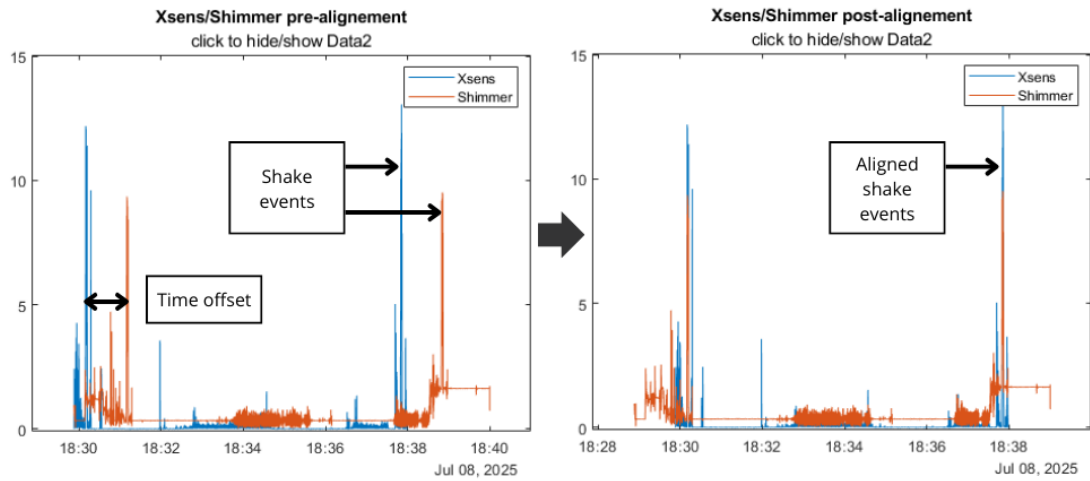


Figure 2.7: Visualisation of the temporal synchronization for a given session

and plotted alongside the Shimmer button output for validation.

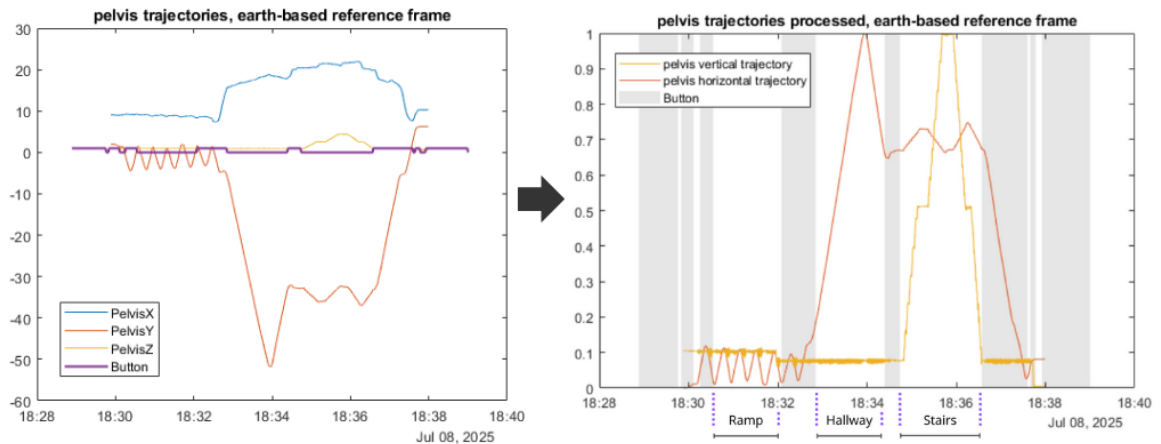


Figure 2.8: Example of a vertical and horizontal pelvis trajectories during a session where the button is available. We can see respectively the ramp, hallway, and stairs exercises.

Based on these plots, distinct characteristics of each task became evident, allowing for clear visual separation:

- **Ramp:** Characterized by a low-magnitude sinusoidal shape in the horizontal plane (representing upward and downward movement on the ramp) and a small vertical movement.
- **Hallway:** Defined by a strong amplitude peak in the horizontal plane and a constant in the vertical plane.
- **Stairs:** Primarily identified by a prominent peak in the vertical plane (indicating ascending and descending stairs), accompanied by a wave in the horizontal plane (reflecting multiple stair landings) or multiple vertical peaks with a similar horizontal wave (for repeated stair climbs/-descents).

2.7.3 Uphill and Downhill Segmentation

Finally, an additional segmentation step was implemented for the ramp and stairs to differentiate between uphill and downhill movements. For the ramp, this involved separating ascent and descent while removing the turning phases at the top and bottom. For the stairs, it entailed isolating up and down motions and discarding stair landings and turning at the top and bottom. Finally for the hallway, sometimes the subject was turning before the end of the task and coming back. This task was therefore segmented by removing turn and keeping straight lines.

Each of these segmentations was achieved using a simple algorithm detecting slope change : For the hallway, the pelvis motion in the horizontal plane was used as the primary signal. Between the X and the Y component, the one capturing most of the motion was chosen and smoothed to reduce noise. For ramp segmentation, the same algorithm was used to detect change of direction in the horizontal plane. As the change in the Z direction was not detectable, it was assumed that the patient was first going up the ramp then down, and it was verified using the video. For stairs, the movement of the pelvis in the vertical plane (Z) was used. Parameters for this algorithm were adapted as necessary for optimal performance across different trials.

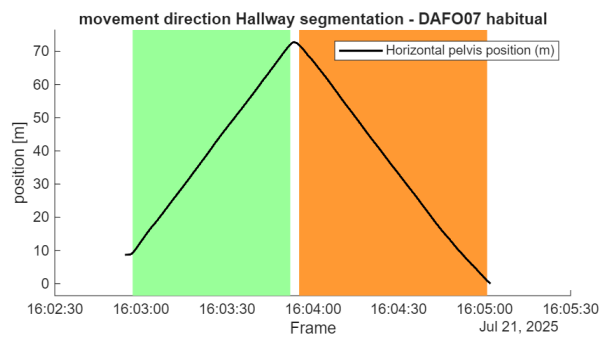


Figure 2.9: Hallway

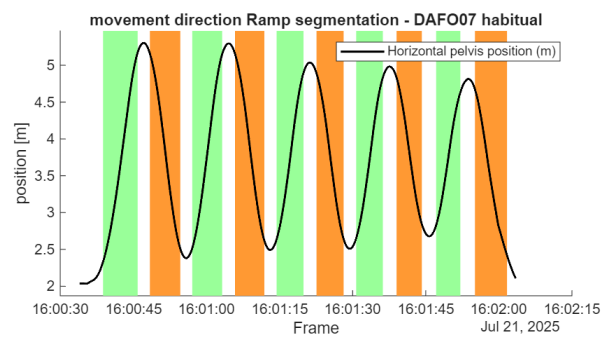


Figure 2.10: Ramp

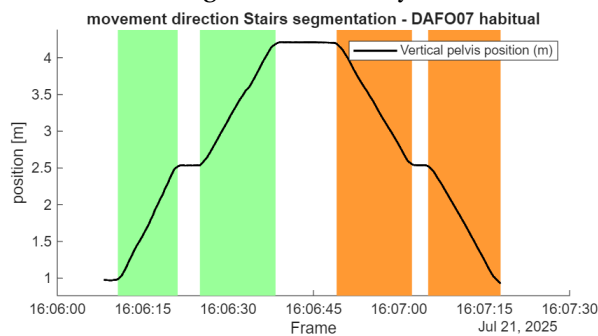


Figure 2.11: Stairs

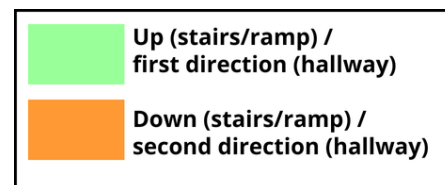


Figure 2.12: Example of direction segmentation for each ambulation task

2.7.4 Gait cycle segmentation and gait event recognition

For each task, gait cycles were extracted. In the literature, several algorithms have been proposed for gait event recognition based on kinematic data [51, 52, 53, 54, 55]. However, most of these methods were developed using healthy subjects and treadmill or overground walking.

French et al. [51] tested three classic algorithms with stroke survivors during overground walking: the horizontal sacral–heel distance (SHD), the horizontal ankle–heel distance (AHD), and horizontal velocity (HV), and concluded that SHD provided the most accurate results for foot-off detection, while AHD performed best for foot-strike detection.

Based on these findings, we implemented custom MATLAB versions of both methods, adapted for Xsens kinematics.

AHD method The AHD method defines gait events based on the horizontal distance between the ipsilateral heel and contralateral lateral malleolus markers, with maxima indicating foot strike and minima indicating toe off [52]. While implemented for comparison, it was not used in the final analysis due to inconsistent performance in stair and ramp tasks.

SHD method The SHD approach defines gait events based on the relative horizontal distance between the heel and sacral markers in the direction of the movement (so negative is the foot is on the back and positive when in front). [55]. Specifically, foot strike corresponds to the maximum heel–sacrum distance (peak of the sinusoidal profile), while toe off corresponds to the minimum toe–sacral distance. With Xsens, as we don't have heel markers, we used the foot position and the toe position.

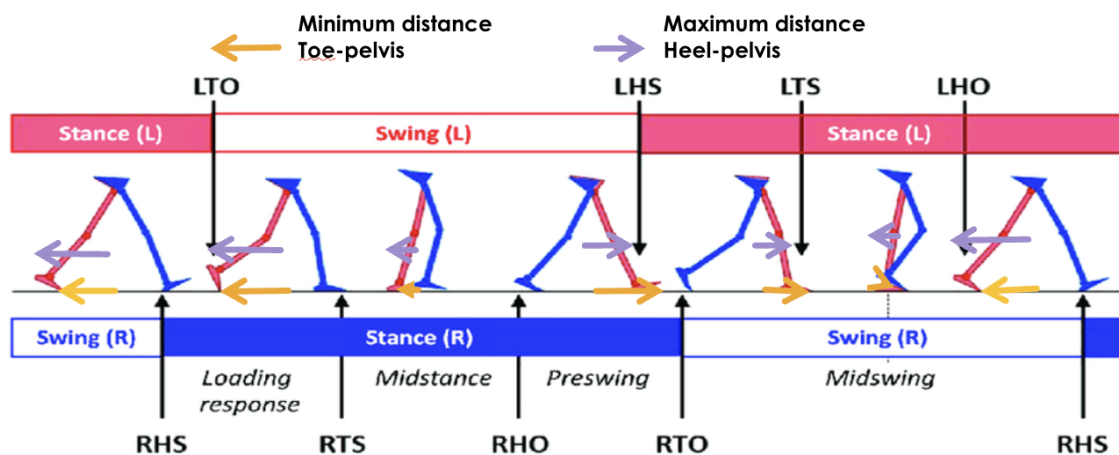


Figure 2.13: General principle of the SHD methods

During visual assessment, however, SHD was found to outperform AHD in stair and ramp tasks, where AHD occasionally produced invalid detections (e.g., foot strike following foot off). Therefore, SHD was ultimately used for both foot strike and foot off detection across all tasks. Automatic outlier

detection was added to improve robustness, and inconsistent events were removed before further analysis.

Implementation of the gait cycle segmentation Both methods assume motion along a principal direction of progression. The MATLAB implementation was similar for both methods, differing only in the relative displacement computation step.

The function was implemented as follows (for the SHD method) :

1. **Estimation of the principal direction of progression:** Xsens provides segment positions in a 3D earth-based reference frame. After task segmentation, all turns are removed. We then assume that the participant is moving in a relatively straight direction and estimate the overall direction of progression.
2. **Marker projection:** Pelvis, foot, and toe trajectories were projected onto the main movement axis. See Figure.2.14
3. **Relative displacement computation:** The relative distances between the projected markers were calculated. For the SHD method, this consisted of the foot–pelvis and toe–pelvis distances at each timestep.
4. **Event detection:** Local maxima and minima in the resulting sinusoidal profiles were identified using MATLAB's *islocalmax* and *islocalmin* functions. These peak values represent the timing of the gait events. See Figure.2.15
5. **Gait cycle construction:** To prevent impossible event sequences (e.g., due to misdetections), events were combined into complete gait cycles and retained only when the full sequence was present: foot strike → opposite foot off → opposite foot strike → foot off → foot strike.
6. **Outlier handling:** This final step removes gait cycles that are likely to be invalid. Outliers were identified when a parameter deviated more than 3 standard deviations from the mean for that specific gait:
 - **Gait length outliers:** For example, when a foot strike is missed or the participant stops mid-gait, resulting in a gait cycle much longer than expected.
 - **Foot-off timing outliers:** Foot-off events can be misdetected due to irregular movements, appearing earlier or later. For example, if a foot-off is missed, the sequence may capture the next foot-off instead.

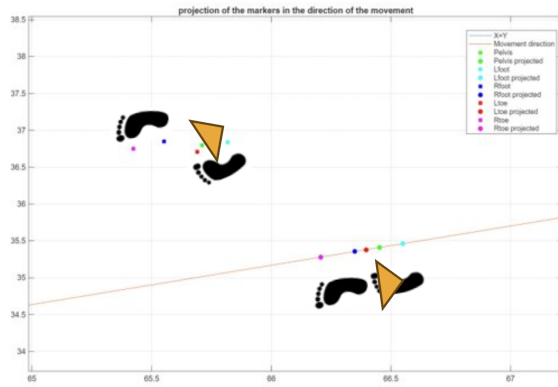


Figure 2.14: Step 2 : projection of the markers in the direction of the movement

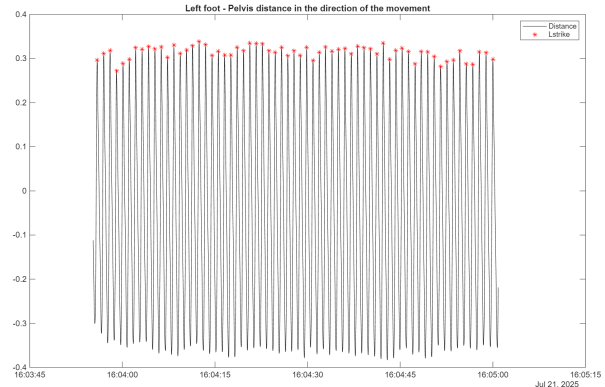


Figure 2.15: Step 4 : Event detection using mini-mas and maximas of sinusoidal trajectories

Figure 2.16: Visualisation of key steps of the gait cycle segmentation

2.7.5 Habitual vs Simulated AFO comparison

As a reminder, in the indoor part we used gait cycle timings extracted manually from the optical data (using the force plate) to compute gait patterns and compare habitual and simulated conditions.

After performing gait cycle segmentation for each ambulation task, we conducted a similar analysis to compare simulated and habitual AFOs on the outdoor data.

Based on the findings from the indoor data comparison, we focused on the ankle angle in the sagittal plane. Specifically, the following assessments were performed :

- Visual comparison of ankle angle trajectories.
- RMSE of the ankle angle during the stance phase between habitual and simulated AFOs.
- Pearson correlation of the ankle angle during the stance phase between habitual and simulated AFOs.
- Differences in ankle range of motion (ROM) between habitual and simulated conditions.
- Key ankle kinematic events, including peak dorsiflexion (DF), peak plantarflexion (PF), foot strike angle, peak DF during stance, and peak PF at push-off.

Chapter 3

Results

3.1 Dataset available after processing

After processing, only patient where data from both motion capture systems (Xsens and Vicon) and both conditions (habitual and simulated AFO) were available were retained for analysis.

Table.3.1 presents the dataset obtained after alignment, segmentation, and quality assessment. In total, 10 subjects were included, equipped with AFOs ranging from flexible to moderately stiff. No solid or highly stiff AFOs were represented. The only subject with solid AFO (subject 6) was excluded due to marker displacement during recording.

Table.3.2 further summarizes the availability of data across all indoor and outdoor tasks.

Type of AFO	Available indoor	Available outdoor	Subject IDs
Carbon Fiber Posterior Lateral Strut	5	5	5, 11, 12, 14, 15
Hinged AFO with posterior stop	2	1	7, (10)
Posterior Leaf Spring	3	3	4, 8, 9

Table 3.1: Number of subject per type of AFOs in the final dataset

	Comparison Habitual-Simulated Summary						NOTE
	Indoor	Hallway	Steps		Ramp		
			Ascent	Descents	Ascent	Descents	
DAFO04	✓	✓	✓	✓	✓	✓	
DAFO05	✓	✓	✓	✓	✗	✗	No static trial no ramp data habitual
DAFO06	✗	✗	✗	✗	✗	✗	No static trial Vicon marker moved
DAFO07	✓	✓	✓	✓	✓	✓	
DAFO08	✓	✓	✓	✓	✓	✓	
DAFO09	✓	✓	✓	✗	✓	✓	Bug in stair descents (stay in the sam height)
DAFO10	✓	✗	✗	✗	✗	✗	No outdoor task recorded
DAFO11	✓	✓	✓	✓	✓	✓	
DAFO12	✓	✓	✓	✓	✓	✓	
DAFO13	✗	✗	✗	✗	✗	✗	No data simulated outdoor, not synchronised indoor
DAFO14	✓	✓	✓	✓	✓	✓	
DAFO15	✓	✓	✓	✓	✓	✓	Small number of gait cycle for the stairs
Final number of patient	10	9	9	8	8	8	

Table 3.2: Summary of data availability for each subject (rows) and task (columns). A cell is highlighted in green when data are available from both conditions (habitual and simulated AFOs) and from both motion capture systems (Vicon and Xsens).

3.2 Validation of the inertial motion capture system inside the laboratory

3.2.1 Within-session Xsens reliability compared to the golden standard

To assess the within-session reliability of Xsens (whether its error remain consistent over time or accumulate) we analyzed the standard deviation of the mean absolute error (MAE) between the joint angle trajectories of Xsens and the golden standard within each session. Figure 3.1 shows, for each joint angle, the distribution of these standard deviations across all sessions.

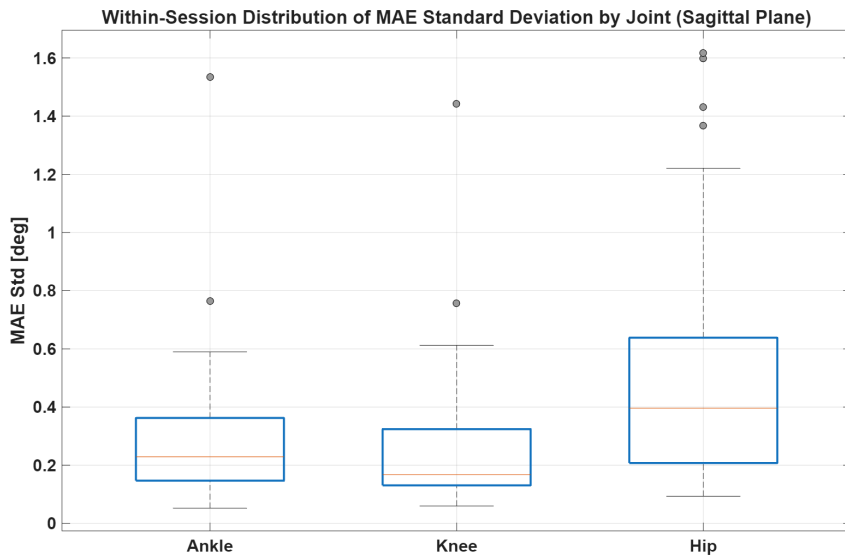


Figure 3.1: Distribution of the standard deviation of the error between Xsens and Vicon within the same session across all sessions, for each angles

The MAE between Xsens-derived joint angles and the Vicon gold standard showed low variability within sessions, with an average standard deviation below 0.5° for all angles and outliers not exceeding 2° .

These findings show that Xsens does not accumulate drift or time-dependent error within a session..

3.2.2 Reliability of Xsens joint angle trajectories compared to the golden standard

Since Xsens does not show building error within a session, the gait pattern of each subject can be estimated using the mean of multiple gait cycles. To assess the reliability of Xsens joint angle trajectories, we compared the mean joint angle during a gait cycle extracted from Xsens with the mean joint angle extracted from the gold standard.

All comparison metrics between the Xsens joint angles and the gold standard are presented in Table 3.3.

		MAE	PEARSON	STD	BIAS
Hip	Sagittal	9.89 (8.41)	0.99 (0.01)	1.59 (0.60)	-7.51 (10.12)
	Coronal	5.09 (2.89)	0.75 (0.23)	2.40 (1.09)	-0.86 (1.78)
	Transverse	9.84 (4.09)	0.82 (0.18)	2.36 (2.05)	1.46 (7.78)
Knee	Sagittal	6.14 (3.63)	0.99 (0.01)	1.55 (0.56)	0.88 (6.40)
	Coronal	3.22 (1.64)	0.51 (0.42)	2.35 (1.62)	0.70 (2.78)
	Transverse	14.08 (10.29)	0.82 (0.18)	2.33 (1.56)	1.94 (4.84)
Ankle	Sagittal	4.83 (1.81)	0.92 (0.08)	1.77 (0.53)	1.10 (4.50)
	Coronal	8.57 (8.50)	0.80 (0.12)	2.24 (0.78)	-1.35 (6.17)
	Transverse	9.26 (6.03)	0.84 (0.14)	1.49 (0.53)	-8.04 (7.28)

Table 3.3: Comparison of raw joint angle trajectories from Xens and Vicon

MAE values ranged from 3° to 15° , with relatively high variability across joints and planes (standard deviation of the same order of magnitude as mean).

Pearson correlation coefficients indicate excellent correlation in the sagittal plane (>0.9 , $SD < 0.1$) and good correlation in the other planes (above 0.7), though with greater variability. The only exception was knee varus–valgus, where the correlation dropped to 0.51 ± 0.42 , indicating extreme variability and low correlation in general.

The standard deviation of the error remained below 3° across all planes, with comparable values between them, indicating no big change in the error accross the gait cycle.

Interestingly, bias values showed high variability across all angles and planes, with standard deviation of the same order of magnitude as the MAE. While the mean bias was close to 0° in most cases, hip flexion–extension (sagittal plane) exhibited a notable mean bias of -7.74° and the largest standard

deviation, indicating that Xsens show constistantly higher extension of the hip than vicon. Additionally, the ankle adduction-abduction (transverse plane) show a mean bias of -8.04° with a high standard deviation.

3.2.3 Static angle investigation and bias removal

In an attempt to capture the bias between the two systems, joint angles during a static pose (referred to as static angles) were extracted for both motion capture systems and compared. These angles and their differences are reported in Table 3.4.

Interestingly, the bias between in static was comparable to the bias dynamic reported in Table.3.3. The difference between systems show high variability for all angles. Again, in the sagittal plane, the hip showed the highest average offsets: $-9.30^{\circ} \pm 8.87^{\circ}$. Both systems show similar static angles in general, except for the hip where the vicon system show a consistent higher value. Vicon static angles showed higher variability across sessions than Xsens.

		Vicon static	Xsens static	Static Bias
Hip	Sagittal	9.73 (9.79)	0.44 (5.91)	-9.30 (8.87)
	Coronal	0.97 (2.22)	0.78 (1.19)	-0.18 (1.81)
	Transverse	-3.25 (7.16)	-0.30 (1.10)	2.95 (7.19)
Knee	Sagittal	2.40 (7.58)	3.54 (3.38)	1.14 (6.17)
	Coronal	-1.01 (2.49)	-0.08 (0.81)	0.93 (2.61)
	Transverse	-1.93 (5.04)	0.42 (0.78)	2.34 (5.10)
Ankle	Sagittal	6.81 (5.55)	5.44 (2.17)	-1.37 (5.41)
	Coronal	0.16 (7.04)	-0.18 (1.57)	-0.34 (6.93)
	Transverse	10.33 (6.02)	-3.35 (6.39)	-13.68 (4.85)

Table 3.4: Comparison of joint angle during the common reference pose from Xens and Vicon

After aligning the two systems, the MAE and Bias were recalculated. Table 3.5 summarizes all metrics after correction. the MAE values decreased below 5° across all joints and planes. As expected, the bias variability dropped, and was found to be below 2° for all angles and plane.

		MAE		Bias	
		Before	After	Before	After
Hip	Sagittal	9.89 (8.41)	3.95 (2.67)	-7.51 (10.12)	1.78 (4.33)
	Coronal	5.09 (2.89)	2.60 (1.11)	-0.86 (1.78)	-0.67 (1.20)
	Transverse	9.84 (4.09)	4.26 (2.26)	1.46 (7.78)	-1.49 (2.61)
Knee	Sagittal	6.14 (3.63)	1.83 (0.67)	0.88 (6.40)	-0.76 (1.26)
	Coronal	3.22 (1.64)	2.55 (1.35)	0.70 (2.78)	-0.09 (1.42)
	Transverse	14.08 (10.29)	4.67 (2.69)	1.94 (4.84)	-0.66 (3.64)
Ankle	Sagittal	4.83 (1.81)	2.39 (1.02)	1.10 (4.50)	-0.80 (1.55)
	Coronal	8.57 (8.50)	2.80 (0.93)	-1.35 (6.17)	-0.28 (2.07)
	Transverse	9.26 (6.03)	4.79 (2.96)	-8.04 (7.28)	0.41 (4.25)

Table 3.5: MAE and Bias between Xsens and Vicon before and after alignment

3.2.4 Joint aggregation investigation

The metrics, after alignment, were also analyzed separately for the impaired side (i.e., the limb affected by the patient's stroke) and the unimpaired side. The results are summarized in Table 3.6.

This separation is motivated by the fact that the two sides may show highly different gait patterns. The unimpaired limb typically shows a larger range of motion and higher speed, while the impaired limb tends to be less predictable and deviates more strongly from normative gait. We will also exclusively focus on the impaired side for the rest of this study.

Overall, no major differences in Xsens performance compared to the golden standard were observed between the impaired and unimpaired limbs. The MAE was slightly higher for the unimpaired limb in the coronal plane, particularly for the hip and ankle.

Correlation values were excellent in the sagittal plane for both limbs, though a small decrease was observed for the impaired ankle compared to the unimpaired side. Aside from these differences, bias and the standard deviation of the error remained comparable between limbs, indicating that impairment status had little effect on the consistency of Xsens measurements.

		MAE	PEARSON	STD	BIAS
Impaired Hip	Sagittal	3.73 (2.61)	0.98 (0.03)	1.50 (0.79)	1.81 (4.10)
	Coronal	2.09 (0.91)	0.76 (0.31)	1.89 (0.79)	0.14 (1.69)
	Transverse	4.36 (2.14)	0.83 (0.17)	2.45 (2.08)	-4.02 (2.08)
Unimpaired Hip	Sagittal	4.18 (3.00)	0.99 (0.01)	1.68 (0.64)	1.76 (4.77)
	Coronal	3.10 (1.67)	0.74 (0.28)	2.91 (1.70)	-1.48 (1.82)
	Transverse	4.16 (3.28)	0.80 (0.20)	2.26 (2.06)	1.03 (4.70)
Impaired Knee	Sagittal	1.88 (0.91)	0.99 (0.01)	1.53 (0.68)	-0.41 (1.80)
	Coronal	2.31 (1.43)	0.50 (0.50)	2.26 (1.87)	-1.09 (1.77)
	Transverse	5.07 (3.29)	0.92 (0.12)	2.23 (1.61)	-0.27 (6.01)
Unimpaired Knee	Sagittal	1.78 (1.09)	1.00 (0.00)	1.57 (0.68)	-1.11 (1.35)
	Coronal	2.80 (2.05)	0.53 (0.50)	2.44 (1.86)	0.92 (3.17)
	Transverse	4.27 (2.79)	0.72 (0.29)	2.43 (1.66)	-1.05 (4.70)
Impaired Ankle	Sagittal	2.28 (1.36)	0.88 (0.12)	1.61 (0.56)	-0.82 (2.29)
	Coronal	2.11 (0.97)	0.86 (0.12)	1.55 (0.46)	-0.16 (2.06)
	Transverse	5.97 (4.70)	0.78 (0.22)	1.35 (0.56)	-0.67 (7.67)
Unimpaired Ankle	Sagittal	2.49 (1.18)	0.96 (0.06)	1.92 (0.86)	-0.78 (2.22)
	Coronal	3.48 (1.56)	0.74 (0.20)	2.94 (1.42)	-0.41 (3.19)
	Transverse	3.62 (2.57)	0.90 (0.11)	1.63 (0.73)	1.50 (4.13)

Table 3.6: Comparison metrics between Xsens and Vicon joint angle trajectories. Metrics were calculated for all patients and all limbs, with impaired and unimpaired side aggregated separately to account for the substantial differences in joint kinematics between limbs

3.3 Comparing the habitual and simulated AFO inside of the laboratory

As previously noted, Xsens present session-specific offset. Therefore, the following comparison was done after the alignment.

The following results summarize the comparison between habitual and simulated AFOs across both motion capture systems. To present the data clearly, two types of graphs were used:

1. **Parameter visualisation :** Vicon (blue) and Xsens (red) results are displayed separately for each parameter (and session, when applicable). Each subject is represented by a unique color, with connecting lines used to highlight agreement or disagreement between the two systems.
2. **Inter-system difference visualisation :** To assess the level of agreement between the two motion capture systems, we then evaluated the difference in parameter estimates directly, computed as $P_{Xsens} - P_{Vicon}$. This quantifying how close Xsens measurements are to the golden standard.

The comparison followed a structured sequence:

- **RMSE in the stance phase:** As a first step, we quantified the overall similarity in joint angle trajectories by analyzing the RMSE between habitual and simulated AFOs for each joint angle and each motion capture system (Figure 3.2). This provides a global measure of agreement between habitual and simulated AFO.
- **ROM and kinematic parameters:** Next, we examined more specific gait characteristics, including range of motion and selected kinematic parameter values for both motion capture systems and both conditions (habitual vs. simulated AFO) (Figure 3.3).

Since our primary focus is the comparison between habitual and simulated AFOs, we are more interested in the difference between the two sessions gait pattern than the absolute values of the joint angles. The following comparison focuses on these differences.

- **Condition differences:** We calculated $P_{Simulated} - P_{Habitual}$ for each motion capture system (Figure 3.5). This isolates the differences between the habitual and simulated AFOs.
- **Consistency across motion capture system:** Finally, to evaluate whether Xsens and Vicon capture gait pattern differences in a similar way, we calculated the “difference of differences”: $(P_{Xsens,Simulated} - P_{Xsens,Habitual}) - (P_{Vicon,Simulated} - P_{Vicon,Habitual})$ (Figure 3.6). This final step directly addresses whether Xsens can replicate the AFO-dependent findings observed with the gold standard.

RMSE between habitual and simulated AFOs

At the group level, the distribution of RMSE between habitual and simulated AFOs was highly similar for Xsens and Vicon across all joint angles (Figure 3.2a). At the subject level, the outputs of the two motion capture systems showed nearly identical RMSE values, with connecting lines in the plots being almost perfectly horizontal. Agreement was particularly strong at the ankle, where RMSE values for the same subject were visually really close between the two systems, with the exception of a single subject in the cohort, DAFO09.

To quantify these differences more explicitly, we examined the inter-system differences, defined as $RMSE_{Xsens} - RMSE_{Vicon}$ (Figure 3.2b). Here, the agreement between systems was confirmed. All mean differences were close to zero, indicating no systematic bias in RMSE between the two systems. Small deviations at the ankle and knee were attributable to isolated outliers. Specifically, mean differences were 0.79° , 1.7° , and 0.6° for the ankle, knee, and hip, respectively.

The variability of these differences was also very low: the standard deviation was below 1° for both the ankle and hip, and below 2° for the knee. In practical terms, this means that for the vast majority of subjects, the RMSE values computed with Xsens and Vicon differed by less than 1° at the ankle and hip, and by less than 2° at the knee. Such small discrepancies are within the expected measurement noise for human motion analysis.

Only one clear outlier was observed at the ankle, where subject DAFO09 showed a difference greater

than 2° (highlighted with a red circle in the figure). Examination of the gait trajectories for this subject revealed an unusual drift in dorsiflexion within the gait cycle for Xsens, a behavior not present in the Vicon data. This was therefore considered an artefact.

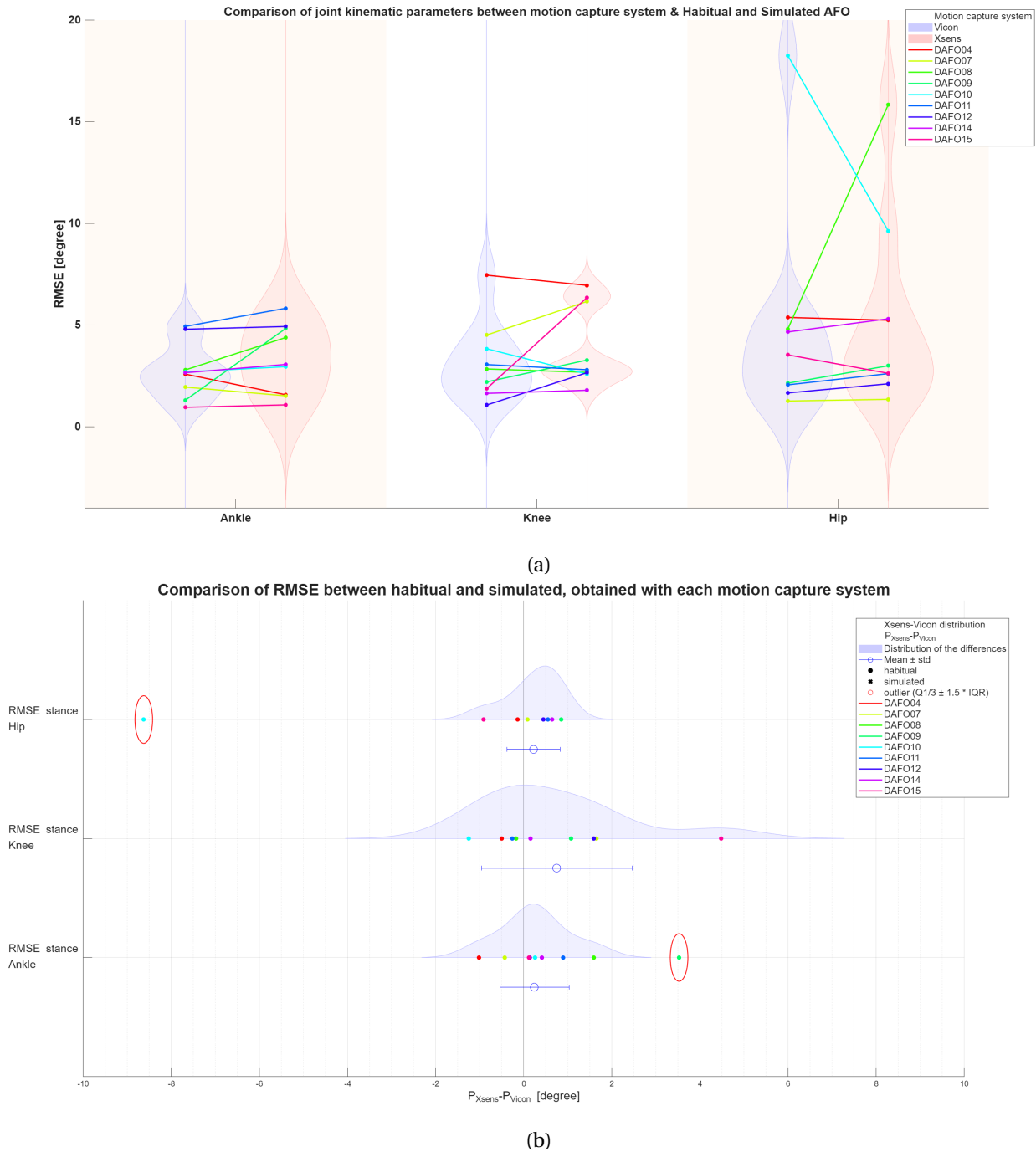


Figure 3.2: RMSE between habitual and simulated AFOs for both motion capture system : (a) Comparison of the RMSE (a) RMSE difference between Xsens and Vicon

These results confirm that Xsens can be reliably used to estimate the RMSE between habitual and

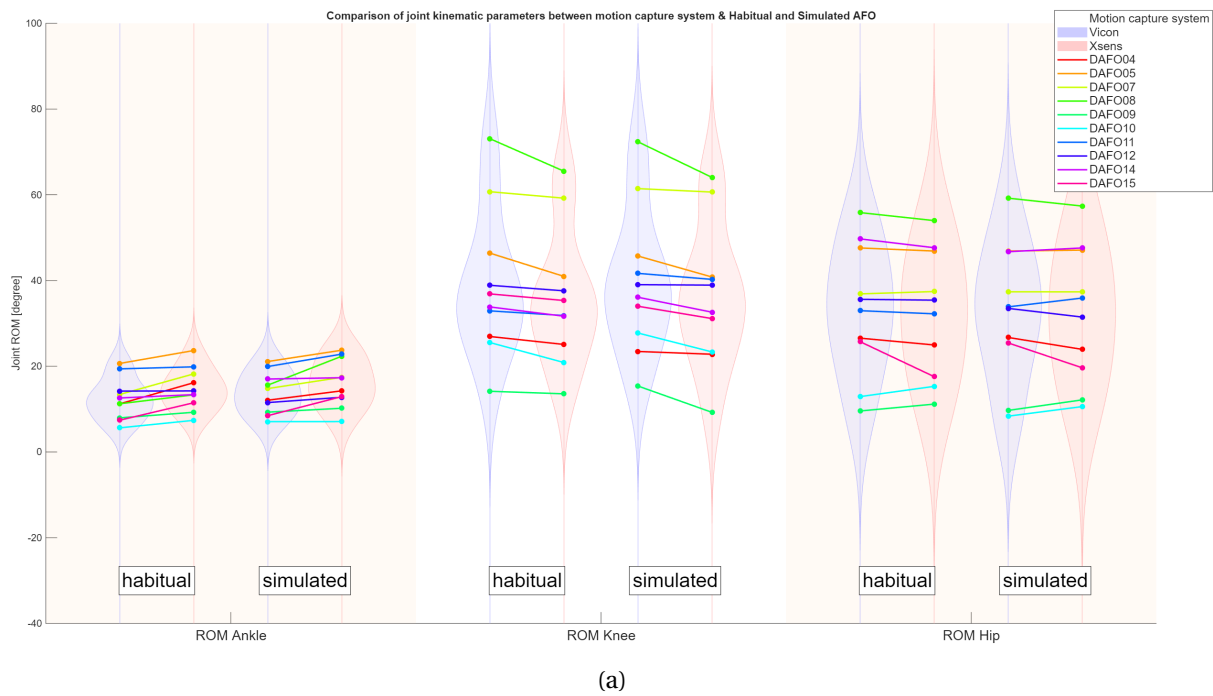
simulated AFO joint angles.

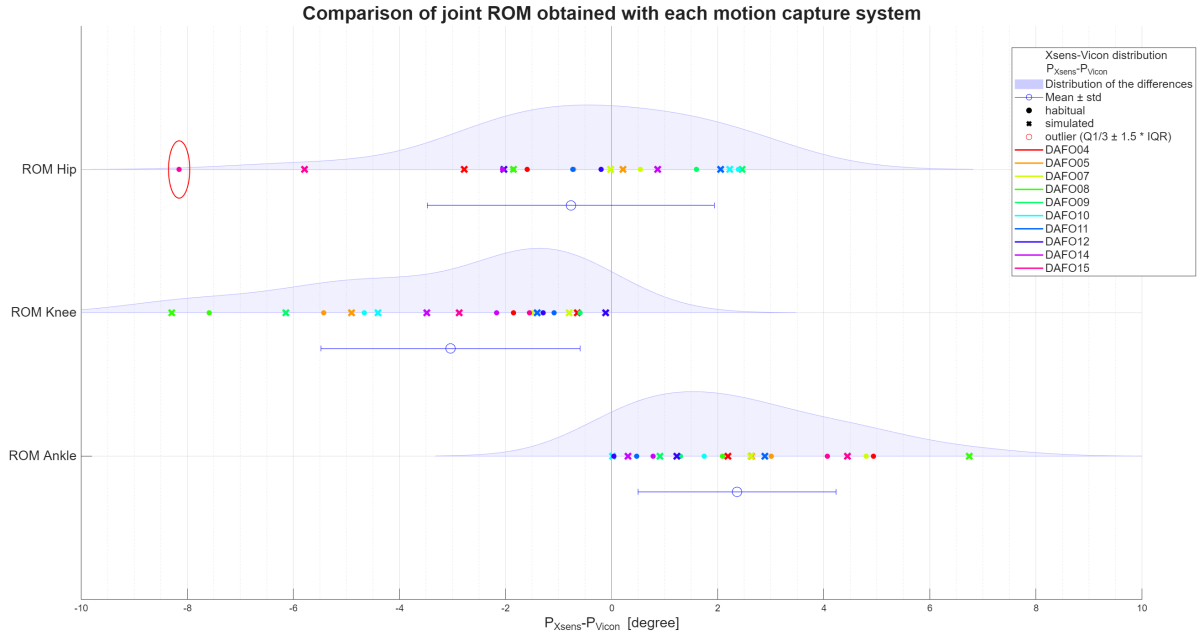
Additionally, we observe that the RMSE at the ankle is below 5° for most subjects. This provides an initial indication that, in general, the gait pattern with the habitual and simulated AFOs are close, therefore meaning that the Dephy is capable of simulating some type of AFOs. The variability observed across patients further suggests that certain types of AFOs may be simulated more accurately than others.

joint range of motion (ROM) across session and motion capture system

At the group level, the ROM distributions were highly consistent between motion capture system, for both habitual and simulated conditions (Figure 3.3a).

When examining inter-system differences $ROM_{Xsens} - ROM_{Vicon}$ (Figure.3.3b), Xsens consistently overestimates ankle ROM ($+2.36^\circ \pm 1.86^\circ$) and underestimates knee ROM ($-2.80^\circ \pm 2.46^\circ$). No consistent bias is observed at the hip ($SD = 2.25^\circ$). Importantly, these biases do not appear related to the absolute magnitude of ROM.





(b)

Figure 3.3: ROM for all joint angle for both session and both motion capture system : (a) Comparison of the ROM (a) ROM difference between Xsens and Vicon

Overall, both systems show comparable ROM distribution. Additionally, ROM distributions between habitual and simulated AFOs look also comparable.

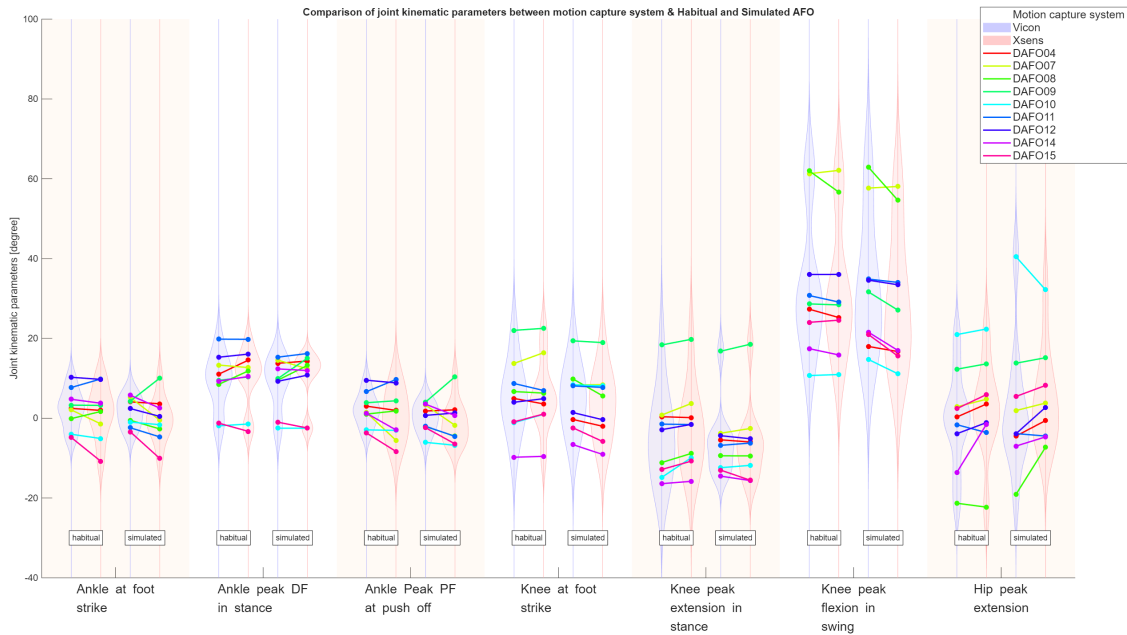
joint kinematic characteristic values across session and motion capture system

Across all kinematic parameters, the distributions are again highly similar between Xsens and Vicon (Figure 3.4a). At the subject level, results are consistent across systems, as indicated by the nearly horizontal connecting lines. In particular, we observe close agreement for the ankle characteristic peak angles (foot strike, maximum dorsiflexion in stance, and plantarflexion at push-off) and for the knee characteristic peak angles (foot strike, maximum extension in stance, and maximum flexion in swing). Greater variability is observed, however, for hip extension.

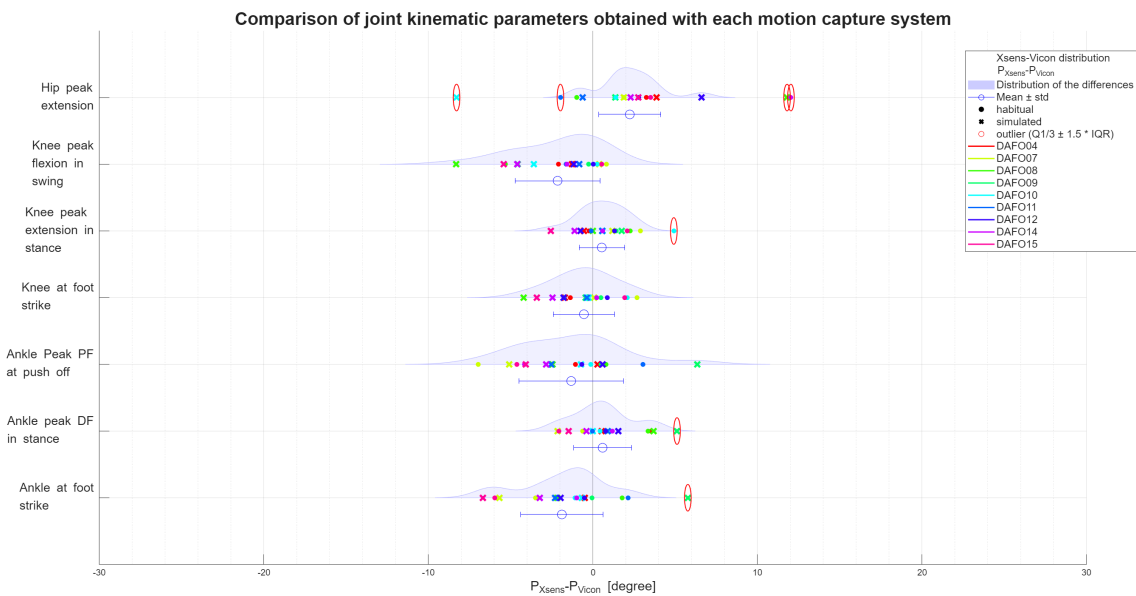
To quantify agreement more precisely, we analyzed the differences in each parameter between motion capture systems (Figure 3.4b). The mean differences cluster around zero, indicating the absence of systematic bias between Xsens and Vicon. An exception is the hip peak extension, where the positive mean difference suggests that Xsens may underestimate hip extension, even after alignment. Moreover, this parameter shows a higher number of outliers, with differences exceeding 10° for two subjects, highlighting that Xsens may be unreliable for hip extension in some cases.

Excluding these outliers, the standard deviation of inter-system differences is small ($SD < 3.5^\circ$) for all other peak values, confirming good overall agreement of Xsens with Vicon as the gold standard. At

the ankle, Xsens shows a slight tendency to overestimate both dorsiflexion and plantarflexion, which is consistent with the ROM overestimation previously observed. At the knee, Xsens tends to slightly underestimate flexion and extension, matching the ROM underestimation noted earlier. These tendencies remain subtle and should be interpreted cautiously given the limited dataset.



(a)



(b)

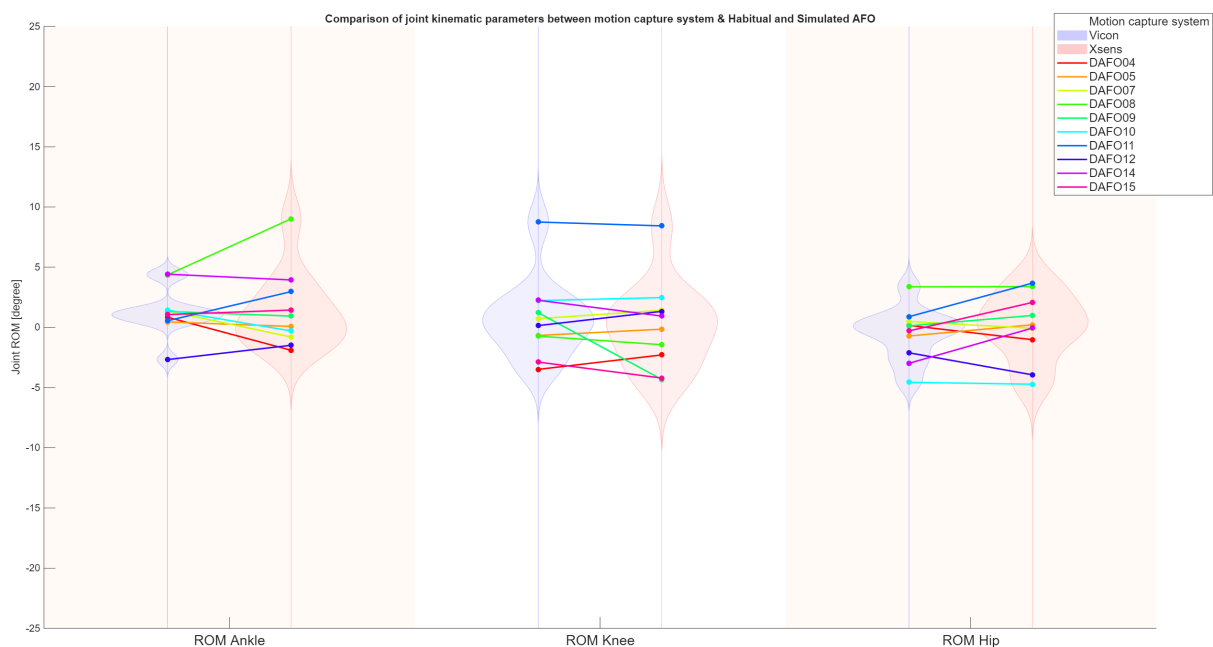
Figure 3.4: Kinematic features for all joint angles for both sessions and both motion capture systems: (a) Comparison of the kinematic features, (b) Feature difference between Xsens and Vicon

Overall, Xsens demonstrated strong agreement with Vicon for peak kinematic values, particularly at the ankle, where differences remained below 3° , a range generally considered negligible for gait analysis. Additionally, we also see similar distribution between habitual and simulated AFO.

Difference between habitual and simulated AFOs across motion capture systems

In this section, we assess both the gait kinematic differences between habitual and simulated AFOs, and the reliability of Xsens in capturing these differences compared to the golden standard (Vicon). Figures 3.5a display the ROM differences between habitual and simulated AFOs ($ROM_{Simulated} - ROM_{Habitual}$) for both motion capture systems. Positive values indicate higher ROM in the simulated condition. Overall, Xsens and Vicon show highly consistent patterns across all joints. In particular, subject-level reliability was strong, with similar values for the same participant across systems. At the ankle, Vicon data indicate that ROM differences were greater than zero for all but one subject. Xsens showed a slightly more variable spread around zero, but values were still consistent overall. The observed differences were always below 5° , with 8/10 subjects showing discrepancies under 3° . These results suggest that the Dephy generally reproduces ROM similar to the habitual AFO, with small differences that tend to favor higher ROM in the simulated condition.

Figure 3.5b further evaluates the accuracy of Xsens in detecting these differences. Distributions are centered around zero, confirming that Xsens neither systematically over- nor underestimates ROM differences compared to Vicon. The standard deviation was 2° for all joints. At the ankle, this discrepancy may represent a non-negligible fraction of the total ROM difference (often $<5^\circ$), but remains small in absolute terms.



(a)

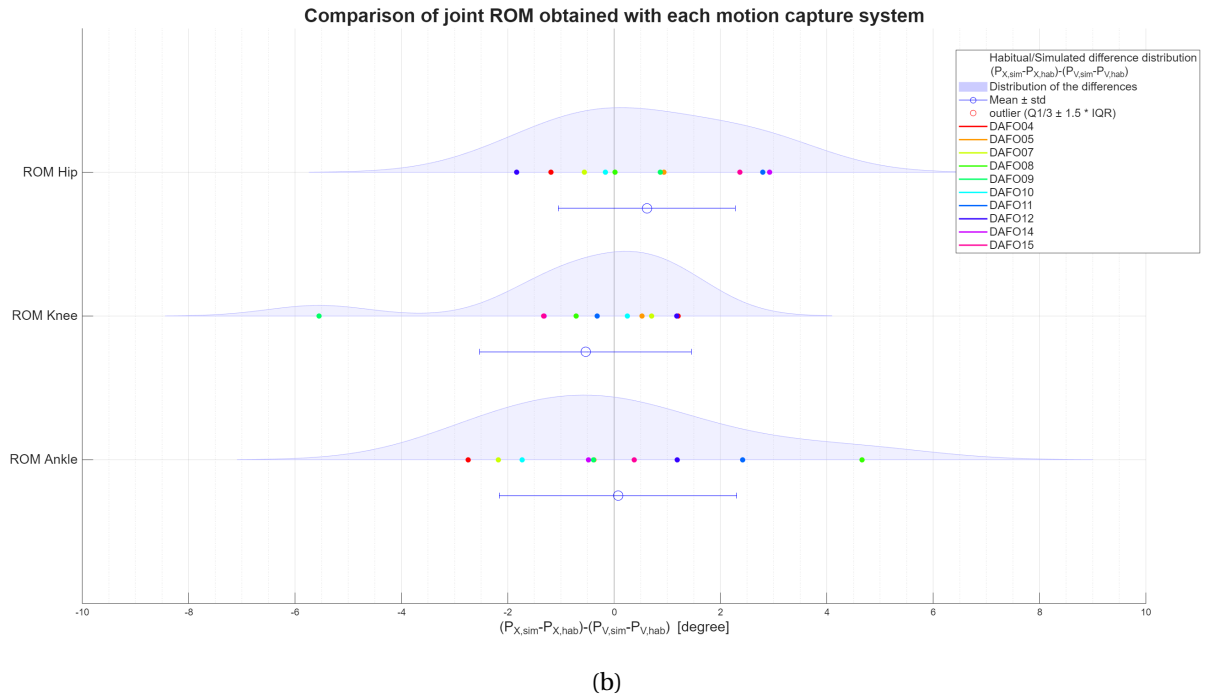
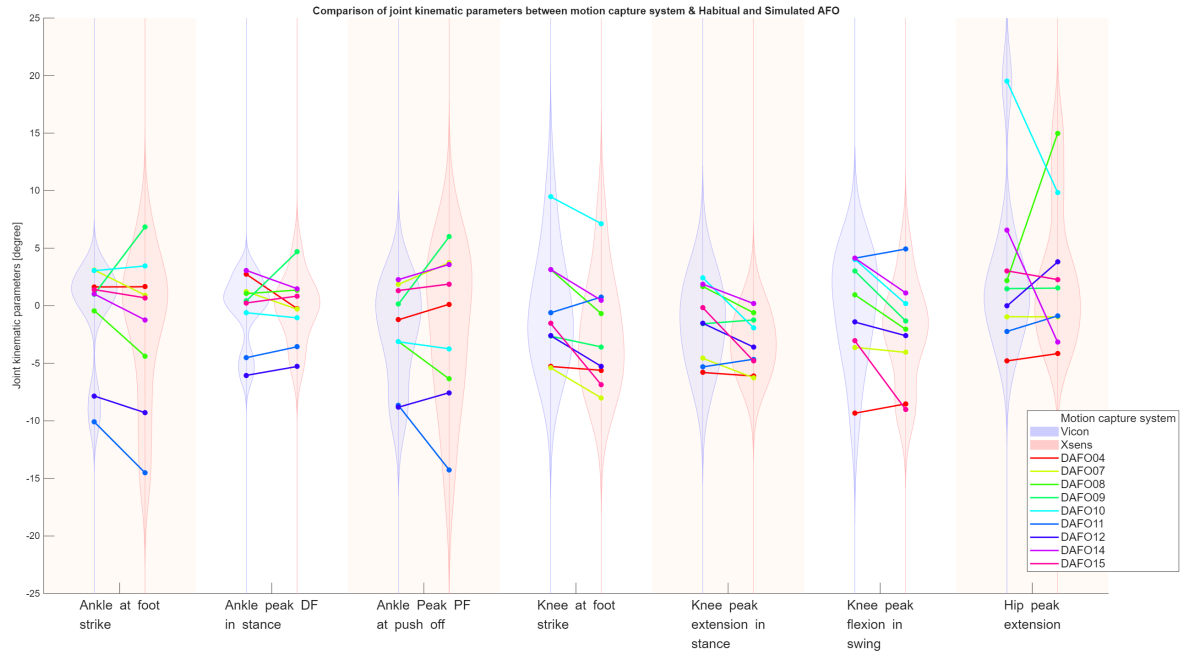
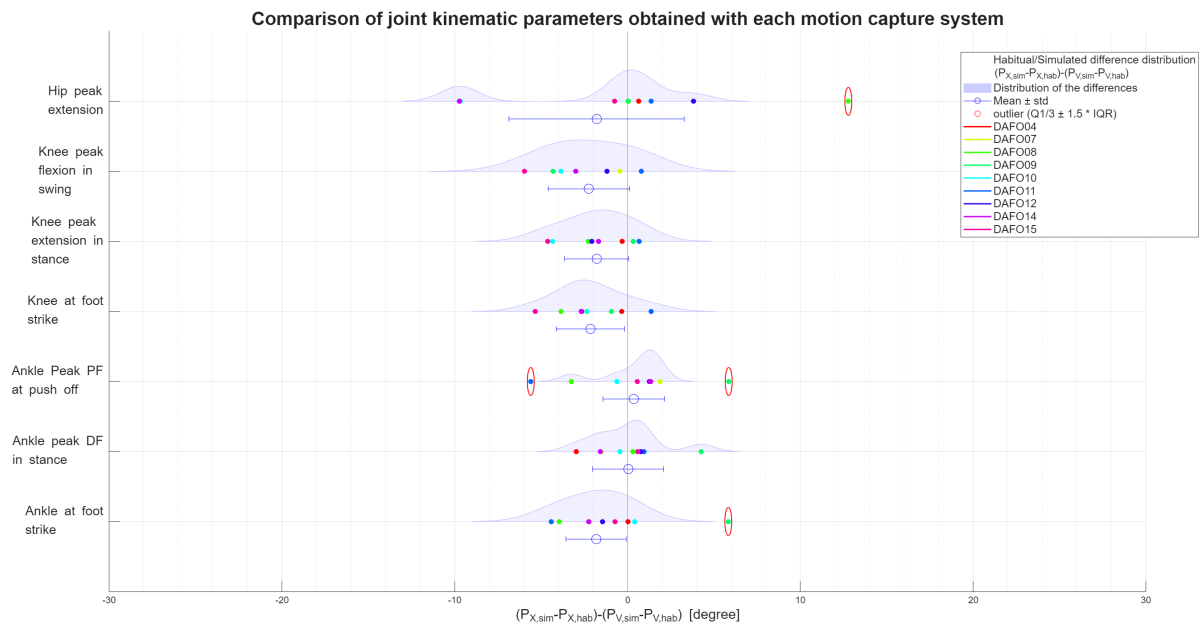


Figure 3.5: ROM difference between habitual and simulated AFO for both motion capture systems: (a) Difference of ROM between habitual and simulated AFO, (b) Difference between motion capture systems

Figures 3.6a report the differences in kinematic features between habitual and simulated AFOs for both systems ($P_{Simulated} - P_{Habitual}$). Again, Xsens and Vicon show similar trends: when Vicon reports small differences, Xsens does too, and vice versa. Figure 3.6b confirms this consistency, showing mean differences close to zero and standard deviations around 2°.



(a)



(b)

Figure 3.6: Kinematic features difference between habitual and simulated AFO for both motion capture systems: (a) Kinematic features difference between habitual and simulated AFO, (b) Difference between motion capture systems

Overall, these findings demonstrate that Xsens reliably captures gait differences between habitual and simulated AFOs. Across RMSE, ROM, and kinematic parameters, Xsens showed strong agreement with Vicon, with only minor biases (ankle overestimation, knee underestimation). Both sys-

tems consistently detected differences between habitual and simulated AFOs. Importantly, Xsens proved particularly reliable at the ankle, the joint most impacted by AFO intervention.

3.3.1 Effect of stiffness mismatch on simulation accuracy

Figure 3.7 shows the RMSE plotted as a function of the stiffness difference between habitual and simulated AFOs, obtained from the mechanical assessment (Habitual – Simulated). Higher values indicate that the Dephy was less stiff than the habitual AFO. Almost all x-values are greater than 0, meaning that the Dephy was consistently less stiff than the habitual AFO in both plantarflexion and dorsiflexion. In some cases, however, the values were very close to zero, which may have occurred by coincidence.

A weak but consistent correlation was observed between RMSE and stiffness difference. This suggests that a mismatch in stiffness between the Dephy and the habitual AFO generally leads to differences in gait pattern, supporting our hypothesis that stiffness plays a central role.

One exception was DAFO07, which exhibited distinct behavior. In dorsiflexion, the Dephy was stiffer than the habitual AFO (the only x-value below zero), because the hinged AFO was highly flexible while the dephey retained its inherent baseline stiffness. In plantarflexion, however, the Dephy failed completely to match the habitual AFO: the stiffness difference was extremely high because the device could not reproduce the posterior stop of the hinge. Despite this mismatch, the RMSE remained low, likely because gait differences appeared mainly in plantarflexion. Thus, while RMSE did not reflect major errors here, the biomechanical characteristics were nonetheless substantially different, as highlighted earlier.

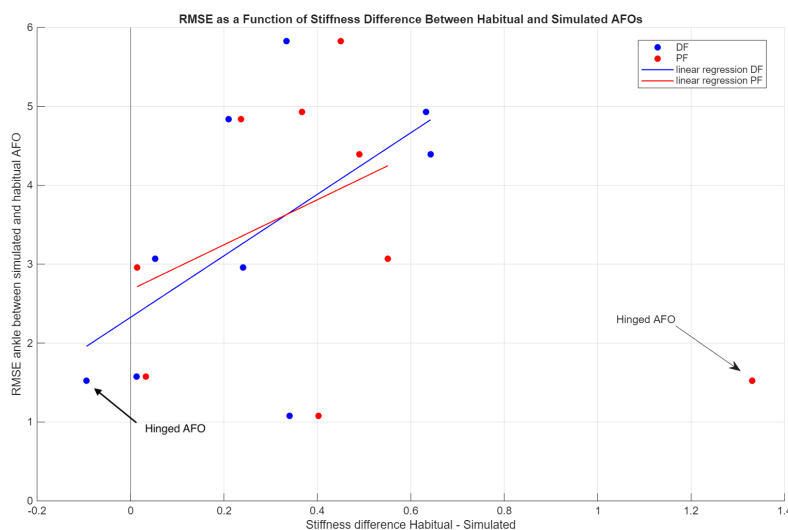


Figure 3.7: Stiffness difference vs RMSE (obtained with the Xsens data) for all tasks

Overall, these findings confirm that stiffness is a key characteristic in AFO, influencing gait pattern. When stiffness is closely matched, the resulting gait trajectories are highly similar, and smaller stiff-

ness differences are generally associated with reduced RMSE.

However, the large spread in the data also highlights that stiffness factor influencing gait pattern, suggesting that AFOs possess more complex mechanical characteristics beyond static stiffness. These may include abrupt stops (as in hinged AFOs), dynamic spring-like responses, or damping effects, properties that the Dephy cannot currently reproduce.

3.4 Assessment of the DEPHY performance outside of the laboratory

In the previous section, we assessed the DEPHY's performance inside the laboratory (overground walking) and demonstrated that Xsens can capture gait differences between habitual and simulated AFOs, both in terms of range of motion (ROM) and kinematic parameters.

Building on these findings, we now use Xsens exclusively to evaluate DEPHY's ability to simulate different AFO types during ambulation tasks performed outside the laboratory, including long straight hallway walking, ramp ascent and descent, and stair ascent and descent. This analysis follows the same approach as the indoor evaluation but extends the comparison to a broader range of tasks.

For all subjects and across all tasks, data were segmented into gait cycles, and gait reports were generated in the sagittal plane for the ankle, knee, and hip. As previously shown, Xsens is particularly reliable at the ankle, and ankle kinematics are also the most directly affected by AFOs. Therefore, this analysis focuses primarily on similarity of the ankle joint angles, as the main indicator of device performance.

Annex A contains, for each subject, detailed gait reports showing ankle joint angles over the gait cycle across all tasks for both the habitual AFO and the simulated AFO. Additionally, the annex includes the mechanical characterization of the habitual AFO and the DEPHY (torque–angle relationship), allowing assessment of their stiffness and comparison of the depHY and the habitual AFO stiffness. Subjects are grouped by AFO type.

3.4.1 RMSE between habitual and simulated AFOs

Figure 3.8 shows the RMSE of the ankle joint angle during the stance phase between habitual and simulated AFOs across all outdoor tasks.

Overall, the patterns are consistent with the indoor results: subjects who exhibited higher RMSE indoors also tended to show higher RMSE across outdoor tasks. For hallway walking, RMSE values are nearly identical to the indoor results (Figure 3.2a). Ramp ascent and descent also display similar distributions, whereas stair tasks produce the largest discrepancies. Subjects with higher RMSE in other tasks generally show even larger errors during stair climbing.

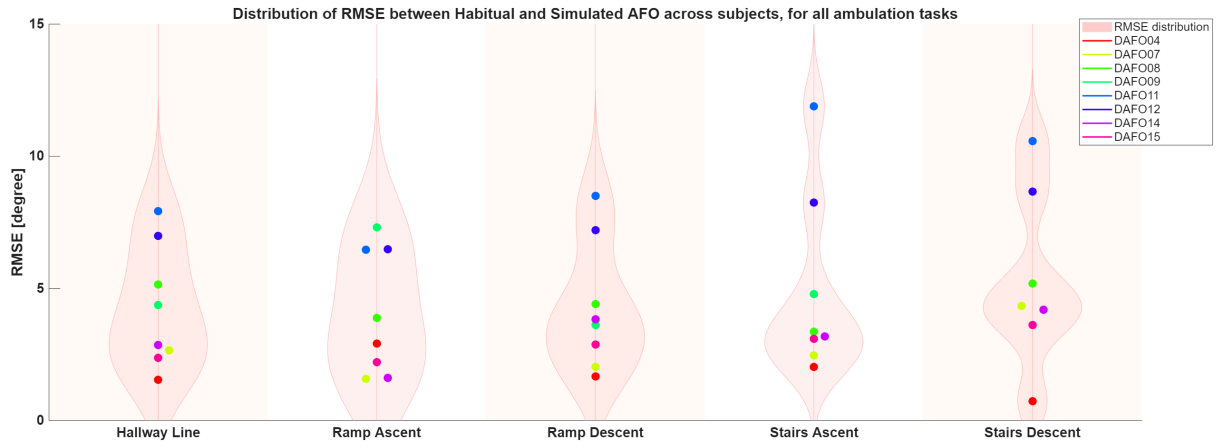


Figure 3.8: RMSE between the ankle joint trajectories in the stance phase of the habitual and simulated AFOs across all tasks

Across the cohort, 6 out of 8 subjects showed RMSE values consistently below 5°, with little variation across tasks, indicating overall good simulation by the dephy. The two exceptions, DAFO11 and DAFO12, displayed consistently higher RMSE (around 8°), which increased further during stair tasks (10 to 12°). This is expected, as stair negotiation is highly demanding and requires greater ankle range of motion. Consequently, if the DEPHY failed to accurately simulate habitual AFOs in simpler tasks, its performance is even less reliable in the stair condition. Both subjects wore carbon fiber posterior lateral strut AFOs. Although their stiffness values were not higher than other AFOs of the same type, mechanical characterization (Figures A.8, A.9) clearly showed that DEPHY did not reach the stiffness of their habitual AFOs. Additionally, for both subjects, we observed differences in ankle angle trajectories during swing, that may indicate a different neutral angle of the devices, and a systematic offset between habitual and simulated AFOs.

One subject (DAFO09) did not show consistent RMSE across tasks : if showed almost double RMSE during ramp ascent compared to other conditions. However, as noted in the indoor analysis, DAFO09 exhibited an unusual offset in the Xsens data. For the ramp task (Figure A.7), the increased error again appears to stem from an offset that may represent an artifact. This subject's results should therefore be interpreted with caution.

3.4.2 Difference of ankle range of motion (ROM) between habitual and simulated AFOs

Figure 3.9 reports the range of motion (ROM) differences between habitual and simulated AFOs across all outdoor tasks. While ROM differences vary across subjects, they are generally consistent within each individual: subjects who show larger differences in one task tend to exhibit similarly larger differences in other tasks.

Across most tasks (except stair ascent) the simulated AFO produces a higher ankle ROM than the ha-

bitual AFO. This effect is most pronounced in cases where the habitual AFO is relatively stiff, or where the Dephy could not replicate its stiffness. For instance, DAFO11 and DAFO08 both show larger ROM with the simulated AFO, consistent with their mechanical characterization, which indicated lower stiffness of the simulated device compared to the habitual AFO.

During stair ascent, the opposite trend emerges: six out of eight subjects display equal or reduced ROM with the simulated AFO relative to the habitual AFO. The two exceptions are again DAFO11 and DAFO08. Video analysis confirmed that these subjects exhibited foot drop at push-off when using the simulated AFO, linked to insufficient stiffness of the device (see Figure 3.10).

This highlights a key limitation of the device: it cannot generate more than X Nm at maximum dorsiplantarflexion, corresponding to a maximum stiffness of 2 Nm/deg. Several habitual AFOs, particularly in plantarflexion, exceeded this stiffness. As a result, even when tuning the device parameters, the Dephy could not fully reproduce the mechanical properties of the stiffer habitual AFOs.

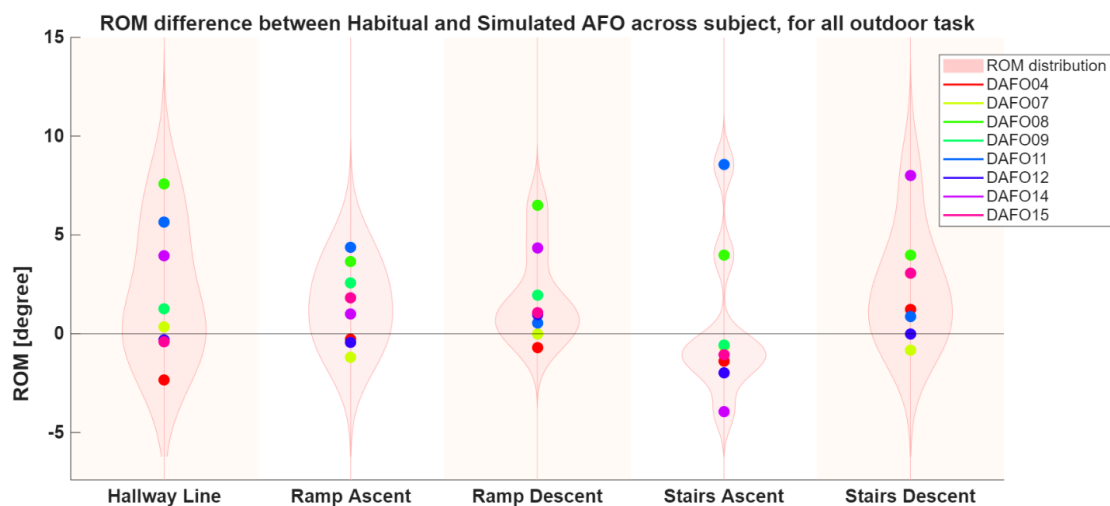


Figure 3.9: ROM difference between the habitual and simulated AFOs across all tasks

3.4.3 Result per task

Hallway.

Biomechanics during hallway walking were consistent with those observed in the instrumented gait analysis inside the laboratory. The same patterns found indoors were reproduced outdoors, confirming that hallway walking is a reliable extension of the laboratory analysis. For some subjects, a slightly higher ROM was observed outdoors compared to indoors, which may be explained by the larger walking space available in the hallway, allowing for more natural movement.

Ramp.

The ramp tasks also showed results very similar to those from overground walking. Ramp ascent

was characterized by greater overall dorsiflexion, while ramp descent involved greater plantarflexion. Interestingly, during ramp ascent, the Dephy generally performed better at simulating habitual AFOs than in overground or ramp descent. By contrast, during ramp descent, the same limitations in plantarflexion stiffness observed in overground walking became more apparent. As noted earlier, several subjects reached the stiffness limit of the Dephy, resulting in compensatory plantarflexion during early stance and foot-off. In these cases, multiple subjects also displayed foot drop at early stance, further highlighting the impact of insufficient plantarflexion stiffness.

Stairs. The stair tasks were particularly insightful, as their biomechanics differ substantially from walking on level ground or ramps. Patients exhibited a wide variety of gait patterns, reflecting different compensatory strategies.

During stair ascent, gait patterns diverged depending on patients' available range of motion. Individuals with higher ROM showed dorsiflexion during stance and strong plantarflexion at foot-off, consistent with normative patterns [56]. In contrast, patients with reduced ROM maintained a nearly constant neutral-to-dorsiflexed ankle angle throughout the cycle.

Similarly, stair descent revealed highly variable strategies across patients. While dorsiflexion demands were generally greater than in other tasks, the way each patient achieved this differed considerably, underlining the biomechanical challenge of stairs. Overall, the stair tasks provided a clear demonstration of the Dephy's limitations, especially in cases where high stiffness or rapid transitions were required.

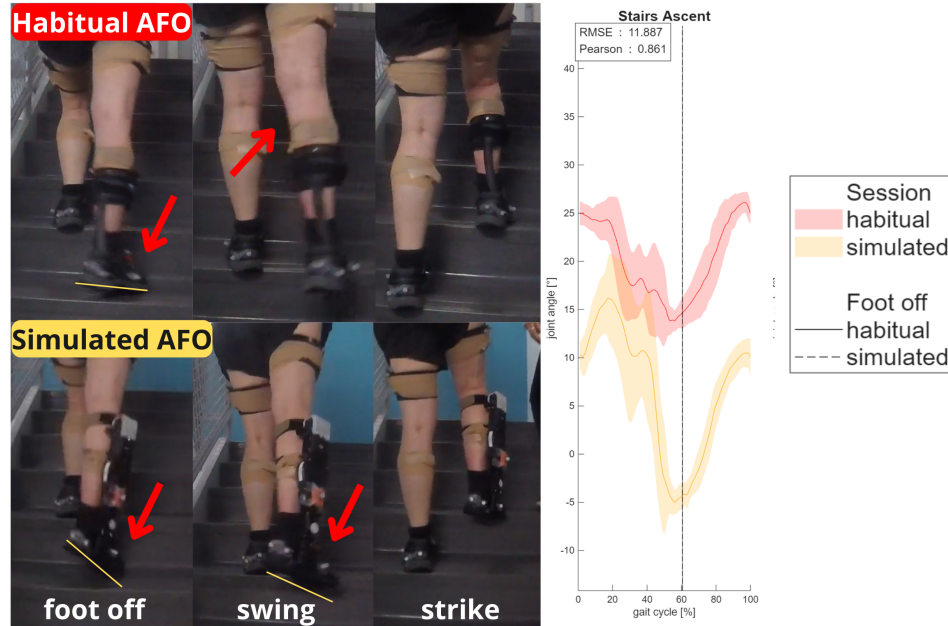


Figure 3.10: DAFO11 – Example of plantarflexion during foot-off. For this subject, the Dephy did not reach the stiffness of the habitual AFO in either plantarflexion or dorsiflexion. Left: foot-off, swing, and strike for both AFOs. Right: ankle joint angle trajectories for both AFOs.

For subject DAFO11, the Dephy failed to match the stiffness of the habitual AFO in both plantarflexion and dorsiflexion. With the habitual AFO, the patient maintained an almost constant ankle angle throughout the gait cycle. After foot-off and during swing, the foot remained stable. To advance the leg, the patient relied on compensatory strategies such as lifting from the hip, leaning sideways, and bending the knee once the foot was cleared.

With the simulated AFO, however, the foot dropped further into plantarflexion at foot-off and during swing. This change altered gait mechanics substantially compared to the habitual AFO, demonstrating how the Dephy's stiffness limitation can directly influence compensatory behavior during stair negotiation.

3.4.4 Result per type of AFOs

For the most flexible posterior leaf-spring AFOs (e.g., DAFO04 and DAFO09), the Dephy was able to reproduce both stiffness and joint angle trajectories with good accuracy. In these cases, the simulated and habitual AFOs showed very similar performance. For DAFO08, the device did not meet the stiffness of the AFO in both plantar and dorsiflexion, leading to foot drop during early stance and foot-off. This discrepancy was particularly visible during overground walking and ramp descent, tasks that demand higher ankle control.

Figure 3.11 illustrates an example with DAFO04, a posterior leaf-spring AFO characterized by relatively low stiffness. In this case, the Dephy replicated the habitual AFO well across tasks, with joint angle trajectories overlapping closely. The only noticeable difference occurred during the swing phase, with deviations of less than 2°. Overall, the patient exhibited a limited range of motion across all tasks, with almost no change in ankle angle during stair walking.

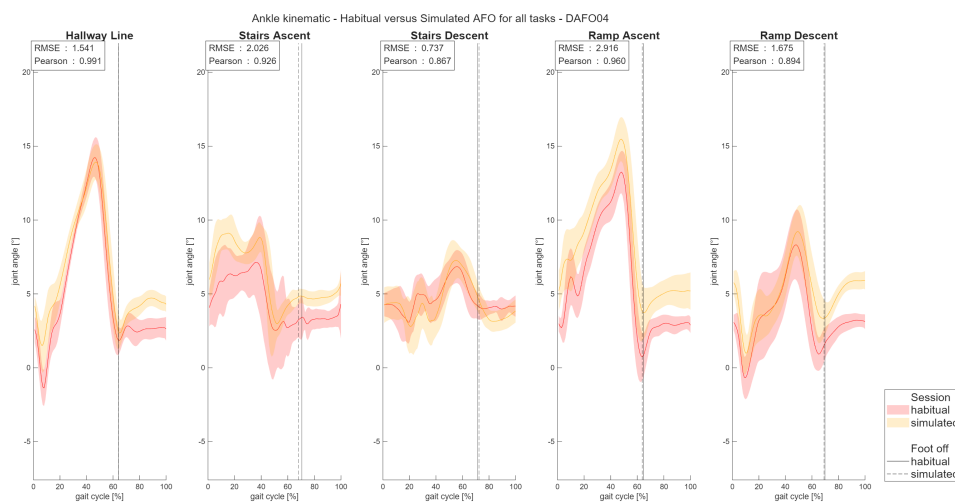


Figure 3.11: Good simulation example : Flexible AFO (DAFO04)

Carbon fiber posterior lateral strut AFOs exhibited higher stiffness than posterior leaf-spring AFOs.

With these stiffer models, the Dephy often reached its plantarflexion limit across subjects, who in turn displayed greater ankle ROM in all tasks. Several patients showed foot drop during stair ascent as well as increased plantarflexion during swing and at foot strike. In one case (DAFO14), the device also exhibited dorsiflexion limitation during stair descent.

Figure 3.12 presents an example with DAFO11, a stiff carbon fiber AFO where the Dephy failed to reproduce the habitual gait pattern. Across all tasks, we observed a consistently different range of motion between the habitual and simulated AFOs. One key discrepancy was the ankle angle during swing: the patient experienced foot drop, with consistent plantarflexion in swing for hallway walking and ramp descent. The patient also exhibited pronounced foot drop during stair ascent, coming close to hitting the step, as highlighted earlier. For stair descent and ramp ascent, where the foot does not move into dorsiflexion, the differences were less pronounced but still evident. In particular, the neutral angle appeared misaligned, as the swing and early stance/foot-strike positions differed between the simulated and habitual AFOs.

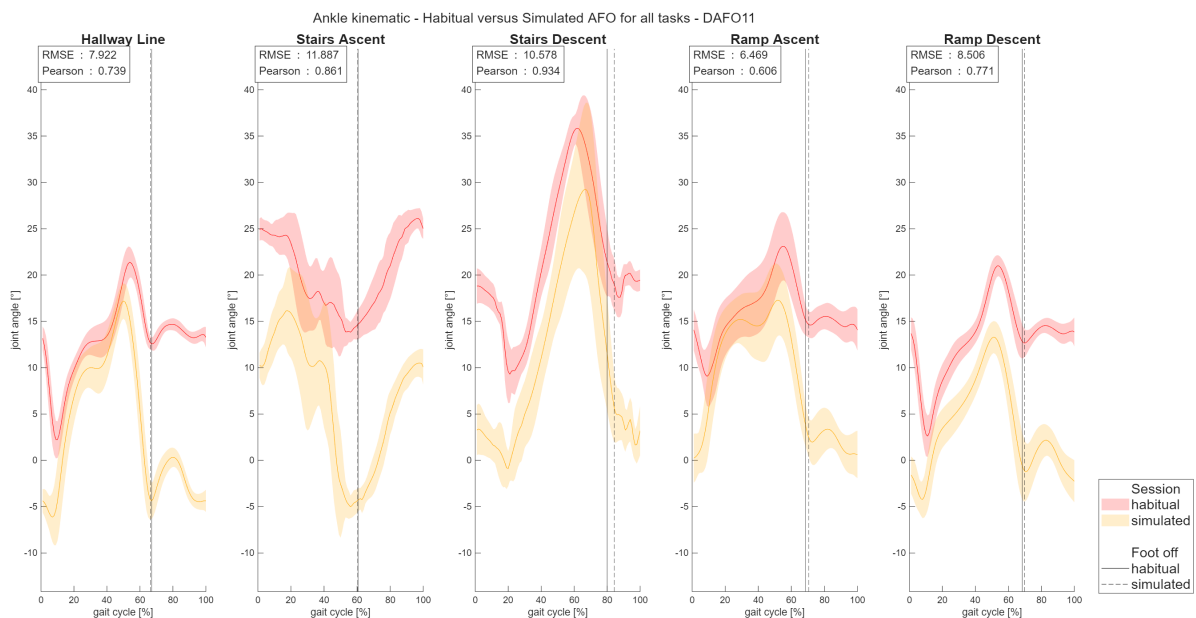


Figure 3.12: Example of AFO limitation : Carbon fiber dynamic AFO (DAFO11)

The hinged AFO (DAFO07 Figure.3.13) was really interesting as it show more complex mechanical characteristic. The habitual AFO is highly flexible in dorsiflexion, with a stiff stop around $-3/-4^{\circ}$ plantarflexion. The Dephy fail to reproduce this behavior. Instead, it began resisting earlier but with less stiffness overall, producing similar torque at a given angle but through a more gradual response. Consequently, the simulated AFO showed a higher neutral angle across task, but foot drop in early stance when the patient is in plantarflexion, particularly during overground walking and ramp descent.

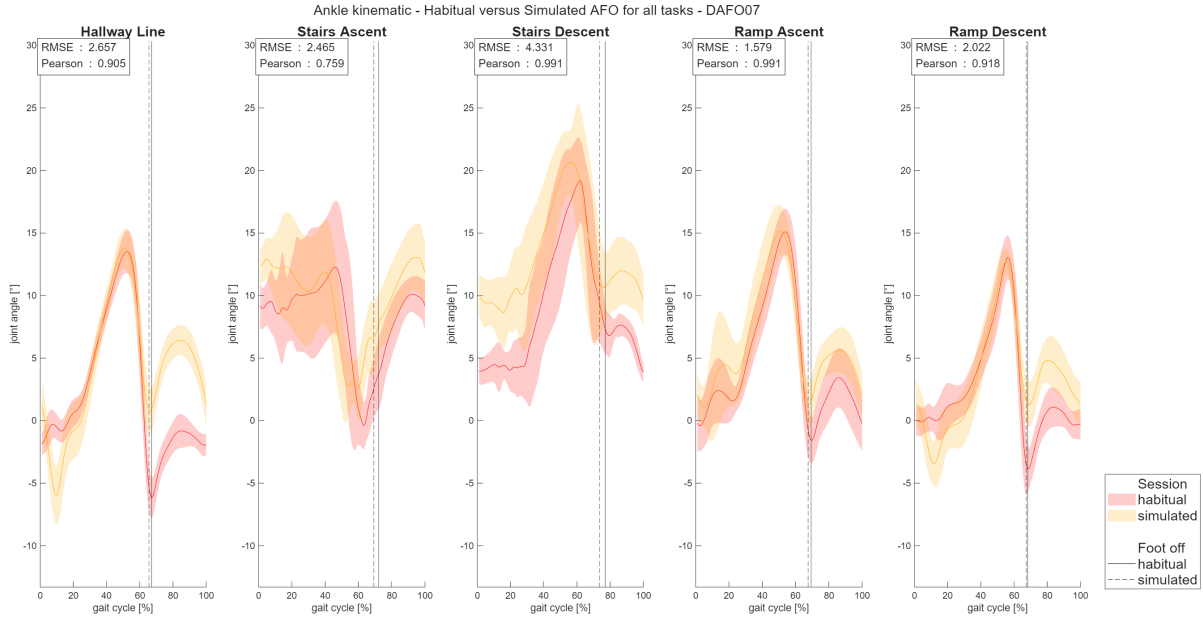


Figure 3.13: Example of AFO with a different mechanical behavior : hinged AFO (DAFO07)

3.4.5 Effect of stiffness mismatch on simulation accuracy

Overall, stiffness appears to play a decisive role in shaping the resulting joint angles. However, stiffness and AFO type are intrinsically linked (for example, solid AFOs are by design stiffer than posterior leaf-spring models). The precision in the stiffness tuning of the Dephy seems to be a key factor for an accurate simulation. By this, we mean that for some AFOs with similar stiffness, the Dephy reproduced the habitual device well, while in other cases limitations were observed. Looking at the mechanical characterization, we found that sometimes the stiffness of the Dephy was similar to the habitual AFO, while in other cases the Dephy did not reach comparable stiffness levels, and that accuracy the result of the simulation.

Figure 3.14 shows the RMSE plotted as a function of the stiffness difference between the habitual and simulated AFOs, obtained during the mechanical assessment (Habitual – Simulated). A higher value means that the Dephy was less stiff than the habitual AFO. Interestingly, the all x-values were always greater than 0, meaning that the Dephy was consistently less stiff than the habitual AFO both in plantar than in dorsiflexion. In some cases, however, the values were very close, which could have been coincidental.

We also note that DAFO07 was removed from these plots as it represented an outlier, though the case remains interesting to mention. Its behavior was similar to the indoor condition (Figure 3.7). In dorsiflexion, the Dephy had higher stiffness than the habitual AFO, since the hinged AFO was highly flexible while the Dephy retained its inherent stiffness. In plantarflexion, however, the difference in

stiffness between the simulated and habitual AFO was extremely high because the Dephy could not accurately replicate the posterior stop. Despite this large stiffness difference, the RMSE was still low, although the biomechanical characteristics were markedly different, as highlighted earlier.

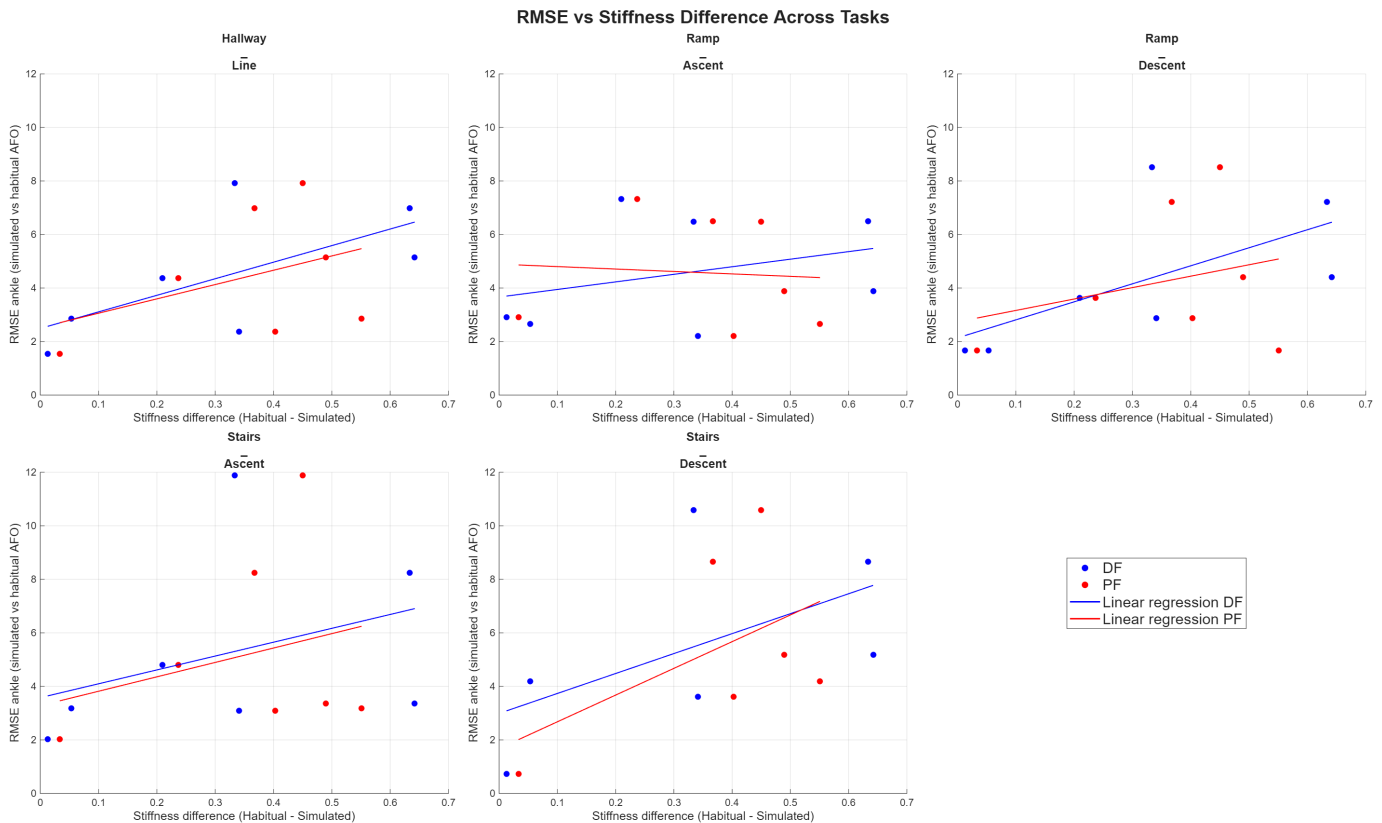


Figure 3.14: Stiffness difference vs RMSE for all tasks

There is a weak but consistent correlation between RMSE and stiffness difference across all tasks, except during ramp ascent. In this task, patients often did not go into plantarflexion or high dorsiflexion, which explains why the Dephy's limitation in plantarflexion did not affect performance and RMSE values remained low.

Overall, these findings confirm that stiffness, and more specifically, the Dephy's ability to reproduce the stiffness of the habitual AFO, is a key characteristic of AFOs. Just matching the stiffness can lead to similar gait pattern, and smaller stiffness differences are generally associated with reduced RMSE. However, the large spread in the data also suggests that stiffness is not the only determinant of accurate simulation. AFOs may present more complex mechanical characteristics beyond static stiffness, such as abrupt stops (hinged AFOs), dynamic spring-like behavior, or damping effects, which

Chapter 4

Discussion

4.1 Motion capture system estimation differences

When comparing data from both systems, it is important to notice several key differences in their biomechanical models, joint angle definitions, and reference frame conventions (Section 2.4). Consequently, discrepancies observed in the resulting joint angle trajectories may arise not only from errors in the inertial measurement system but also from differences in modeling approaches. These discrepancies can be summarized as follows:

1. **Sensor-to-segment calibration :** The methods for defining segment orientation differ significantly. The sensor-to-segment calibration of Xsens depends on accurate static N-pose and correct sensor placement. Small errors in posture or misalignment can propagate across the model. [41] The Vicon system, in contrast, uses marker clusters and doesn't rely on a pose assumption to map the markers to the body. The optical system is therefore more susceptible to errors from marker misplacement, a human factor that is more controllable. The use of clusters, filtering and interpolation further decreases the impact of individual marker misplacement on the global joint angle estimation.
2. **Joint center definition :** The joint centers are defined differently. The Vicon method uses digitalized markers placed with a wand on anatomical landmarks, whereas the Xsens method estimates them using inverse kinematics [45], which is inherently subject to error.
3. **IMU drift :** IMU pose estimation may drift overtime, at contrario optical marker tracking is drift free. However, the Xsens proprietary fusion algorithm, along with additional interpolation and filtering, tends to limit the error rate, especially during long recordings. [41]
4. **Recording frequency :** The recording frequencies are different, with the Vicon system operating at 120 Hz and the Xsens system at 100 Hz. This difference provides greater temporal precision for the marker-based optical pose estimation.

4.2 Reliability of Xsens joint angle trajectories compared to the golden standard indoor

This study confirmed that Xsens joint angles align closely with the optical golden standard, particularly in the sagittal plane.

4.2.1 Within session consistency

Within each session, the MAE between the two systems was consistent across time for all joints. (Figure 3.1) This shows that, within the time of a session and under the laboratory settings, potential drift and accumulative error are well handled by Xsens systems. This finding is consistent with the literature assessing the Xsens system [41] but is not trivial. In fact, IMUs orientation estimation is prone to drift in time due to offset in the angular velocity estimation provided by the gyroscope, ultimately influencing segment position estimation and therefore joint angle estimation. This result shows that the Xsens system calibration and inner algorithm is well handling this limitation of IMUs.

As the Xsens estimate is not losing reliability within the recording, we can represent the gait pattern of a subject by the mean and the standard deviation of several gait cycles extracted across the session.

4.2.2 Xsens joint angle general reliability

The Pearson coefficient indicates excellent correlation in the sagittal plane for all angles. The error was of a similar range across all planes. Of course, as the range of motion is higher in the sagittal plane a error of the same range will be less impactful as in other planes.

High bias standard deviation, of the same magnitude as the error, but high Pearson correlation coefficient indicates that most of the error between the two systems is due to offsets. The fact that the static bias shows similar value as the dynamic bias confirms the presence of an offset between the two systems.

Interestingly, the bias was around 0 for most angles and planes, with high standard deviation. This would mean that the difference is not systematic between the two systems, but depends on the session. Exception was seen for the hip flexion-extension and the ankle adduction-abduction: the Xsens motion capture system shows mean higher hip extension and ankle abduction. However, as the standard deviation is higher than the mean in both cases, this is more showing high discrepancies of both systems for these angles than a systematic bias.

(Table 3.3)

The session-dependent variability suggests that the observed offsets are likely caused by a combination of poor subject calibration in Xsens MVN and marker placement errors in the optical system, as previously noted in the literature [41]. Additional contributing factors may include differences in biomechanical models and reference frame definitions [38].

Interestingly, the variance of the static pose angles accross subject was higher in the Vicon system than in Xsens (Table.3.4). This may indicate that Xsens reduces apparent variability between sessions by relying on the N-pose calibration. Because the N-pose assumes perfectly straight legs and an ideal posture, any deviation during calibration (e.g., slightly flexed knees) is effectively corrected in the model. While this reduces variability in static poses across subjects, it also introduces offsets between Xsens and Vicon.

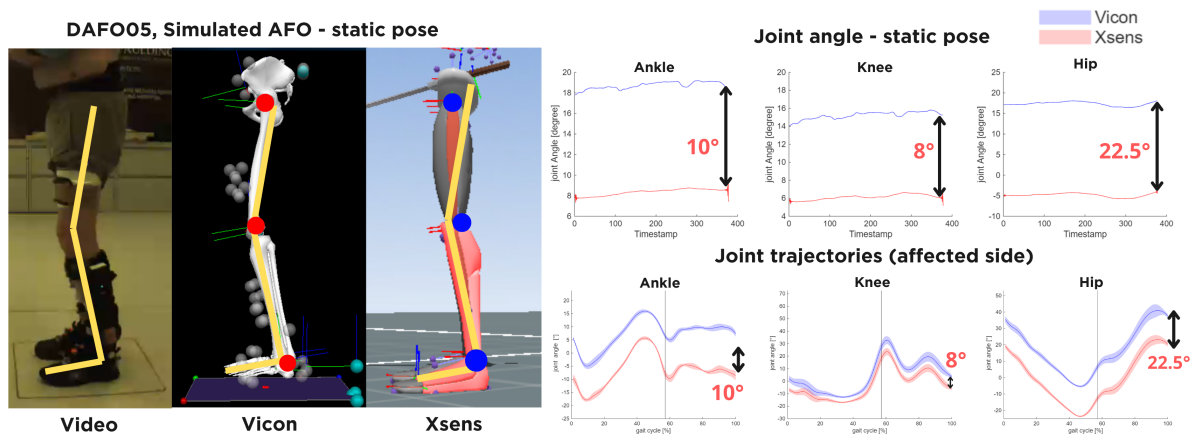


Figure 4.1: Example of a bias introduced by Xsens calibration - The pictures (left) show the subject during the static calibration. The subject (left) is bending the knee and slightly dorsiflexing. The optical motion capture system (center) reliably represent this behavior. However, the Xsens system (right) show straight legs. These differences are visible in the joint angle (top right), where Xsens joint angles show less dorsiflexion and more knee extension. This offset is also visible in motion (bottom right)

After alignment, the MAE lower under 5 degrees in all angles and planes, with a standard deviation of less than 3 degrees, and the variability of the bias drop. This further support the hypothesis that the bias arise from from both biomechanical model difference and qualibration (Table.3.3). This findings is also consistant with the litterature. We found error in the same range as in the paper of Robert-Lachaine & all, where they assessed Xsens both using the Xsens provided biomechanical model, and a custom biomechanical model closer to the model of the golden standard. [38, 41]

4.2.3 Knee artefact and joint-specific observations

Knee coronal angles showed a low correlation between the two systems. However, the knee coronal movement is low and is mainly dominated by noise, due for example to motion artefacts. This is also possible that the optical system is more subject to the knee varus valgus artefact due to crosstalk between the sagittal and the coronal plane, as already described in the litterature as a limitations of skin-mounted markers. [57, 58]. Visually, Xsens showed less motion in this plane. This could be

expected, as the Xsens system use fonctionnal calibration to update the position of the joint center and to define the reference frame of each joint, it may minimise the artefact as already showed in the litterature for optical motion system [59, 60].

The consistently higher Xsens hip extension angles (-6° bias) and larger hip rotation errors should be further explored. These may stem from differences in joint definition and model assumptions.

Similar discrepancies at the knee varus valgus and hip flexion extention have also already been observed by Schepers & al [41] when comparing the Xsens model to to optical data processed with OpenSim.

4.2.4 Error variation between affected and unaffected side

Overall, the error of the Xsens estimates compared to the golden standard does not appear to be systematically influenced by whether the limb is impaired or unimpaired (Table 3.6). This is an important finding, as post-stroke gait is often highly asymmetric, with the impaired limb typically showing reduced range of motion, lower speed, and less predictable trajectories compared to the contralateral side. One could therefore expect that these altered and more variable kinematic patterns might challenge the accuracy of an inertial measurement system. However, our results suggest that the Xsens system is robust to these differences, maintaining a comparable level of accuracy across both limbs. Minor discrepancies were observed. For instance, the mean absolute error (MAE) was slightly higher for the unimpaired limb in the coronal plane, particularly at the hip and ankle. (Table 3.6). Similarly, correlation values in the sagittal plane remained excellent for both sides, but a small decrease was noted for the impaired ankle compared to the unimpaired side. These variations are modest, may be due to noise, and are unlikely to impact the overall interpretation of gait analysis.

This imply that neither joint angle amplitude nor movement speed appeared to systematically influence the accuracy of the estimates, in the context of our study. This is consistent with prior validation studies of the Xsens system, which have reported stable performance across a range of walking speeds and movement conditions [36, 49]. Together, these findings reinforce the reliability of Xsens for use in clinical populations, even when gait patterns deviate substantially from normative kinematics.

4.2.5 Limitations

Not all sessions have been used for the analysis due to failure in alignment or Xsens data for the static trial, necessary for static offset removal. The small sample size may affect generalizability. The number of extracted gait for each sessions was extremely variable, between 3 and 15.

Each session recording was pretty short (around 10 mins). In longer session, the drift of the IMU may be less well handeled. However, Schepers & all showever similar result with longer sessions[41]. Additionnaly, the laboratory environnement is well controlled. If the environnement change a lot

during the session, which will be the case in the outdoor part of the study, the error may change accordingly within a session. However, the magnetometer may be more reliable outside of the laboratory than inside, because it will be less likely to encounter magnetic disturbance.

4.3 Reliability of Xsens for detecting gait differences across sessions in-door

4.3.1 Comparing kinematic features between Xsens and Vicon

Our results demonstrate that Xsens is consistent with Vicon for the evaluation of gait in patients walking with AFOs.

Across ROM (Figure 3.3) and kinematic parameters (Figure 3.4), the differences between Xsens and the golden standard were generally small, typically below 3°, and showed no evidence of systematic bias. This level of agreement is well within the expected error tolerance for human motion analysis (3–5°) [61, 62] and is therefore unlikely to affect clinical interpretation. These findings support the use of Xsens as a valid alternative to laboratory-based optical systems.

At the knee, Xsens systematically underestimated the range of motion, while at the ankle it slightly overestimated it. These biases were modest (around 2 to 3°) and not dependent on the absolute magnitude of the motion, suggesting they are inherent to model differences between the systems rather than linked to patient-specific gait characteristics.

Overall, a strong agreement was particularly evident at the ankle in the sagittal plane, the joint most directly impacted by AFO function. This reinforces the suitability of Xsens for monitoring orthosis-related gait modifications.

4.3.2 Assessing the capacity of Xsens to capture gait differences

Importantly, both Xsens and Vicon captured the same patterns of difference between habitual and simulated AFOs.

This was evident across all metrics: RMSE (Figure 3.2), ROM differences (Figure 3.5), and peak parameter differences (Figure 3.6). In each case, Xsens showed similar result as Vicon, with no systematic bias and a standard deviation of approximately 2°.

These results indicate that Xsens not only reproduces joint trajectories accurately, but also detects clinically meaningful gait adaptations induced by orthotic interventions. Such reliability is essential for practical applications, as the primary goal is often to evaluate relative changes between conditions (e.g., habitual vs simulated AFO) rather than the absolute values of joint angles themselves.

4.3.3 Limitations

However, it must be noted that the strong agreement reported between Xsens and the golden standard is highly dependent on the alignment procedure applied between the two systems. Without this alignment, the differences in RMSE and kinematic parameters between Xsens and Vicon nearly doubled. More importantly, because calibration-related offsets are session-specific, the comparison between habitual and simulated AFOs sometimes produced different results than those obtained with Vicon. This introduces additional uncertainty, with the standard deviation of the error rising from 2° up to 5°.

In the present study, we were able to correct for this limitation by using the indoor optical motion capture as a reference. The offset correction could then be transferred to the outdoor data, under the assumption that the offset remains constant between the two sessions, since markers and sensors were not repositioned. While valid here, this assumption is fragile: if recalibration is performed, or if sensors shift between sessions, the offset will differ, and such correction would no longer be possible. This represents a major limitation for standalone use of inertial motion capture systems.

Moreover, aligning Xsens sessions against each other (and not with the optical golden standard) is not a reliable workaround, since some offsets may correspond to real biomechanical changes. For example, if the Dephy induces greater plantarflexion in static stance compared to the habitual AFO, subtracting this offset would artificially remove a genuine effect.

Therefore, when relying only on inertial motion capture, calibration-related uncertainty remains a key limitation. Potential strategies to reduce this issue include the use of markerless optical motion capture to reconstruct the calibration pose, manual static joint angle measurements to correct offsets, visual inspection of calibration artefacts from video, or ensuring rigorous sensor alignment when processing data outside of MVN Analyze. Nonetheless, until such solutions are standardized, users must account for the uncertainty introduced by calibration offsets when interpreting comparisons between AFO conditions.

4.4 Comparing the habitual and simulated AFO, inside and outside of the laboratory

4.4.1 Assessment of the dephy capacity to simulate habitual AFO inside the laboratory

The comparison between habitual and simulated AFOs revealed that the Dephy is capable of simulating multiple habitual orthoses, though the accuracy depends on the difference of stiffness between the habitual and the simulated AFO and on the type of AFO.

At the group level, RMSE values between habitual and simulated AFOs were low, <5° at the ankle

for most subjects (Figure.3.2a), suggesting that gait pattern remained mostly similar. ROM (Figure.3.5a), and kinematic features (Figure.3.6a) showed similarly small differences. These results indicate that the Dephy can reproduce functional characteristics of several habitual AFOs.

Flexible AFO such as posterior leaf spring generally showed lower RMSE (ex. DAFO04 and DAFO08). The highest RMSEs value was reaching for stiffer carbon fiber models (ex. DAFO11 and DAFO12) (Figure.3.2a).

When stiffness differences were examined directly (Figure.3.7), a weak but consistent correlation with RMSE was observed: the greater the mismatch in stiffness between habitual and simulated AFOs in plantarflexion or dorsiflexion, the larger the deviation in gait pattern. This supports the hypothesis that stiffness is a key determinant of AFO behavior. However, the spread in the data also showed that stiffness alone cannot fully explain gait differences. AFOs may have additional mechanical features, such as abrupt posterior stops (hinged AFOs), spring-like compliance, or damping properties, that the Dephy cannot currently reproduce.

Therefore, while stiffness matching is necessary to achieve accurate simulation, it is not sufficient to account for the full spectrum of AFO mechanics. These insights underscore the importance of both refining the Dephy device and incorporating more complex orthotic properties into future simulations.

4.4.2 Extension of the depky analysis on multiple ambulation tasks outside the laboratory

The outdoor evaluation confirmed and extended the findings from the indoor gait analysis. On one hand, the Dephy can reproduce habitual biomechanics well in low-to-moderate stiffness conditions, especially for tasks such as hallway walking and ramp ascent. On the other hand, its maximum stiffness limit prevents the device from replicating stiffer AFOs. This limitation is particularly evident in plantarflexion where multiple AFOs were stiffer. In consequence, discrepancies were more apparent for tasks that allow plantarflexion, such as stair and ramp descent. (Figure.3.8)

Hallway and ramp walking produced kinematic patterns closely aligned with those observed indoors. When walking in the hallway, patients showed nearly identical patterns as indoors as expected, with a slightly higher range of motion at the level of the ankle for some patients. (Figure.3.9). Although the number of patients doesn't allow to drive strong conclusions, this change may be explained by the larger walking space available in the hallway, allowing for more natural movement.

Ramp ascent was characterized by greater dorsiflexion, while ramp descent emphasized plantarflexion, across all the gait trials. The Dephy reproduced these patterns reasonably well similarly to indoors, although its limitation in plantarflexion stiffness again became evident during ramp descent.

Importantly, stair ascent and descent revealed the clearest discrepancies between simulated and habitual AFOs. Stair ascent demands high ankle ROM and rapid transitions, amplifying the impact of insufficient stiffness in the simulated condition, especially for subject where the lack of stiffness was already inducing discrepancies in gait pattern in other task.

During the descent patient usually reached more dorsiflexion than during other task. On other task, the Dephy tend to simulate pretty well in dorsiflexion. however, for this task, we saw limitation in the device capacity. For example, for one subject (DAFO08 Figure.A.6) the device was not stiff enough to prevent the foot in late stance.

In both cases (ascent and descent), the Dephy's reduced stiffness capacity limited its ability to replicate habitual biomechanics, particularly for carbon AFOs.

4.4.3 Influence of stiffness and AFO type

Across all tasks, stiffness emerged as a critical factor shaping the accuracy of the simulation. The Dephy performed best with flexible AFOs, where it could replicate both the stiffness and joint angle trajectories of the habitual devices. In contrast, stiffer AFOs exceeded the maximum stiffness capacity of the Dephy, leading to higher ROM in the simulated condition, systematic offsets, and functional deficits such as foot drop.

The combined indoor and outdoor analyses (Figures 3.7, 3.14) confirmed a weak but consistent correlation between stiffness mismatch and RMSE. In nearly all cases, the Dephy was less stiff than the habitual AFO, particularly in plantarflexion, which partly explains the higher ROM observed in the simulated condition. Smaller stiffness differences were generally associated with reduced RMSE, underscoring the importance of precise stiffness tuning for accurate simulation. However, the broad data spread suggests that stiffness is only one determinant, and more complex mechanical behaviors must be considered.

Carbon fiber AFOs illustrated this point clearly (Figure.3.12). Despite similar stiffness values, the dynamic properties of carbon, such as energy storage and release, contributed to preventing foot drop and support the movement at foot off, effects absent with the Dephy. This mismatch led to functional deviations, especially in tasks requiring rapid plantarflexion at foot-off.

The hinged AFO (DAFO07) (Figure.3.13) provided another illustrative case. The habitual device allowed free dorsiflexion but had a sudden stiff stop in plantarflexion, whereas the Dephy produced resistance earlier and more gradually. This altered torque–angle profile shifted the neutral angle and produced foot drop in early stance. Simulating such devices would require not only higher stiffness capacity but also the ability to reproduce abrupt transitions in torque response.

Overall, these findings show that stiffness tuning is necessary but not sufficient for accurate simulation. AFO type, material properties, and mechanical design (e.g., rigid stops, spring-like elasticity,

damping) also play decisive roles in shaping gait outcomes.

4.4.4 limitations

This outdoor analysis focused primarily on the ankle joint angle, as it is the parameter most directly influenced by changes in AFOs. However, including knee and hip kinematics in the sagittal plane would provide additional insight, particularly into compensatory strategies. This is especially relevant for complex tasks such as stair negotiation. For example, the patient illustrated in Figure 3.10 compensated for limited ankle stiffness by lifting the hip and bending the torso to bring the foot upward. Evaluating knee and hip angles would allow us to better capture such compensatory mechanisms. In addition, several patients exhibited knee hyperextension during stance, which could also be important to quantify.

Another limitation is that some observed differences may stem from patient adaptation to the AFO design rather than the device mechanics alone. Changing to a different AFO may temporarily alter gait strategies or reduce confidence compared to the habitual device. A longer-term study, or at least repeated sessions, would be required to confirm the stability and representativeness of gait patterns.

Chapter 5

Conclusion

5.1 General findings

This study evaluated both the reliability of the Xsens system for joint angle estimation and the performance of the Dephy device in simulating different types of ankle-foot orthoses (AFOs) during indoor and outdoor ambulation tasks.

Regarding motion capture system, the results demonstrated that Xsens provides joint angle estimates that are consistent within a session and in close agreement with the Vicon golden standard, particularly in the sagittal plane.

Nevertheless, offsets between the systems were observed, hypothesized to be due to subject calibration as previously highlighted in the literature [41]. Where offsets related to biomechanical model differences [38] are likely to remain consistent across sessions, calibration-related offsets may vary from one session to another, therefore limiting the accuracy of comparisons across sessions or in that case AFOs.

Importantly, once offsets were corrected by aligning joint angles with the golden standard (Vicon), Xsens proved to be reliable for detecting relevant gait differences between habitual and simulated AFOs, confirming its suitability for real-world gait assessment.

The performance of the Dephy device varied depending on the type of task and the characteristics of the orthosis being simulated. It successfully replicated a wide range of ankle-foot orthoses (AFOs) across multiple tasks, performing particularly well for more flexible models such as posterior leaf-spring AFOs. However, it showed limitations when simulating stiffer devices, as the maximum achievable stiffness was insufficient to fully replicate the behavior of habitual AFOs.

During overground walking and ramp ascent, the simulated and habitual AFOs generally produced similar gait patterns, with joint angle differences remaining within clinically acceptable ranges. In contrast, stair negotiation highlighted more pronounced limitations. Performance on stairs posed a greater challenge for the devices, as stair ascent demands greater joint mobility and control. Users often relied on compensatory strategies such as increased hip flexion, trunk bending, hip hiking and

show less consistency in their gait pattern. In some cases, insufficient device stiffness led to excessive foot plantarflexion during ascent, highlighting the limitations of the devices when engaging in more complex and demanding tasks.

While stiffness was a major determinant of simulation accuracy, it was not the only factor. Material properties and more complex behaviors, such as abrupt stops in hinged AFOs or the dynamic spring-like effects found in carbon fiber models, also played significant roles. These findings underscore the need for more nuanced simulation parameters to accurately replicate the biomechanical demands of dynamic, multidirectional tasks.

Overall, these findings confirm the value of IMU-based gait analysis for orthosis research and the potential of AFO simulators such as the Dephy. At the same time, they highlight methodological and technical challenges that must be addressed to improve the robustness of both measurement and simulation.

5.1.1 Limitation

Several limitations of this study must be acknowledged.

The session-specific offset of Xsens represents a major limitation of the motion capture system compared to the optical golden standard, mainly due to subject calibration (N-pose accuracy). While offsets were corrected during the indoor/outdoor transition using optical motion capture alignment, such correction is not feasible if Xsens is used alone. This introduces uncertainty in inter-session comparisons and represents a key limitation for clinical use.

The dataset was relatively small and did not include any solid AFOs, which limits the generalizability of the findings. Furthermore, since stiffness was identified as a key factor in the Dephy's ability to simulate AFOs, testing stiffer orthoses would be particularly valuable.

There were instances where the Dephy appeared capable of matching the stiffness of an AFO, based on its performance with other subjects, but mechanical characterization revealed that it did not. In such cases, the discrepancies observed in gait patterns between the habitual and simulated AFO may not reflect a limitation of the device itself, but rather a mismatch in stiffness.

Patients may also have walked differently simply because of reduced comfort or unfamiliarity with the simulated AFO. Such adaptation effects, particularly during complex tasks like stair ascent, may have amplified the observed differences. A longer acclimatization period or repeated sessions would be required to confirm gait differences over time. In addition, patients showed wide variability in gait strategies and impairment, making conclusions difficult to generalize, as some effects may have been subject-specific rather than consistent across the group.

Additionally, this study focused exclusively on joint kinematics, as joint kinetics (moment and power) data were not available in the laboratory. It was assumed that the majority of the effects of the AFOs would be reflected in joint kinematics. However, this remains a simplification and joint kinetics could provide important complementary information about joint loading and compensatory mechanisms. To support this assumption, it would have been useful to examine the joint kinetics recorded inside the laboratory (indoor) to determine whether any changes occurred that were not apparent in the kinematics. Additionally, estimating kinetics using approaches such as inverse dynamics with OpenSim could have provided further insight. A simpler alternative would be to use pressure sensors inside or around the foot, as demonstrated by Sie & al. [63] for studying kinetics during stair negotiation.

5.2 Next steps

Several potential next steps could be undertaken to address these limitations and broaden the scope of the analysis.

First step would be to expand the outdoor gait evaluation to the knee and hip joint, as this would help capture compensatory strategies such as knee hyperextension or hip hiking, which are particularly relevant during demanding tasks like stair negotiation. In parallel, providing outdoor gait reports to physiotherapists for clinical interpretation would strengthen the assessment of functional outcomes and reveal additional patient-specific strategies.

Beyond kinematics, integrating kinetic estimation would provide valuable additional insight. Using the already available data, this could be achieved through inverse dynamics. For example, by exporting Xsens data via Opensense into OpenSim to perform inverse kinetics analyses. Then, a validation similar to the one performed for kinematics could be conducted. However, this would be challenging in ramp and stairs due to level change and inclinaison.

Another important direction is to extend the outdoor assessment of the Dephy to its optimal and active modes. This could emphasize the role of patient-specific stiffness tuning in shaping gait patterns during more challenging tasks and provide a clearer understanding of the effects of active support during push-off.

Bibliography

- [1] Valery L Feigin, Michael Brainin, Bo Norrving, Sheila O Martins, Jeyaraj Pandian, Patrice Lindsay, Maria F Grupper, and Ilari Rautalin. World Stroke Organization: Global Stroke Fact Sheet 2025. 20(2):132–144.
- [2] Joy Adamson, Andy Beswick, and Shah Ebrahim. Is stroke the most common cause of disability? 13(4):171–177.
- [3] Sandra J. Olney and Carol Richards. Hemiparetic gait following stroke. Part I: Characteristics. 4(2):136–148.
- [4] Henrik S. Jørgensen, Hirofumi Nakayama, Hans O. Raaschou, and Tom S. Olsen. Recovery of walking function in stroke patients: The copenhagen stroke study. 76(1):27–32.
- [5] G. Stoquart, C. Detrembleur, and T. M. Lejeune. The reasons why stroke patients expend so much energy to walk slowly. 36(3):409–413.
- [6] Marina M. Platts, Daniel Rafferty, and Lorna Paul. Metabolic Cost of Overground Gait in Younger Stroke Patients and Healthy Controls. 38(6):1041.
- [7] Jessica L. Allen, Steven A. Kautz, and Richard R. Neptune. Step length asymmetry is representative of compensatory mechanisms used in post-stroke hemiparetic walking. 33(4):538–543.
- [8] Sylvie Nadeau, Denis Gravel, A. Bertrand Arsenault, and Daniel Bourbonnais. Plantarflexor weakness as a limiting factor of gait speed in stroke subjects and the compensating role of hip flexors. 14(2):125–135.
- [9] Nancy E. Mayo, Sharon Wood-Dauphinee, Sara Ahmed, Gordon Carron, Johanne Higgins, Sara Mcewen, and Nancy Salbach. Disablement following stroke. 21(5–6):258–268.
- [10] George Chen, Carolynn Patten, Dhara H. Kothari, and Felix E. Zajac. Gait differences between individuals with post-stroke hemiparesis and non-disabled controls at matched speeds. 22(1):51–56.
- [11] Bruce Carse, Roy Bowers, Barry Meadows, and Philip Rowe. Improving independence in the community for stroke survivors: The role of biomechanics visualisation in ankle-foot orthosis

- tuning. In *2011 5th International Conference on Pervasive Computing Technologies for Healthcare (PervasiveHealth) and Workshops*, pages 399–403.
- [12] Aliyeh Daryabor, Toshiki Kobayashi, Sumiko Yamamoto, Samuel M. Lyons, Michael Orendurff, and Alireza Akbarzadeh Baghban. Effect of ankle-foot orthoses on functional outcome measurements in individuals with stroke: A systematic review and meta-analysis. 44(22):6566–6581.
 - [13] Aliyeh Daryabor, Mokhtar Arazpour, and Gholamreza Aminian. Effect of different designs of ankle-foot orthoses on gait in patients with stroke: A systematic review. 62:268–279.
 - [14] Deema Totah, Meghna Menon, Carlie Jones-Hershinow, Kira Barton, and Deanna H. Gates. The impact of ankle-foot orthosis stiffness on gait: A systematic literature review. 69:101–111.
 - [15] Foundations for Ankle Foot Orthoses.
 - [16] AFO | Optec USA.
 - [17] Articulated AFO.
 - [18] Össur. Life Without Limitations.
 - [19] SpryStep® Plus AFO - PDAC Approved L1932/L1933.
 - [20] Bruce Carse, Roy Bowers, Barry C Meadows, and Philip Rowe. The immediate effects of fitting and tuning solid ankle-foot orthoses in early stroke rehabilitation. 39(6):454–462.
 - [21] Rob Green-Buckley, Ingrid Hamlet, and Steve Seccombe. A guide to tuning AFOs and its relevance to stroke rehabilitation. 13:S27–S33.
 - [22] ExoBoot.
 - [23] Different types of motion capture.
 - [24] Award winning motion capture systems.
 - [25] Motion capture systems.
 - [26] Home | movella.com.
 - [27] About us.
 - [28] Andrea Cereatti, Ugo Della Croce, and Angelo M. Sabatini. Three-dimensional human kinematic estimation using magneto-inertial measurement units. In *Handbook of Human Motion*, pages 1–24. Springer, Cham.
 - [29] Marco Caruso, Angelo Maria Sabatini, Daniel Laidig, Thomas Seel, Marco Knaflitz, Ugo Della Croce, and Andrea Cereatti. Analysis of the accuracy of ten algorithms for orientation estimation using inertial and magnetic sensing under optimal conditions: One size does not fit all. 21(7):2543. Number: 7 Publisher: Multidisciplinary Digital Publishing Institute.

- [30] Julian Jeyasingh-Jacob, Mark Crook-Rumsey, Harshvi Shah, Theresita Joseph, Subati Abulikemu, Sarah Daniels, David J Sharp, and Shlomi Haar. Markerless motion capture to quantify functional performance in neurodegeneration: Systematic review. 7:e52582.
- [31] Logan Wade, Laurie Needham, Polly McGuigan, and James Bilzon. Applications and limitations of current markerless motion capture methods for clinical gait biomechanics. 10:e12995. Publisher: PeerJ Inc.
- [32] Jonathan Williams. SMPL.
- [33] Daniel Debertin, Anna Wargel, and Maurice Mohr. Reliability of Xsens IMU-Based Lower Extremity Joint Angles during In-Field Running. 24(3):871.
- [34] Jeroen Hm Bergmann, Ruth E Mayagoitia, and Ian Ch Smith. A portable system for collecting anatomical joint angles during stair ascent: A comparison with an optical tracking device. 8(1):3.
- [35] Matthew P Mavor, Gwyneth B. Ross, Allison L. Clouthier, Thomas Karakolis, and Ryan B. Graham. Validation of an IMU Suit for Military-Based Tasks. 20(15):4280.
- [36] Eline M. Nijmeijer, Pieter Heuvelmans, Ruben Bolt, Alli Gokeler, Egbert Otten, and Anne Benjaminse. Concurrent validation of the Xsens IMU system of lower-body kinematics in jump-landing and change-of-direction tasks. 154:111637.
- [37] Eduardo Palermo, Stefano Rossi, Francesca Marini, Fabrizio Patanè, and Paolo Cappa. Experimental evaluation of accuracy and repeatability of a novel body-to-sensor calibration procedure for inertial sensor-based gait analysis. 52:145–155.
- [38] Xavier Robert-Lachaine, Hakim Mecheri, Christian Larue, and André Plamondon. Validation of inertial measurement units with an optoelectronic system for whole-body motion analysis. 55(4):609–619.
- [39] Jun-Tian Zhang, Alison C. Novak, Brenda Brouwer, and Qingguo Li. Concurrent validation of Xsens MVN measurement of lower limb joint angular kinematics. 34(8):N63–69.
- [40] Michael Alexander Wirth, Gabriella Fischer, Jorge Verdú, Lisa Reissner, Simone Balocco, and Maurizio Calcagni. Comparison of a new inertial sensor based system with an optoelectronic motion capture system for motion analysis of healthy human wrist joints. 19(23):5297. Number: 23 Publisher: Multidisciplinary Digital Publishing Institute.
- [41] Martin Schepers, Matteo Giuberti, and Giovanni Bellusci. Xsens MVN: Consistent tracking of human motion using inertial sensing. Publisher: Unpublished.
- [42] A Cappozzo, F Catani, U Della Croce, and A Leardini. Position and orientation in space of bones during movement: anatomical frame definition and determination. 10(4):171–178.

- [43] Angelo Cappello, Aurelio Cappozzo, Pier Francesco La Palombara, Luigi Lucchetti, and Alberto Leardini. Multiple anatomical landmark calibration for optimal bone pose estimation. 16(2):259–274.
- [44] Marker sets.
- [45] Xsens MVN: Full 6dof human motion tracking using miniature inertial sensors.
- [46] Roelien Smit-Russcher, Domitille Princelle, and Ibrahim Bilal. MVN white paper: Enhancing motion tracking accuracy with novel gender-specific models.
- [47] Andrea Cereatti, Reed Gurchiek, Annegret Mündermann, Silvia Fantozzi, Fay Horak, Scott Delp, and Kamiar Aminian. ISB recommendations on the definition, estimation, and reporting of joint kinematics in human motion analysis applications using wearable inertial measurement technology. 173:112225.
- [48] Alberto Ferrari, Andrea Giovanni Cutti, and Angelo Cappello. A new formulation of the coefficient of multiple correlation to assess the similarity of waveforms measured synchronously by different motion analysis protocols. 31(4):540–542.
- [49] Howard Chen, Mark C. Schall, and Nathan Fethke. Effects of Movement Speed and Magnetic Disturbance on the Accuracy of Inertial Measurement Units. 61(1):1046–1050.
- [50] R. G. Burdett, D. Borello-France, C. Blatchly, and C. Potter. Gait comparison of subjects with hemiplegia walking unbraced, with ankle-foot orthosis, and with Air-Stirrup brace. 68(8):1197–1203.
- [51] Margaret A. French, Corey Koller, and Elisa S. Arch. Comparison of three kinematic gait event detection methods during overground and treadmill walking for individuals post stroke. 99:109481.
- [52] Jacob J. Banks, Wen-Ruey Chang, Xu Xu, and Chien-Chi Chang. Using horizontal heel displacement to identify heel strike instants in normal gait. 42(1):101–103.
- [53] Gwang Tae Kim, Myunghyun Lee, Yongcheol Kim, and Kyoungchul Kong. Robust Gait Event Detection Based on the Kinematic Characteristics of a Single Lower Extremity. 24(6):987–1000.
- [54] Shuaijie Wang, Kazi Shahrukh Omar, Fabio Miranda, and Tanvi Bhatt. Automatic gait EVENT detection in older adults during perturbed walking. 22(1):40.
- [55] J. A. Zeni, J. G. Richards, and J. S. Higginson. Two simple methods for determining gait events during treadmill and overground walking using kinematic data. 27(4):710–714.
- [56] Daiki Shimotori, Kenji Kato, Tatsuya Yoshimi, and Izumi Kondo. Validation of Gait Kinematics With Ramp and Stair Ascent and Descent Revealed by Markerless Motion Capture in Simulated Living Space: Test-Retest Reliability Study. 12(1):e66886.

- [57] “dynamic knee valgus” – are we measuring what we think we’re measuring? an evaluation of static and functional knee calibration methods for application in gait and clinical screening tests of the overhead squat and hurdle step - ClinicalKey.
- [58] The impact of systematic displacement of the lateral knee marker on gait kinematics using the virtual knee alignment device and the plug-in gait model - ClinicalKey.
- [59] A principal component analysis approach to correcting the knee flexion axis during gait - ClinicalKey.
- [60] Hanna Ulbricht, Meijin Hou, Xiangbin Wang, Jian He, and Yanxin Zhang. The effect of correction algorithms on knee kinematics and kinetics during gait of patients with knee osteoarthritis. 2020(1):8854124. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1155/2020/8854124>.
- [61] Matt C. Smith, Phaedra Leveridge, Garry Massey, Jessica Tyrrell, Malcolm Hilton, and Genevieve K. R. Williams. Investigating the effect of motion capture suits on the test–retest reliability of gait parameters. 14(18):8570. Publisher: Multidisciplinary Digital Publishing Institute.
- [62] Kishor Das, Thiago de Paula Oliveira, and John Newell. Comparison of markerless and marker-based motion capture systems using 95% functional limits of agreement in a linear mixed-effects modelling framework. 13(1):22880. Publisher: Nature Publishing Group.
- [63] Descending 13 real world steps: A dataset and analysis of stair descent - ClinicalKey.

Appendix A

Comparison of the habitual and simulated AFOs across tasks

Posterior Leaf Spring

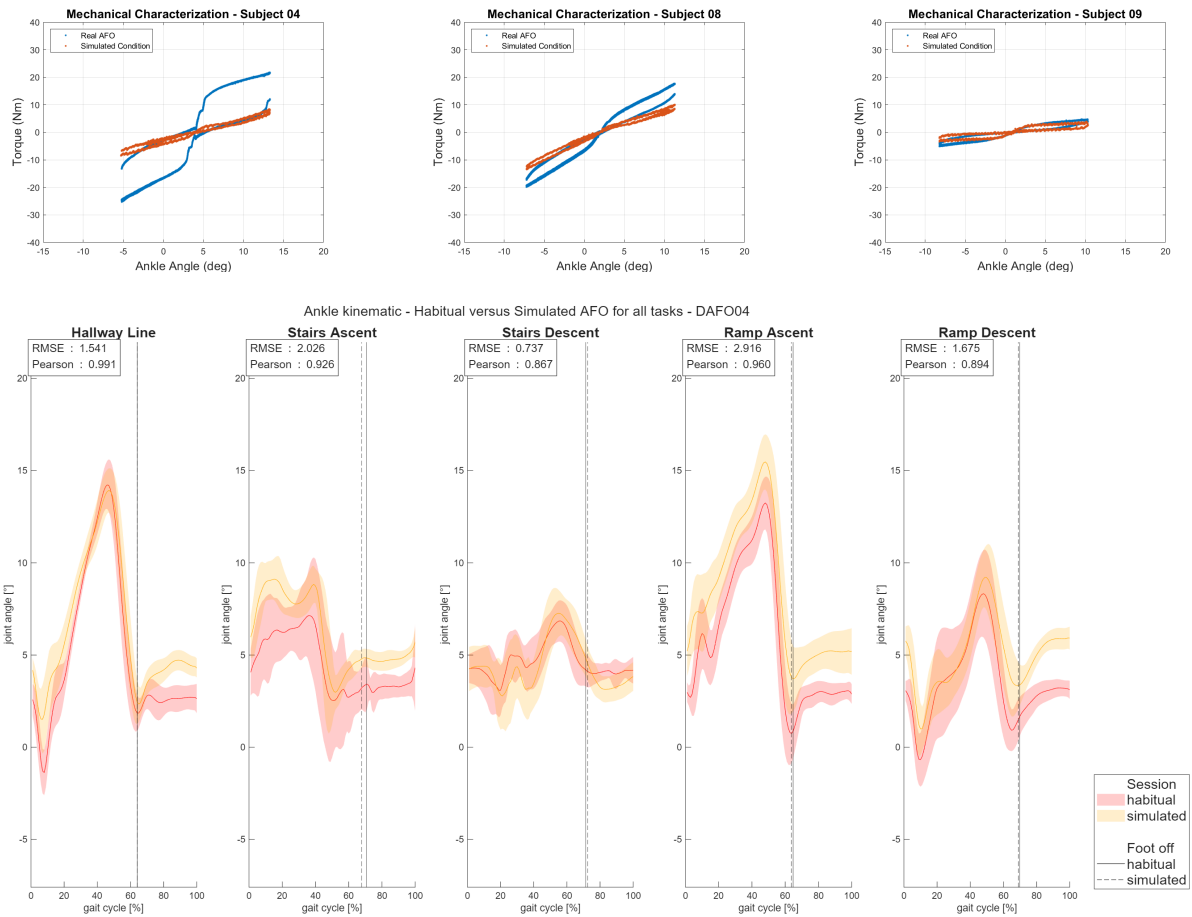


Figure A.5: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO04

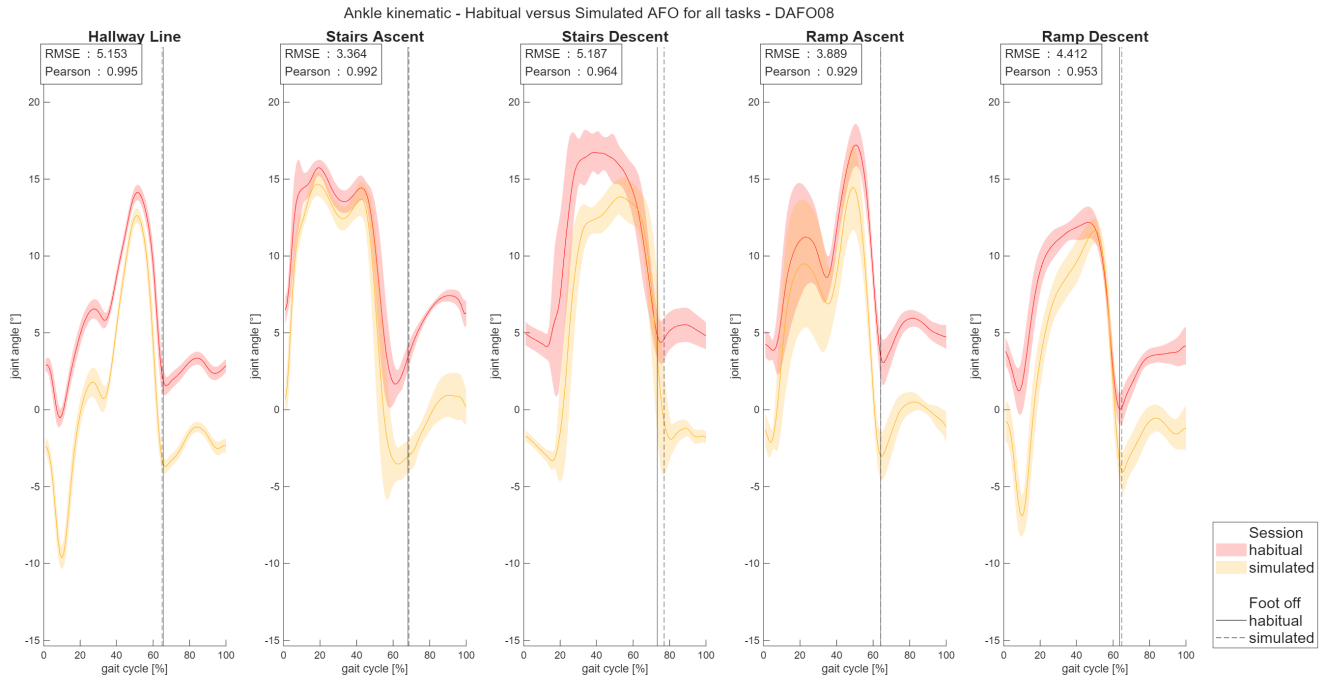


Figure A.6: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO08

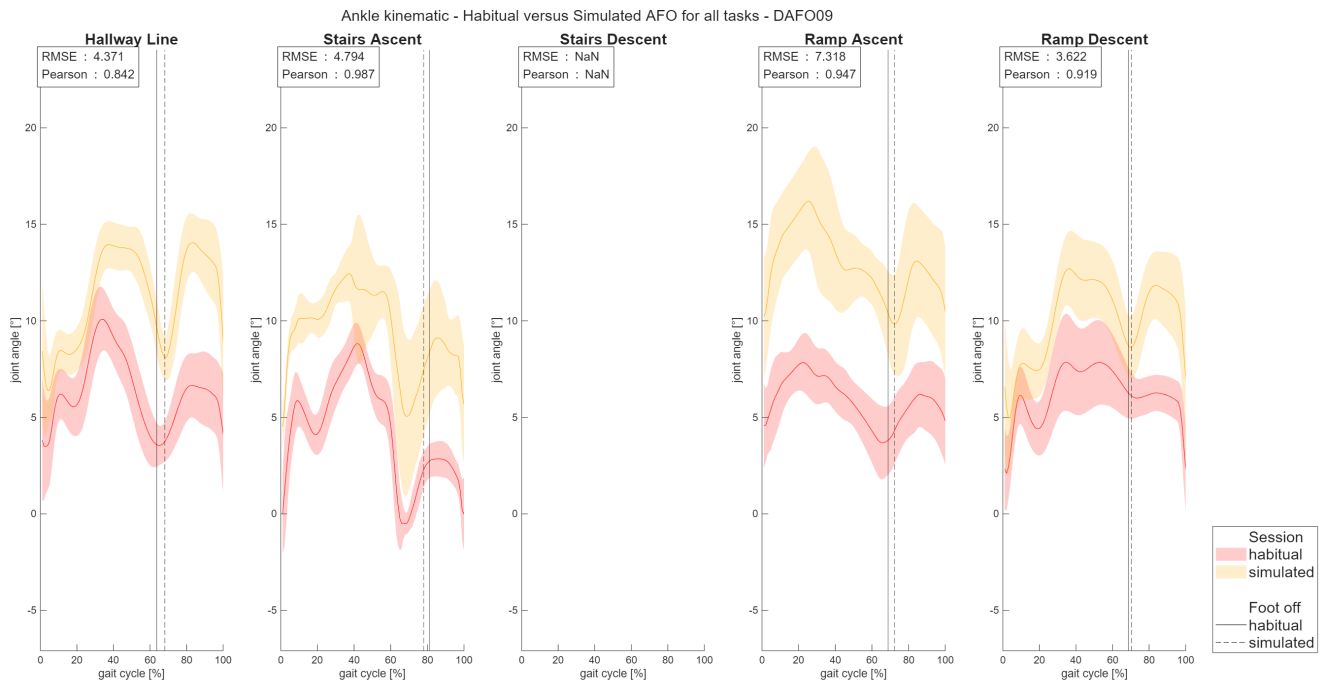


Figure A.7: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO09

Carbon Fiber Posterior Lateral Strut

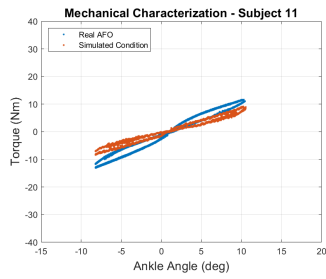


Figure A.8: DAFO11

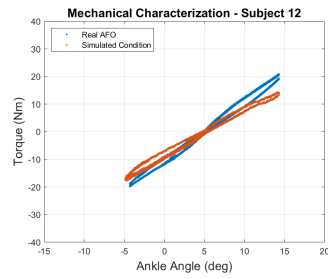


Figure A.9: DAFO12

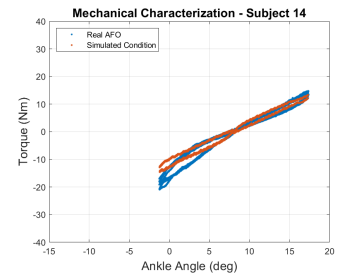


Figure A.10: DAFO14

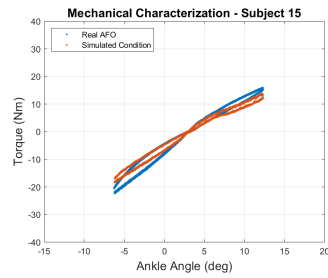


Figure A.11: DAFO15

Figure A.12: RIG data for Carbon Fiber Posterior Lateral Strut

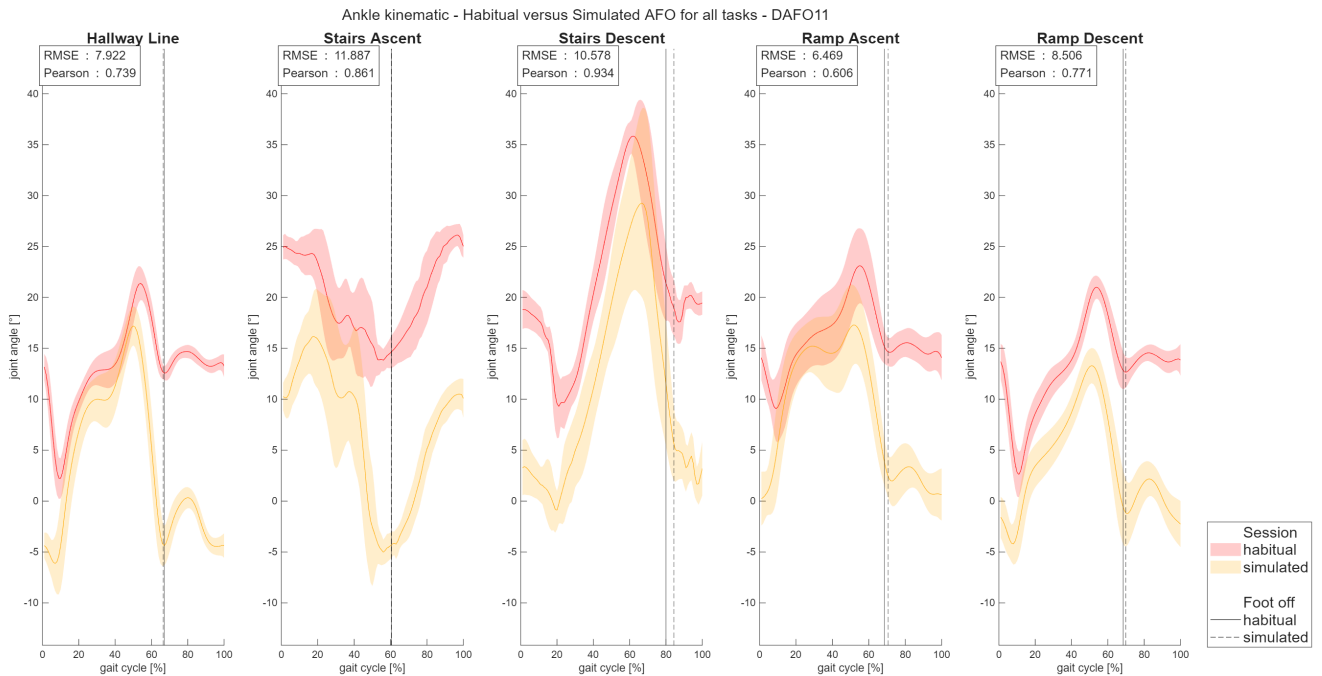


Figure A.13: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO11

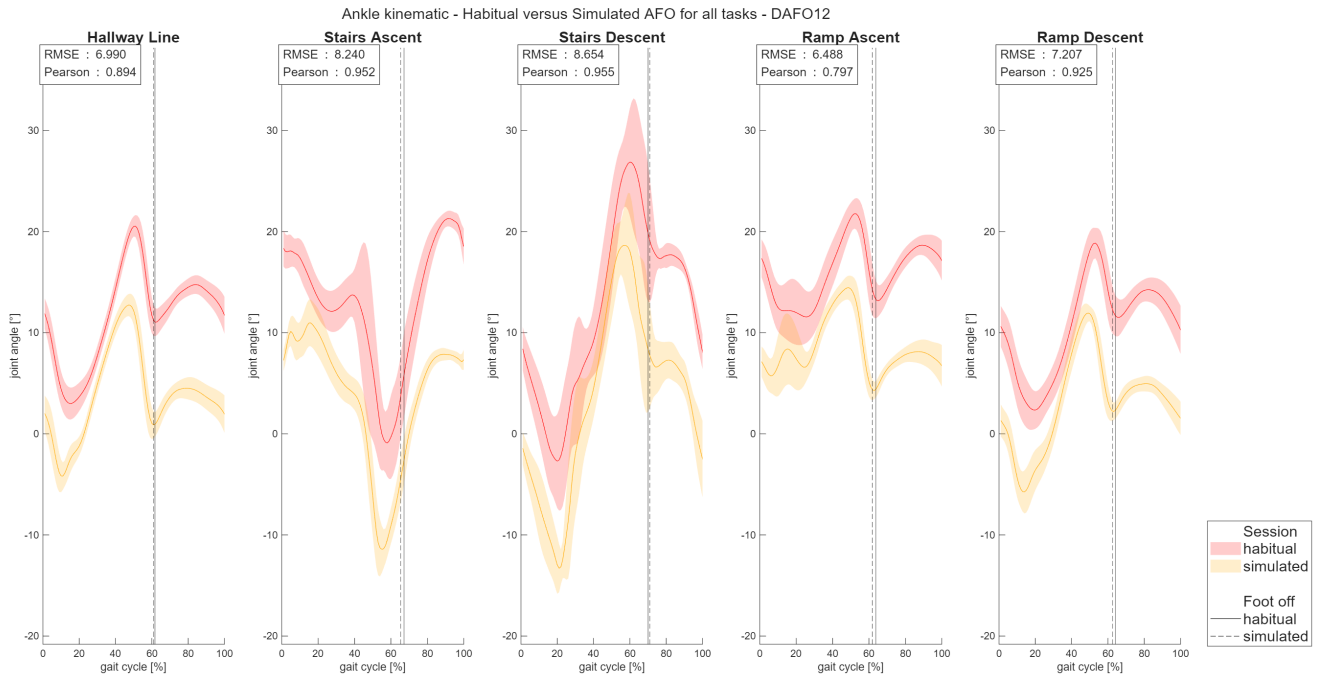


Figure A.14: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO12

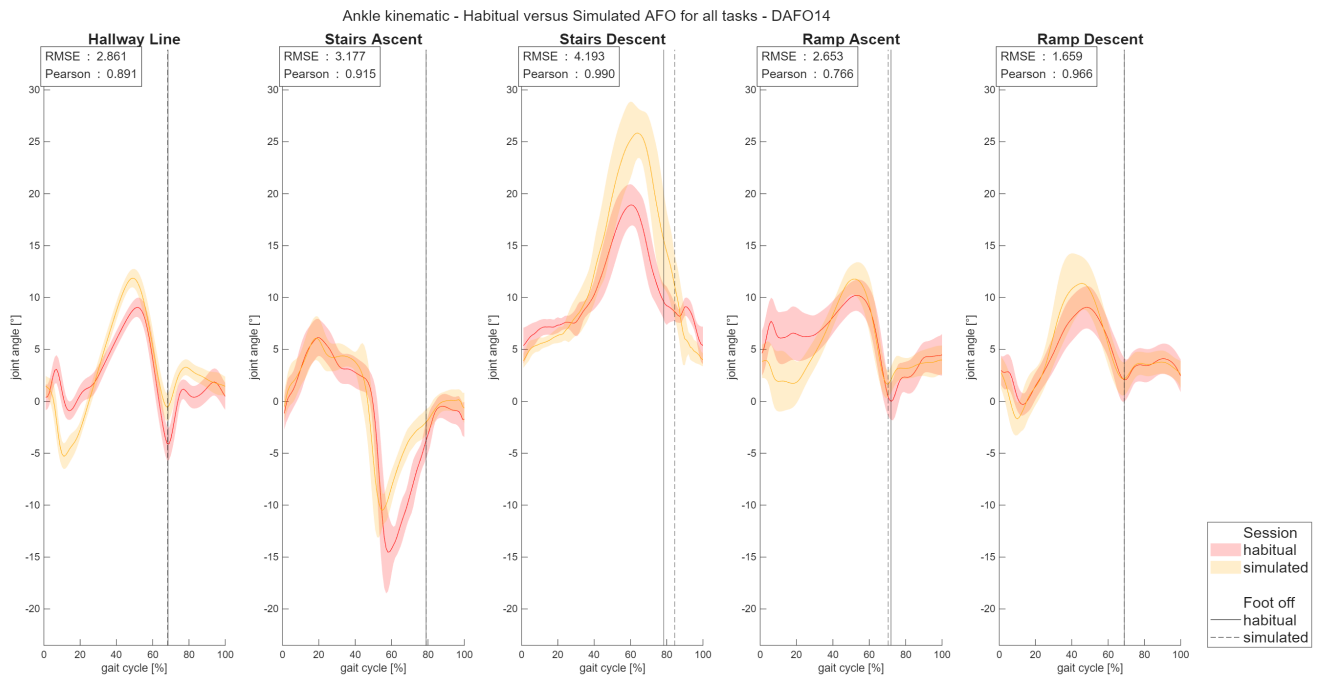


Figure A.15: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO14

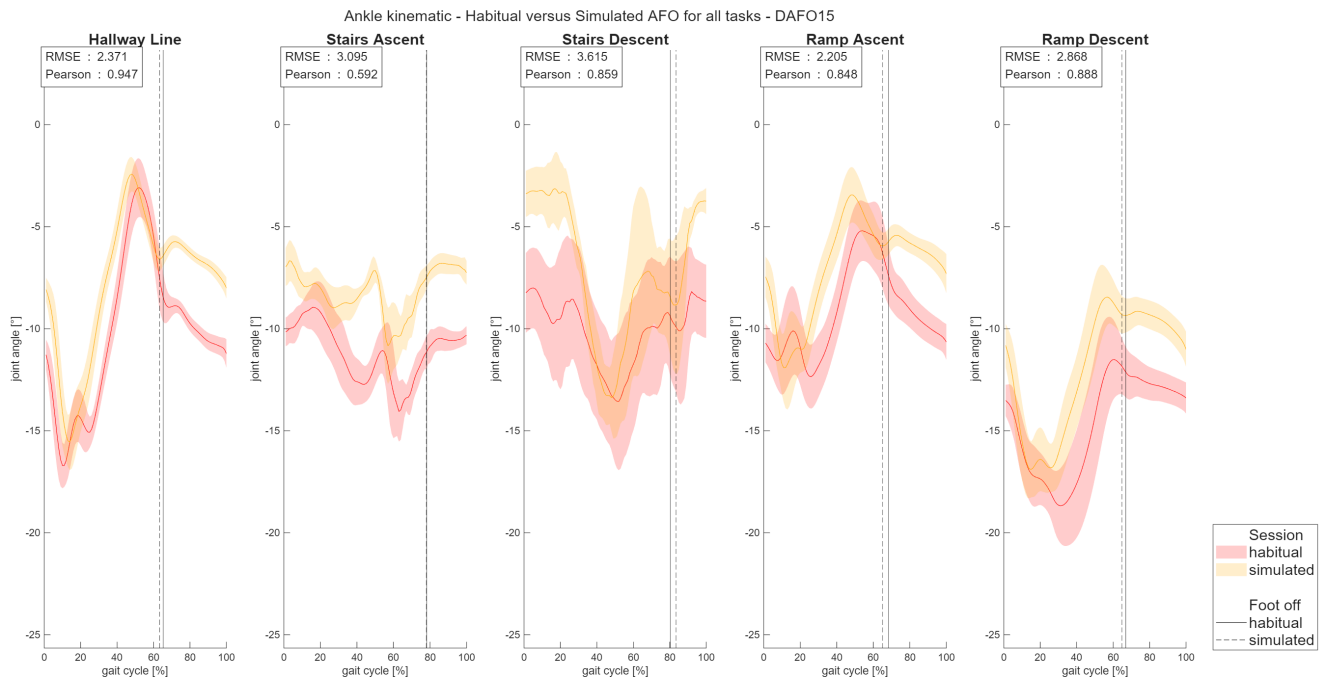


Figure A.16: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO15

Hinged AFO

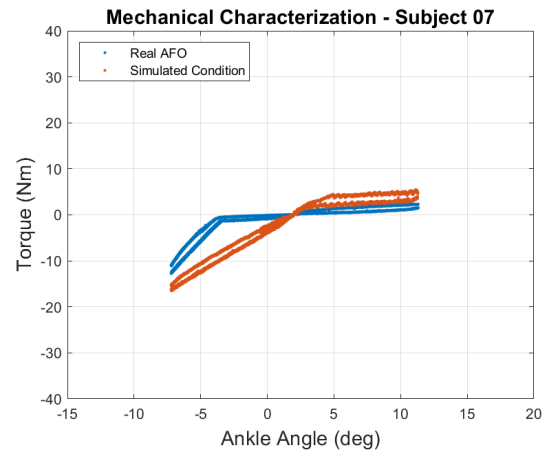


Figure A.17: DAFO07

Figure A.18: RIG data for Hinged AFO

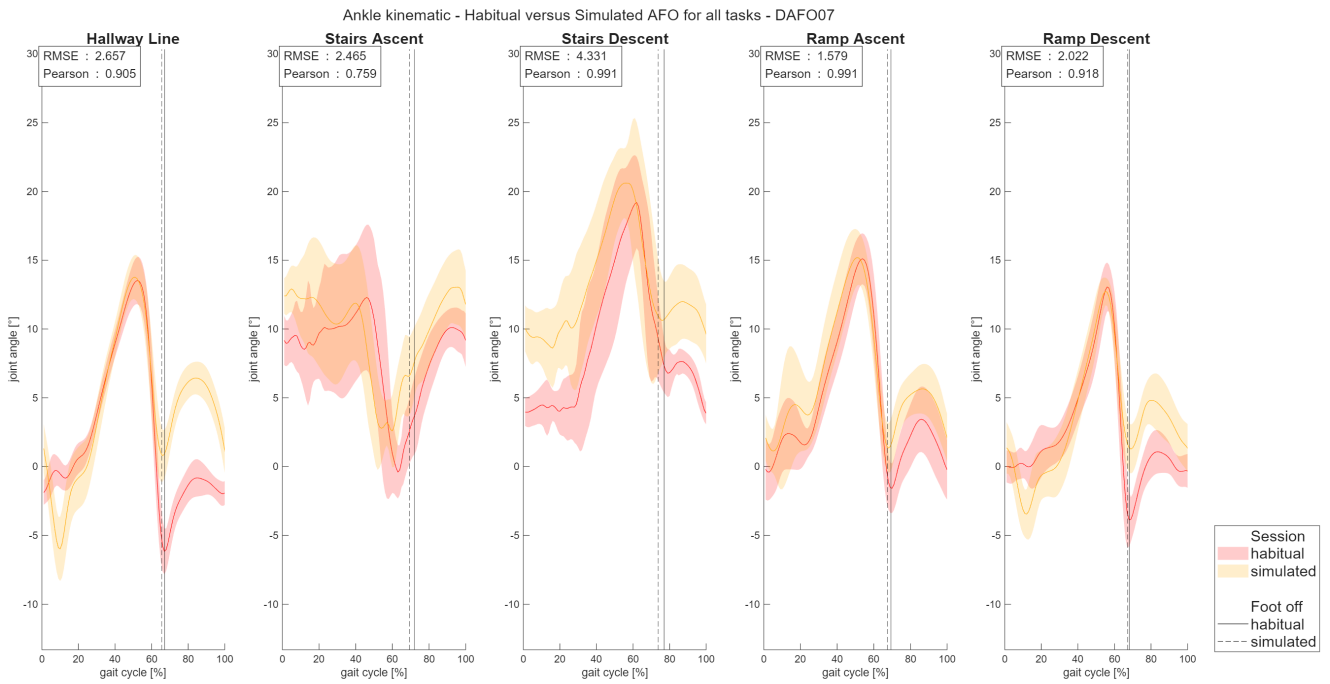


Figure A.19: Comparison of ankle angle when wearing the habitual and simulated AFOs - DAFO07

Characterization of gait progression in Adrenomyeloneuropathy using vicon motion capture - preliminary result

Abstract

Adrenomyeloneuropathy (AMN) is a rare neurodegenerative disease characterized by progressive abnormalities in leg strength, tone, and sensation. [1] Due to a limited number of patients and lack of treatments, the course of AMN-related disabilities over time is poorly or incompletely understood. As part of an observational study aiming to characterize the progression of the disease in adults with AMN, instrumented gait analysis has been performed on multiple patients annually. This report summarize our preliminary findings.

1 Introduction

Adrenomyeloneuropathy (AMN) is an inherited form of spastic paraplegia, occurring in the context Adrenoleukodystrophy (ALD), a progressive neurodegenerative disorder caused by a mutation in the ABCD1 gene localized to the X-chromosome (Xq28).[2, 3]

As the disease is a differential diagnosis of multiple sclerosis (MS) [3], it is hypothesized that AMN patients will exhibit gait characteristics similar to MS. Consequently, a set of parameters known to be impacted in MS were extracted.

For each visit, an individual gait analysis was performed. Subsequently, a longitudinal analysis, incorporating EDSS and SSPROM as proxies of the state of the disease, was conducted.

2 Methods

2.1 Data recording and processing

Data was collected from respectively 3,3 and 2 subjects at each clinical site (Spaulding Rehabilitation Hospital, Boston, MA; Stanford University, Stanford, CA; Cornell University, Ithaca, NY) annually for up to three years following the initial session (Table.1,Appendix).

Walking trials were recorded using a Vicon system and force plates. At Spaulding Rehabilitation Hospital, Vicon Nexus has been used for labeling, gap filling and gait event identification. Visual 3D was used for biomechanical modeling, signal processing and analysis. A custom lower limb bio-mechanical model was used for the analysis.

2.2 Gait parameters

Gait deviations in Adrenomyeloneuropathy (AMN) tend to progress similarly to those observed in the early stages of Multiple Sclerosis (MS). The selected set of spatio-temporal and kinematic gait parameters was chosen to capture the deviations most likely to be observed during the study.

In early-stage MS, patients typically exhibit subtle gait changes associated with instability and balance impairment. Even before the onset of clinically evident functional deficits, individuals may demonstrate reduced walking speed, shorter stride length, and prolonged double support time [4]. Increased step width has also been reported and is correlated with greater severity of impairment [5].

All active joint trajectories and range of motion tend to be affected. Notably, patient show reduced ankle range of motion and increased hip range of motion : hip flexion during terminal swing may increase as a compensatory strategy to overcome distal joint stiffness. [4]

These gait alterations may also affect peak ankle power generation as the disease progresses, although consistent findings in this area are still lacking .

All gait parameters were computed for each session using MATLAB.

2.3 Longitudinal analysis

Two clinical scales (EDSS and SSPROM) were used to measure disability and progression of the disease, enabling longitudinal analysis.

EDSS is a widely used assessment to provide a single measure of disability based on an ordinal rating scale ranging from 0 to 10, originally developed in the context of multiple sclerosis. [6]. An inclusion criterion for the study was an EDSS score between 1.0 and 6.5 at the initial clinical visit. This score was subsequently recorded at each follow-up visit.

SSPROM is a 10-item scale ranged from 0 (total disability) to 100 (normal) designed to assess progressive myelopathies. The domains covered by SSPROM are motor disability, sphincter dysfunction, spasticity, and sensory losses. [7]

3 Results & Discussion

All individual gait report can be found in the Additional materials.

During individual gait analyses, several deviations from normative patterns were observed across joints trajectories.

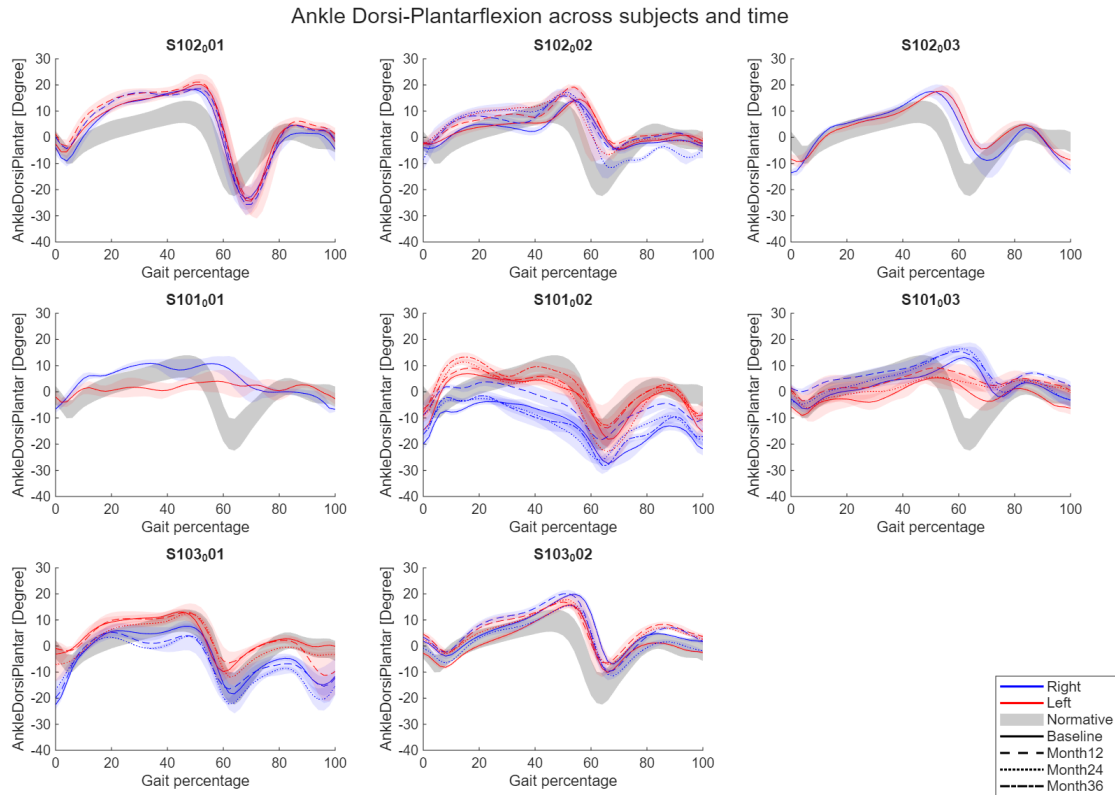
At the ankle joint, five patients showed a reduced range of motion, primarily due to diminished plantar flexion during the gait cycle.Fig.1

At the knee, four patients exhibited hyperextension during the stance phase, and two of them also demonstrated reduced knee flexion during the swing phase..Fig.2

Hip kinematics revealed that four patients walked with a slight but consistent flexion posture, although their range of motion remained within normal limits, while one patient (S101-001) showed a reduced hip range Fig.3

The most pronounced abnormalities were observed at the pelvis. This included inconsistent or exaggerated pelvic tilt (both posterior and anterior), highly irregular pelvic obliquity, and unusual rotation patterns in the transverse plane; three subjects exhibited an inverted rotation while others displayed inconsistent or noisy motion. These findings align with previous literature in Multiple Sclerosis [8]. That compensatory pelvic movements in the sagittal and coronal planes may develop in response to reduced mobility in the lower limb joints.Fig.4&5&6

Exaggerated foot motion during the swing phase, manifesting as either internal or external rotation, was observed in three individuals.Fig.7



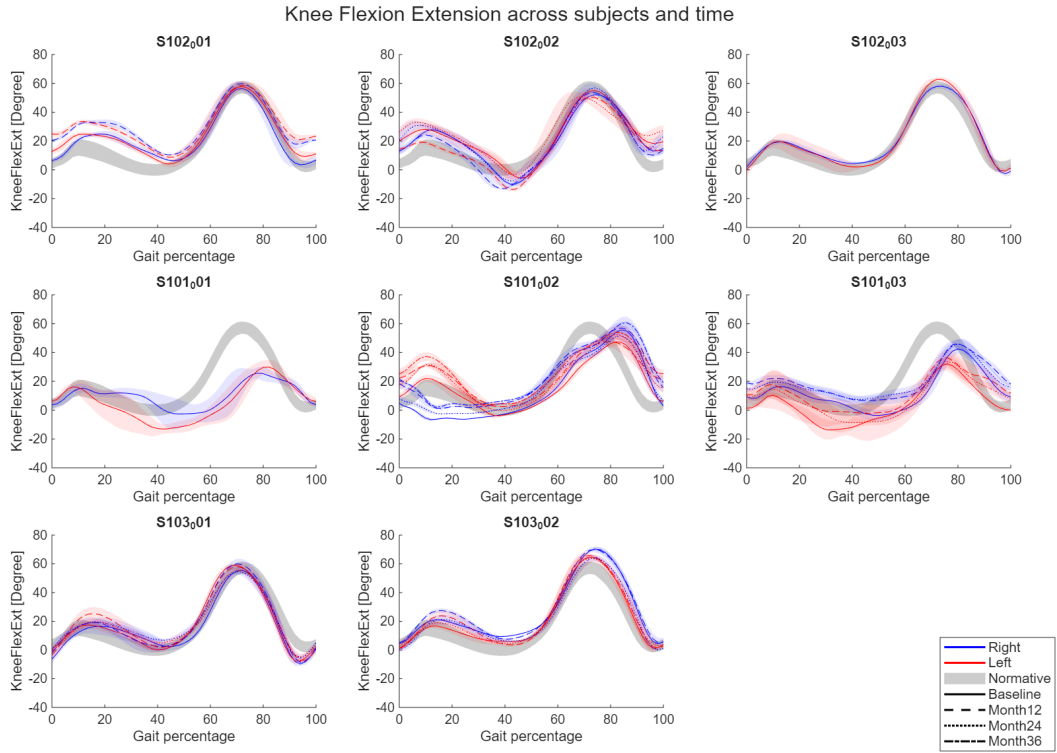


Figure 2: Knee flexion/extension for all subjects and sessions

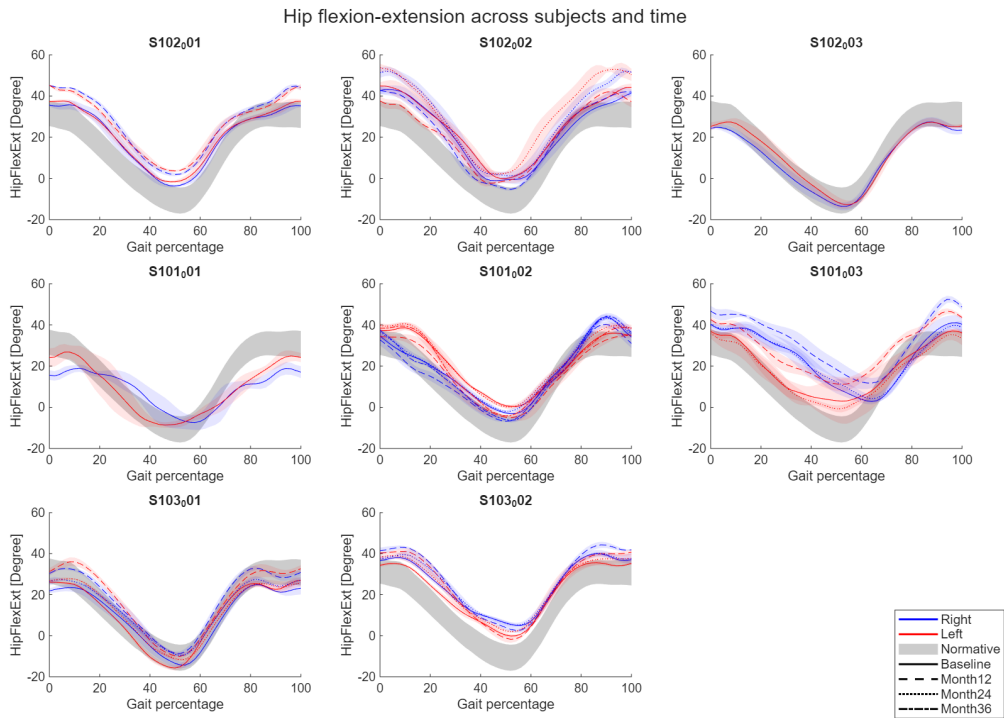


Figure 3: Hip flexion/extension for all subjects and sessions

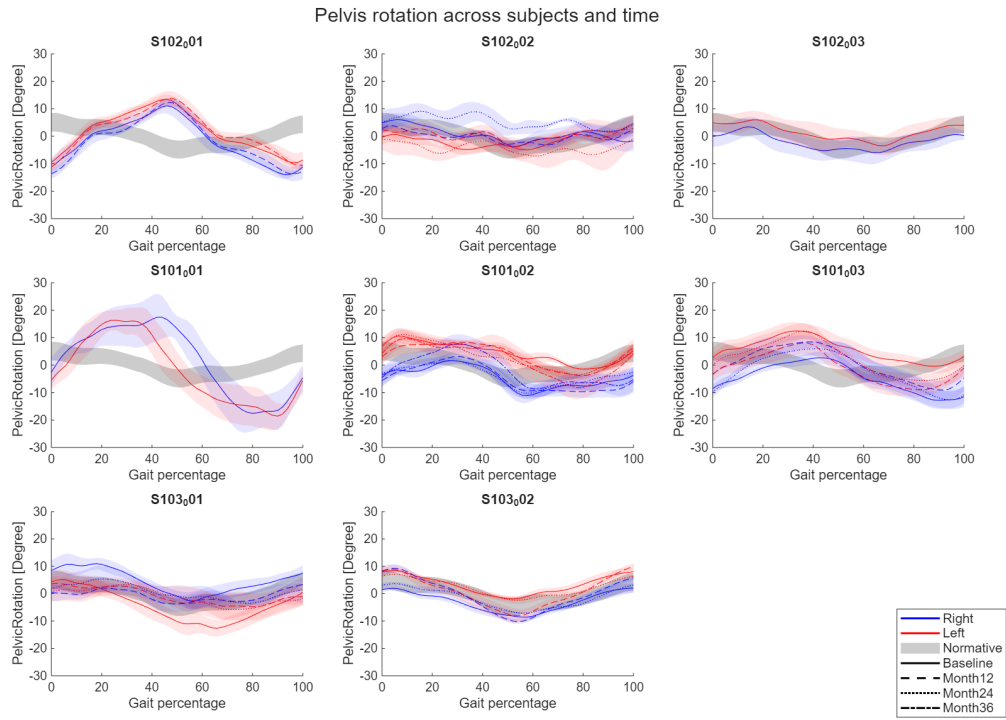


Figure 4: Pelvis rotation for all subjects and sessions

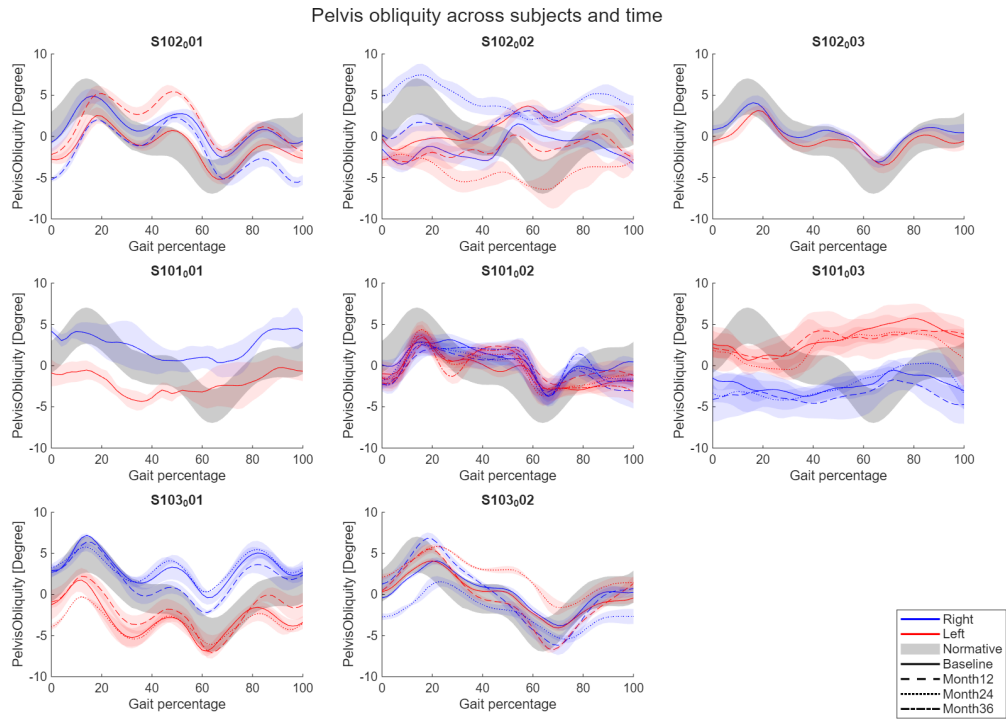


Figure 5: Pelvis obliquity for all subjects and sessions

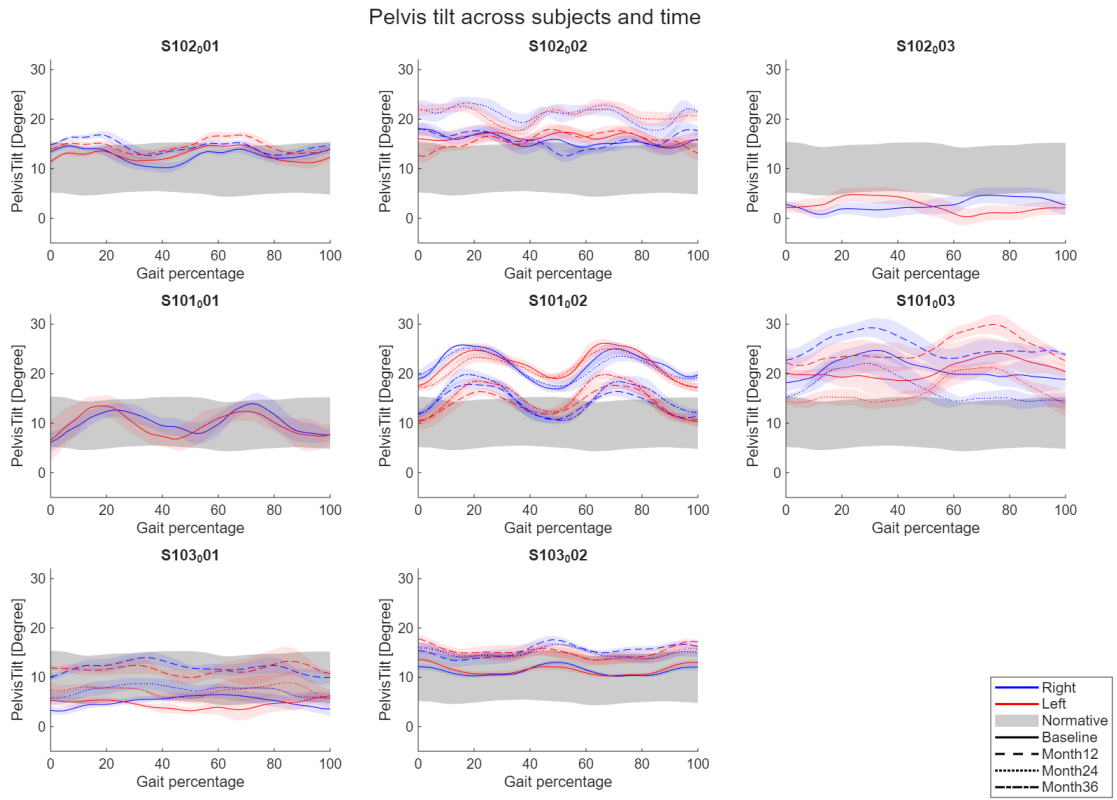


Figure 6: Pelvis tilt for all subject and session

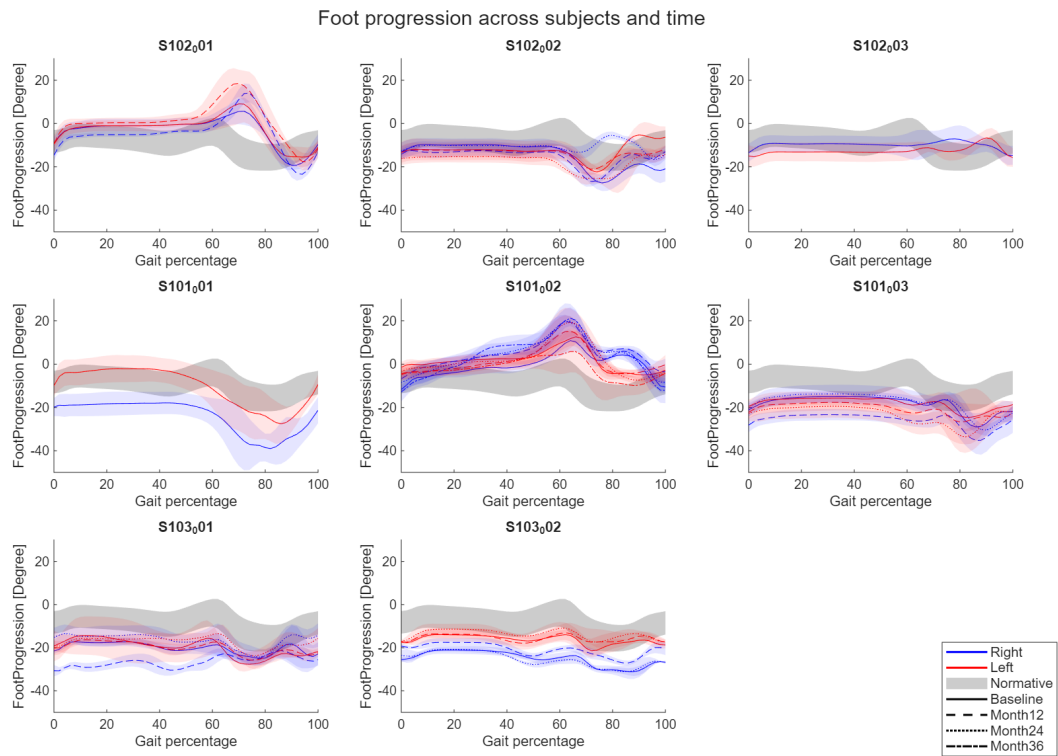


Figure 7: Foot progression, all subject and session

Given the correlation between EDSS and SSPROM scores, EDSS was used as a proxy for disease progression in the longitudinal analyses that follow.

In terms of spatio-temporal parameters, five patient showed a consistently longer double support time than normative data; however, a relationship between EDSS score and double support appeared. Similarly, most patients walked with wider steps than normative, with one individual exhibiting step widths two times and another three times greater than normative values. A slight relationship with disease severity was observed but yet insignificant due to the low number of patient. Walking speed was reduced in nearly all patients but two, and again a relationship between speed and EDSS score was observed. Stride length was symmetrical across limbs in all subjects. Only 2 patients demonstrated consistent shorter stride than normative across session, and five in at least half sessions. Again, this reduction did correlate with disease progression. These parameters (double support, step width, walking speed, and stride length) appeared to be interrelated. Fig.8&9

Joint excursions at the hip, knee, and ankle were generally within normative ranges. One patient (102-001) displayed increased ankle motion. Across the cohort, no clear longitudinal pattern was observed, though patients with greater impairment tended to show increased hip excursion and reduced ankle excursion, still within normative limits. Fig.10&11

Interestingly, peak hip flexion during swing exceeded normative values in five patients and was associated with a general tendency for the hip to remain in a flexed posture throughout the gait cycle. Ankle power generation was notably reduced, approximately half the normative value, in all patients' data collected at Stanford, and longitudinal analysis suggested a potential trend toward decreasing ankle power and increasing hip flexion during swing with advancing disease severity. Fig.12&13

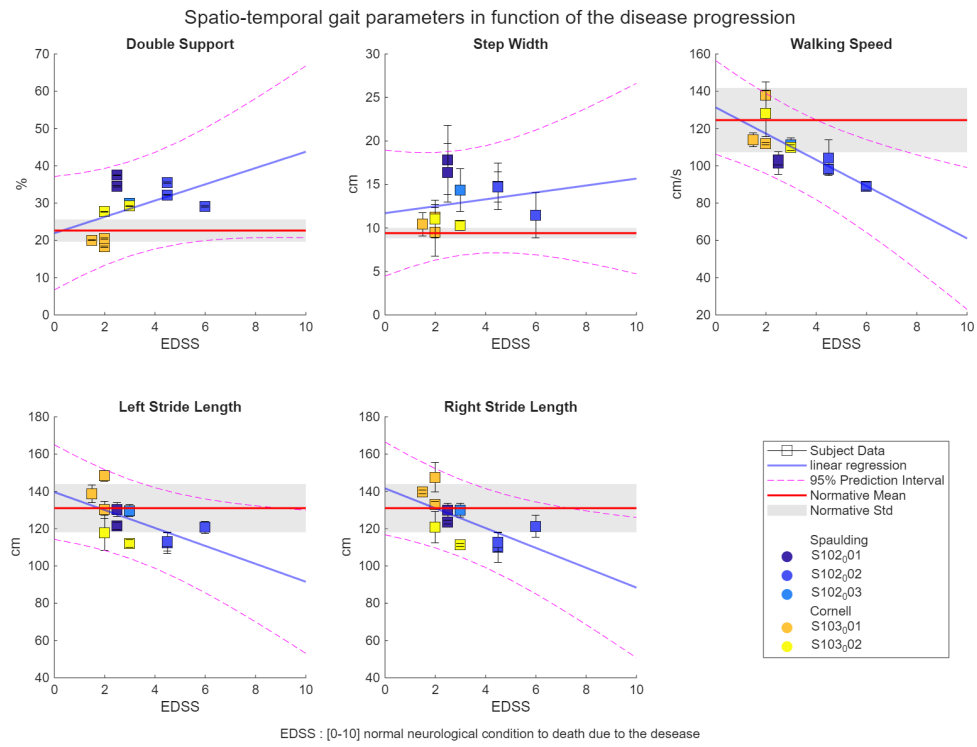


Figure 8: Longitudinal analysis - Spatiotemporal gait parameters in function of EDSS
Subject S102-002 was using a walker in the last session (EDSS 6)

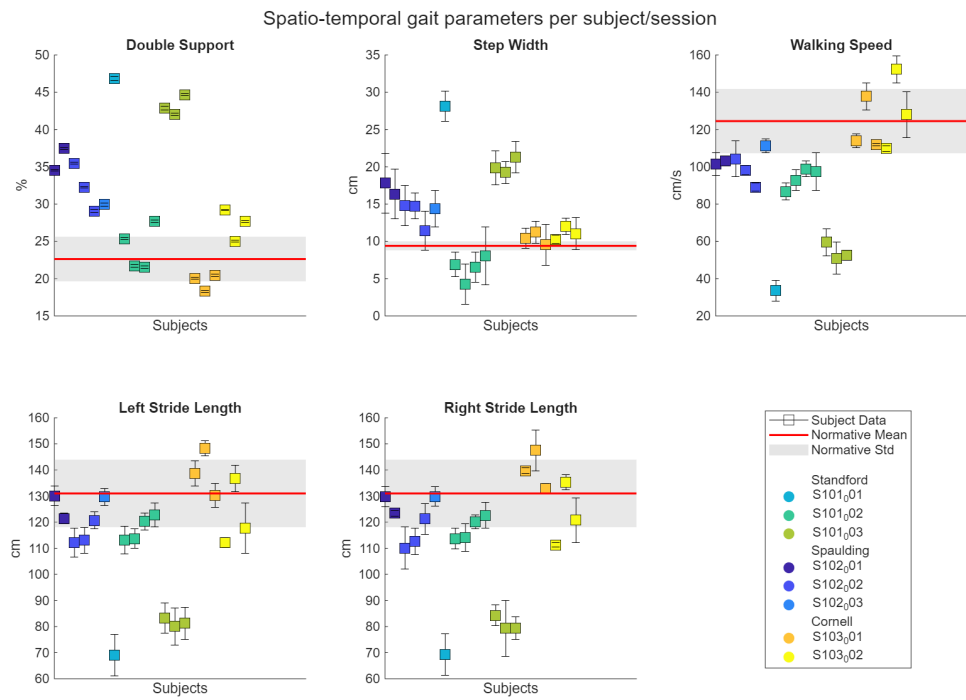


Figure 9: Longitudinal analysis - Spatiotemporal gait parameters

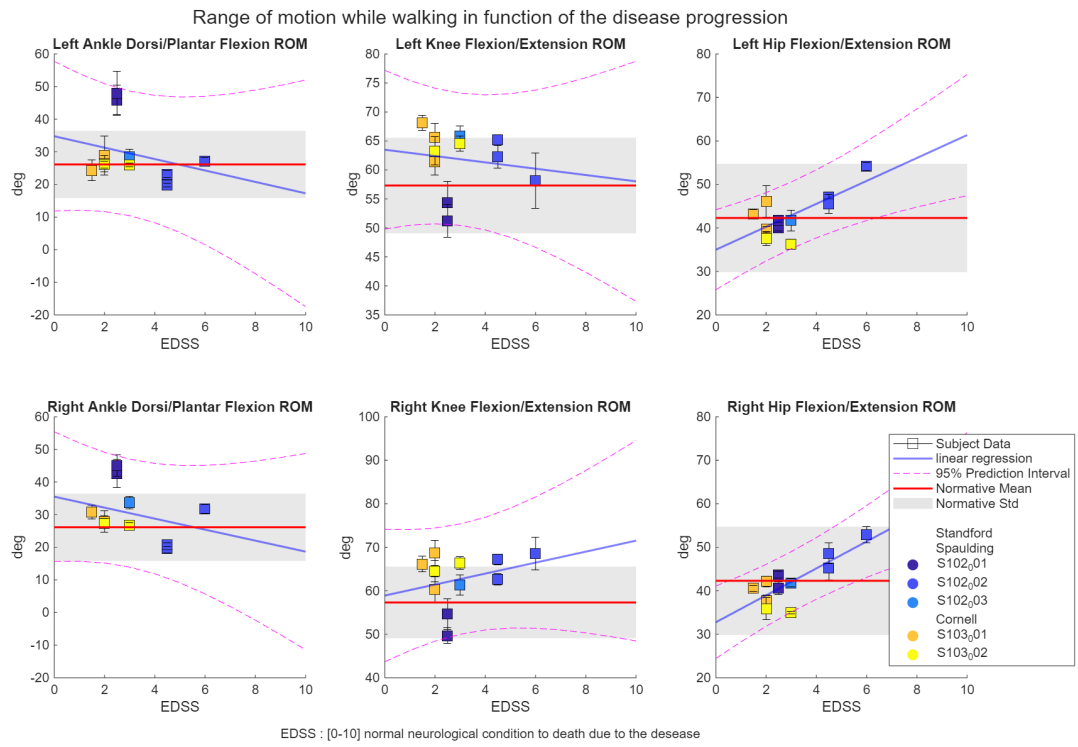


Figure 10: Longitudinal analysis - joint excursion in function of EDSS
Subject S102-002 was using a walker in the last session (EDSS 6)

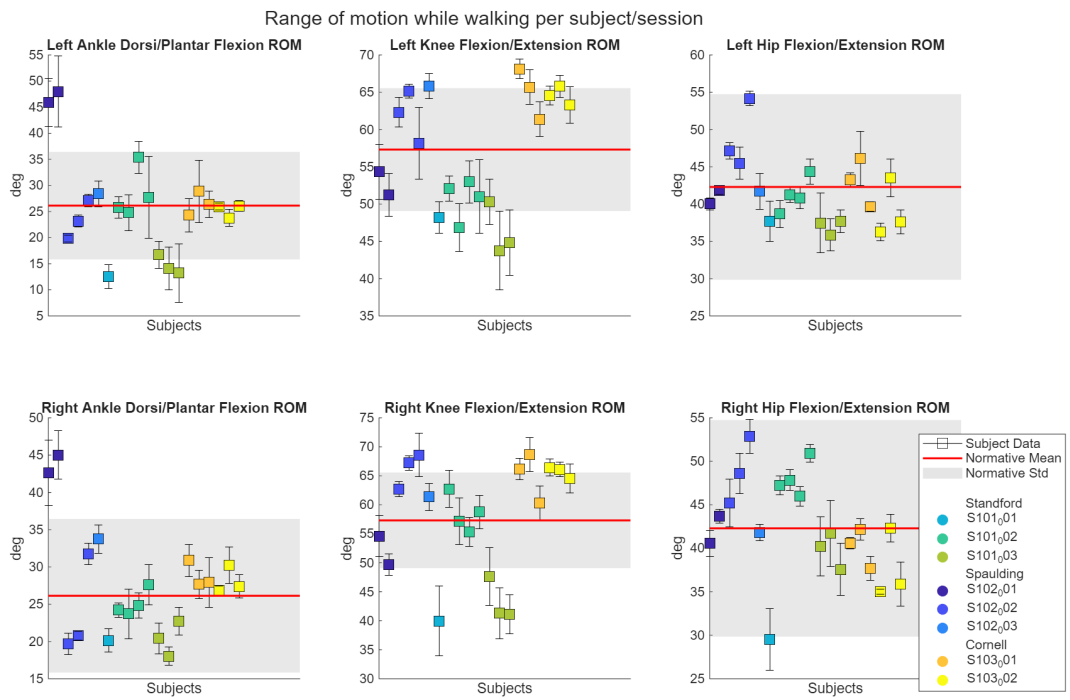


Figure 11: Longitudinal analysis - joint excursion

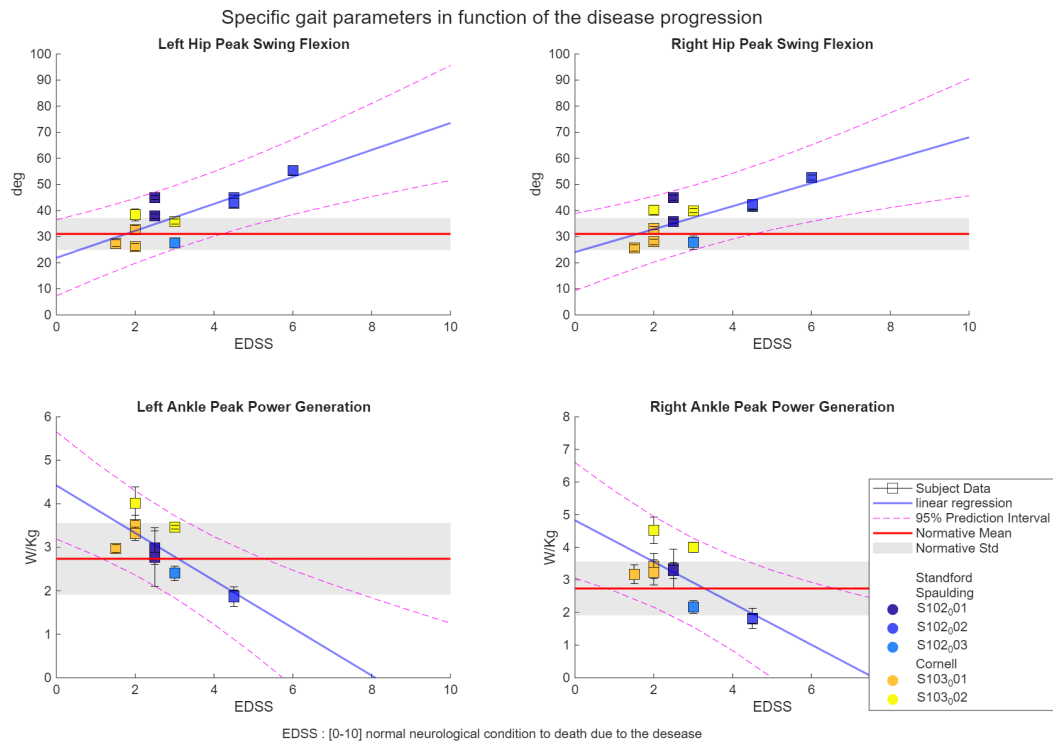


Figure 12: Longitudinal analysis - specific gait parameters in function of EDSS
Subject S102-002 was using a walker in the last session (EDSS 6)

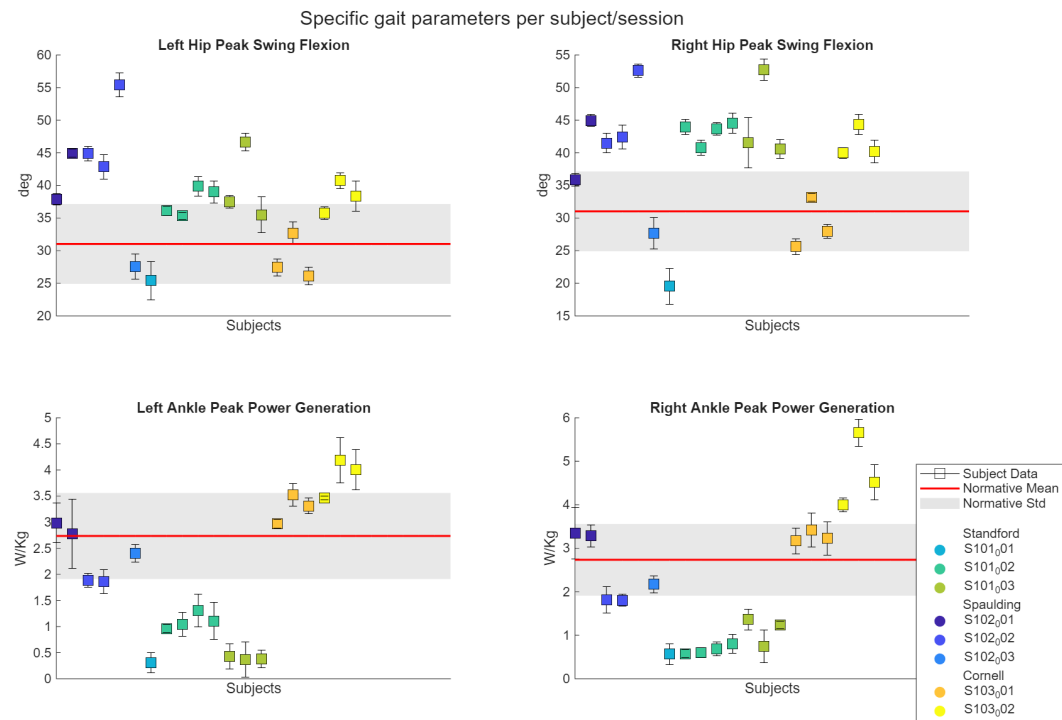


Figure 13: Longitudinal analysis - specific gait parameters

4 Conclusions

All calculated gait metrics appeared largely symmetrical, suggesting that the disease affects both sides of the body in a similar manner. Most patients demonstrated reduced walking speed, prolonged double support phases, wider step widths, and shorter stride lengths. These patterns are consistent with compensatory strategies used to manage gait instability. A persistent hip flexion posture was commonly observed and may serve as a compensation for reduced ankle mobility. Ankle power generation was low in the majority of subjects, in line with our initial hypothesis, and reduced plantar flexion was present in several patients. Compensatory pelvic movements were also observed, potentially reflecting adaptations to restricted motion at distal joints.

The small sample size limits our ability to draw strong conclusions regarding the progression of gait deviations over time. Furthermore, EDSS and SSPROM scores were only available for data collected at our institution (Spaulding Rehabilitation Hospital), which significantly restricts the scope of the longitudinal analysis. Access to EDSS data for all patients will greatly enhance the interpretability and statistical power of future analyses. Finally, future work should focus on assessing correlations between multiple gait parameters to better understand compensatory mechanisms and disease-related trends.

Appendix

Data availability summary

Table 1: Gait data Overview

Clinic	Subject	Baseline	Month12	Month24	Month36	EDSS/SSPROM
Stanford	101 001	X	-	-	-	-
	101 002	X	X	X	X	-
	101 003	X	X	X	-	-
Spaulding	102 001	X	X	-	-	X
	102 002	X	X	X	-	X
	102 003	X	-	-	-	X
Cornell	103 001	X	X	X	-	X
	103 002	X	X	X	-	12 missing

For patient 102-002, data from the most recent session has been included for completeness. However, this data cannot be used reliably in analyses because the subject was using a walker during the session. Despite this, the need for a walker within two years suggests a rapid progression of the disease in this patients.

References

- [1] Jennifer L. Keller, Joseph I. Wang, Jonathan Y. Kang, Joseph A. Hanson, Priya Kamath, Jennifer O. Swain, Gerald V. Raymond, and Kathleen M. Zackowski. Strength: A relevant link to functional performance in the neurodegenerative disease of adrenomyeloneuropathy. 26(9):1080–1088. ISSN 1545-9683, 1552-6844. doi: 10.1177/1545968312441682. URL <https://journals.sagepub.com/doi/10.1177/1545968312441682>.
- [2] Irene C Huffnagel, Wouter J C Van Ballegoij, Björn M Van Geel, Johanna M B W Vos, Stephan Kemp, and Marc Engelen. Progression of myelopathy in males with adrenoleukodystrophy: towards clinical trial readiness. 142(2):334–343. ISSN 0006-8950, 1460-2156. doi: 10.1093/brain/awy299. URL <https://academic.oup.com/brain/article/142/2/334/5233858>.
- [3] Anderson Rodrigues Brandão de Paiva, Carlos Rory Pucci Filho, Alan Monteiro Porto, Fabrício Stewan Feltrin, Fernando Kok, and Carlos Henrique Ferreira Camargo. When multiple sclerosis and x-linked adrenoleukodystrophy are tangled. 8(2):156–158. ISSN 2163-0402. doi: 10.1212/CPJ.0000000000000431. URL <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5914752/>.
- [4] M G Benedetti, R Piperno, L Simoncini, P Bonato, A Tonini, and S Giannini. Gait abnormalities in minimally impaired multiple sclerosis patients. 5(5):363–368. ISSN 1352-4585, 1477-0970. doi: 10.1177/135245859900500510. URL <https://journals.sagepub.com/doi/10.1177/135245859900500510>.
- [5] Uri Givon, Gabriel Zeilig, and Anat Achiron. Gait analysis in multiple sclerosis: Characterization of temporal–spatial parameters using GAITRite functional ambulation system.
- [6] J. F. Kurtzke. Rating neurologic impairment in multiple sclerosis: an expanded disability status scale (EDSS). 33(11):1444–1452. ISSN 0028-3878. doi: 10.1212/wnl.33.11.1444.

- [7] R.M. Castilhos, D. Blank, C.B.O. Netto, C.F.M. Souza, L.N.T. Fernandes, I.V.D. Schwartz, R. Giugliani, and L.B. Jardim. Severity score system for progressive myelopathy: development and validation of a new clinical scale. 45(7):565–572. ISSN 0100-879X. doi: 10.1590/S0100-879X2012007500072. URL <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3854272/>.
- [8] M. M. Mañago, P. W. Kline, E. Alvarez, and C. L. Christiansen. Trunk and pelvis movement compensation in people with multiple sclerosis: Relationships to muscle function and gait performance outcomes. 78:48–53. ISSN 1879-2219. doi: 10.1016/j.gaitpost.2020.03.006.